athenea: A Relight Baker for Gaussian Splatting from Production USD Assets

jesus luque

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ABSTRACT

We describe athenea, a relight baker for 3D Gaussian splatting: it converts production USD meshes and their materials into Gaussians and bakes into each Gaussian the light transport needed to relight it at draw time, with the reference always a path trace of the source mesh. A mesh is converted on the device into anisotropic Gaussians, one per cell of a camera-derived lattice, with coverage-correct opacity, edges held where a surface stops or folds, layered OpenPBR materials, glass classified as solid or thin wall, and roughness widened by the spread of normals under each Gaussian. The renderer's own GPU path tracer then bakes a transfer into every Gaussian: degree-3 spherical harmonics of direct and indirect light, a reflected field, 256 bits of open directions and the material's marks. At draw time these relight the cloud under any sky and any light, with sharp reflections turned per pixel by a stored shape operator. On a Chevrolet Corvette C7 of 192 meshes, doubling the bake resolution from 640 to 1280 px raised the relit car's PSNR against the path trace from 16.55 to 17.88 dB on its crop and from 22.18 to 23.90 dB on the frame, and a 5120 px close-up level of 74.5 million leaves in 24 trees was baked for close views. The baker rests on a USD and Hydra renderer in which every step that touches scene data runs as a GPU compute kernel. Rigged assets keep their skeleton, with a Jacobian that includes the weights' gradients, zonal transfer lobes that turn with a limb and per-part visibility fields. Per-part merge trees with a monotone, bounded screen error, back-face and occlusion culls, and pages ordered by error give levels of detail at a cost of about 0.1 dB. As one application, clustered PCA, vector-quantised visibility and a 40-byte compact record carry the relit car to a WebGPU player on an iPhone.

Keywords: relight baking; 3D Gaussian splatting; precomputed radiance transfer; spherical harmonics; USD; Hydra; mesh to Gaussians; path tracing; level of detail; clustered PCA; GPU-only pipelines; WebGPU.

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A.1 Repositories
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1. Introduction

1.1 The problem

Production 3D assets are authored as meshes: polygons, layered physically based materials, textures, lights and, for characters, skeletons, all composed in USD. They are rendered faithfully by a path tracer, which is too slow for interactive use, and interactively by a rasteriser, which needs the asset's full geometry and texture set resident and still approximates its light transport. 3D Gaussian splatting (3DGS) offers a third route: a cloud of anisotropic Gaussians, sorted and blended in screen space, draws photographic detail at interactive rates on modest hardware. A Gaussian cloud, however, is normally *trained* from photographs of a real scene. It holds the light of that capture, it cannot be relit, and it has no material, no skeleton and no relation to an authored asset.

This paper addresses the converse problem: a **relight baker** for Gaussian splatting. Given a photoreal USD asset (the case followed throughout is a Chevrolet Corvette C7 of 192 meshes with OpenPBR car paint, chrome, glass, rubber and lamps), the baker produces a Gaussian representation that

1. **relights**: each Gaussian carries the light transport baked from the asset, so that the cloud answers any sky and any light at draw time, not only the illumination it was made under;
2. is **faithful to a path trace** of the source mesh under that lighting, judged image against image and region by region;
3. keeps a rig where the asset has one, so that an animated asset still moves and its light turns with it;
4. scales from far views to close-ups through levels of detail.

Neither of the obvious routes meets these. Training a cloud from path-traced views of the asset gives a fixed illumination and spends hours of optimisation per asset. Rasterising the mesh with an approximate shading model gives up the path trace's light transport. The route taken here converts the mesh directly into Gaussians and bakes the asset's light transport into each Gaussian with the renderer's own GPU path tracer; the relit cloud is then compared with the path trace of the mesh. Reducing, compressing and streaming the baked cloud to a WebGPU player on a phone is a parallel application of the same data (§7).

1.2 Goals and constraints

The central requirement is fidelity of the relit Gaussians to the path-traced reference. The design requirements below hold throughout and shape almost every design decision in Sections 2–7.

requirement	statement	where it acts
Fidelity to the path trace	The reference is the path trace of the source mesh under the same lighting. Where quality and frame rate conflict, quality takes precedence.	§3.10, §4.10, §8
GPU-only arithmetic	No arithmetic on scene data runs on the CPU: no CPU renderer, fallback or test oracle. The host reads files and keeps the books (counts, slots, queues).	Every section; §2.1 states the rule
Pure 3DGS at draw time	What is drawn is Gaussians only. The ground is not a mesh: its interaction with the object is converted into Gaussians (a shadow catcher). A mesh is only an input to the bake.	§3.9, §7.9

requirement	statement	where it acts
Automatic, not manual	Per-asset choices are detected from the data: the cell size from a camera, solid versus thin glass, closed shells, the ground for the shadow catcher. Options only override.	§3.3, §3.5, §3.8, §3.9
Small pieces held to the same standard	Badges, grilles, trims and lamps are judged as carefully as panels, close up as well as far.	§3.11, §6.9, §7.12

1.3 Contributions

The contributions described in this paper are the following.

1. **A relightable transfer baked into each Gaussian** (TX, Section 4), the core of the baker: degree-3 spherical harmonics of direct and indirect transfer, a reflected field normalised over closed directions and coupled to the sky at draw time, 256 bits of open directions read as cones by each reflection lobe, and the material's marks; per-pixel sharp lobes driven by a stored shape operator, with the bake's widening undone and a tube distinguished from a fold.
2. **A mesh-to-Gaussian conversion** (mesh2splat, Section 3) that prepares the baker's input, run as an AOFX effect on the device, with cells derived from a camera's pixel footprint, opacity that reproduces a surface's coverage, hard edges and closed shells found on the device, Toksvig widening of roughness by the normals' spread, layered OpenPBR capture, and automatic glass classification.
3. **A GPU-only USD renderer** (athenea, Section 2) on which the baker runs, drawing meshes, points and Gaussian splats through Hydra, with a tile rasteriser and a Gaussian ray tracer that share one Gaussian layout, a MaterialX-to-Slang material compiler checked against MaterialX itself, a path tracer with multiple importance sampling used as ground truth and as the bake's integrator, and a verification discipline in which every image comparison is itself a GPU kernel.
4. **Skinned Gaussian clouds** (Section 5) that carry the rig rather than the frames: a skinning kernel using the full Jacobian including the weights' gradients, zonal transfer lobes that turn with the Gaussian, per-part visibility fields read as a product of table lookups, and decimation that keeps the rig.
5. **Per-part merge trees with a bounded error cut** (Section 6): pair merging under guards that respect parts and materials, a least-squares coverage opacity for merges, an error made monotone with covering spheres so that one parallel test selects the cut, back-face culling of closed shells, a conservative shared occlusion codebook, and pages ordered by the parent's error.
6. **A measured account** (Section 8) of what each step of the bake buys against the path trace, on one production asset, through the relit leaves, the cuts and the 640, 1280 and 5120 px bakes.
7. **An application to mobile playback** (Section 7): a `.athc` v3 container with byte-plane encodings, clustered PCA of the transfer kept compressed on the GPU, vector-quantised open-direction bits, a 40-byte compact record, quantised traversal records of 5–11 bytes, and a frame-time and still-frame policy for a WebGPU player on an iPhone.

1.4 Organisation

Section 2 describes the engine and renderer on which the baker runs. Section 3 describes the conversion from USD meshes to Gaussians, and Section 4 the light transport baked into them and how they are relit. Section 5 covers animated assets: skinning, the sparrow and the pawn. Section 6 covers levels of detail.

Section 7 describes the application to mobile: compression and a WebGPU player. Section 8 consolidates the evaluation from Sections 2–7, Section 9 gathers the limitations and the work left, and Section 10 lists the references, followed by an appendix on the repositories and commands (Figure 1).

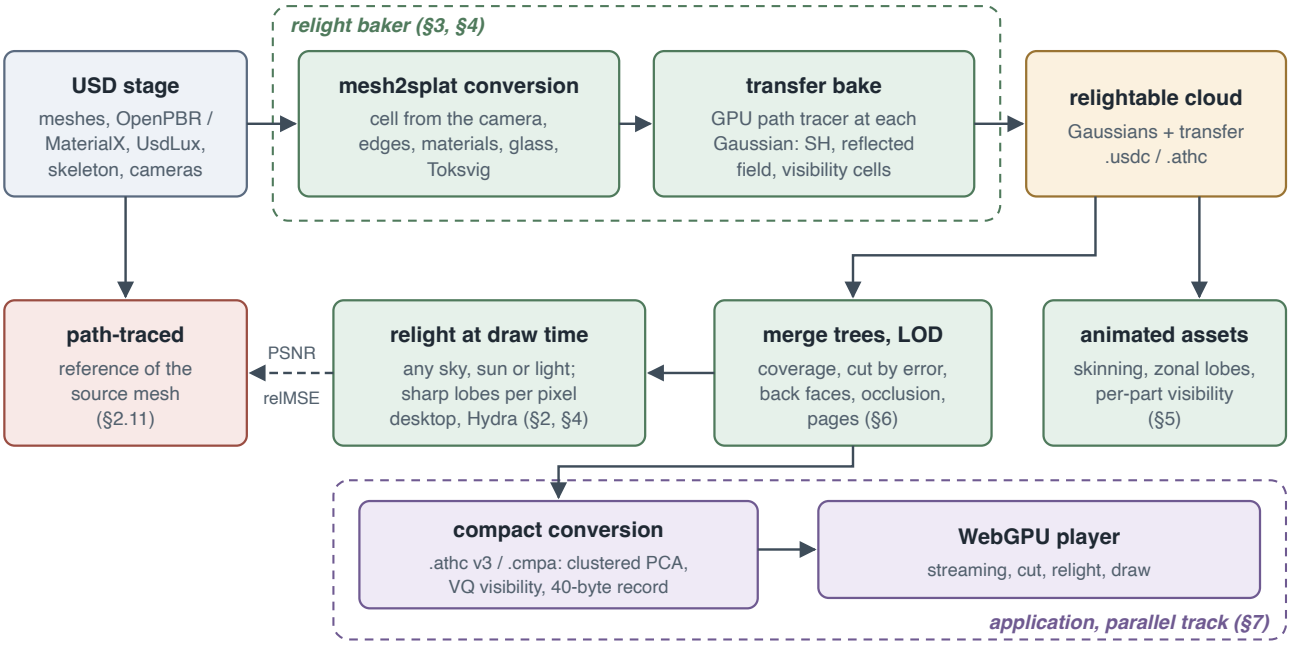


Figure 1. The pipeline. The relight baker converts a USD stage into Gaussians (§3) and bakes light transport into each one with the path tracer (§4); the relit cloud is compared with a path trace of the source mesh. Merge trees (§6) and animated assets (§5) build on the baked cloud; the WebGPU player (§7) is an application.

1.5 Related work

Gaussian splatting. Kerbl et al. (2023) introduced 3D Gaussian splatting: a scene as anisotropic Gaussians with spherical-harmonic colour, optimised from photographs and drawn by a tile-based, depth-sorted rasteriser. Its projection is the EWA splatting of Zwicker et al. (2001, 2002), a local affine approximation of the perspective map. Mip-Splatting (Yu et al., 2024) adds a 3D smoothing filter and a 2D screen filter against aliasing; athenea's 0.3 px² dilation with its opacity compensation is of that family (§2.5.1). 2D Gaussian splatting (Huang et al., 2024) flattens each Gaussian to an oriented disc evaluated by ray–plane intersection, in the line of surfels (Pfister et al., 2000); athenea's surfels follow it (§2.5.2). 3D Gaussian ray tracing (Moënne-Loccoz et al., 2024) traces particles through bounding proxies in a BVH, which athenea's hardware route follows (§2.6).

From meshes to Gaussians. Electronic Arts' mesh2splat converts a mesh into Gaussians by rasterising each triangle in its dominant axis plane, one Gaussian per covered texel; athenea's conversion starts from it (§3.1). Mesh2GS (arXiv 2606.21898) derives a Gaussian's size from a camera's pixel footprint, which athenea's `--cell-from-camera` follows (§3.3).

Precomputed light transport. Precomputed radiance transfer (Sloan, Kautz and Snyder, 2002) stores, per point, the projection of visibility times the cosine onto spherical harmonics, so that relighting is a dot product with the environment's coefficients; Ramamoorthi and Hanrahan (2001) showed that nine coefficients represent irradiance well. Clustered PCA (Sloan et al., 2003) compresses such transfers by clustering and a per-cluster principal basis, which athenea uses both for the file and on the phone's GPU (§7.2). Zonal har-

monics, which rotate by rotating an axis, are fitted after Sloan (2008) (§4.4.5, §5.7). Filtered importance sampling (Křivánek and Colbert, 2008) prefilters the environment (§4.3), and Lagarde and de Rousiers (2014) give the specular occlusion fit that the first transfer used (§4.6.1).

Specular antialiasing. Toksvig (2005) widens a microfacet distribution by the variance of the normals it averages; athenea applies the same idea at conversion, to the spread of normals under one Gaussian (§3.7, §4.9).

Levels of detail for Gaussians. Octree-GS (Ren et al., 2024) organises anchors in an octree selected by view distance. Hierarchical 3D Gaussians (Kerbl et al., 2024) build a merge hierarchy from Gaussian moments and cut it by screen size; athenea's original level of detail follows its moments (§6.1). NanoGS (Xiong et al., 2026) merges mutual nearest neighbours pairwise, which athenea's per-part trees run on the device (§6.2). The cost uses Runnalls' (2007) bound for Gaussian mixture reduction, and the error-band pages follow Nanite's (Karis et al., 2021) use of a monotone error for a parallel cut (§6.3, §6.8).

Engines and formats. The engine is built on OpenUSD and Hydra, MaterialX and OpenPBR materials, Slang shaders compiled through slang-rhi to Metal, Vulkan and CUDA, Intel Open Image Denoise and OpenColorIO. The player is a fork of Spark with a WebGPU backend.

1.6 Conventions

Unless stated otherwise, measurements were taken on an Apple M5 Pro under Metal. Images are compared on the GPU (§2.15). **PSNR** is in dB on display-encoded images; **relMSE** is relative mean squared error on linear values. **The path trace** (PT) of the Corvette is the mesh path-traced from `/World/Camera` at 1920 × 1080 under the `autoshop_01_4k` sky. **Car crop** is PSNR on the 900 × 420 crop of the car, **frame** on the whole frame; a **class ratio** is a region's mean over the path trace's. The validate-corvette harness of the player measures other regions (car, door, sill, bumper, hood), which are not the car crop (§7.12.3). The words *Gaussian* and *splat* are used interchangeably; a *leaf* is a baked Gaussian, a *merge* a node of a merge tree, and a *cut* the set of nodes drawn for one view. *Transfer* is the baked light transport (§4), *TX* its current form.

2. The athenea engine and renderer

athenea is a USD renderer built so that every step that touches scene data runs on the GPU. It draws three kinds of primitive: meshes, with MaterialX materials and UsdLux lights; points; and 3D Gaussian splats. It offers two techniques, an interactive raster and a path tracer that serves as ground truth. Hydra feeds both through the `hdAthenea` render delegate. This section covers the engine's architecture, the algorithms each subsystem runs, how correctness is established without a CPU oracle, and what has been measured. Unless stated otherwise, every number is quoted from `docs/decisions.md`, with its unit and conditions, and was measured on an Apple M5 Pro under Metal.

2.1 Design rules

The codebase holds to a small number of rules. Where possible, each one is enforced by a test.

1. **No CPU arithmetic on data.** athenea has no CPU reference renderer, no CPU fallback and no CPU test oracle. The host reads files, parses headers, decompresses, and keeps the books: counts, slots and queues. Decoding, sorting, merging, culling, mip generation, light power prefixes and image comparison are all compute kernels. Steps that had run on the host were moved to the device; examples are per-element interleaving of `ParticleField` arrays, half-to-float conversion, per-pixel conversion of Hydra render buffers, and each light's cumulative share of the frame's power (`light_prefix.slang`). The one documented exception is Dear ImGui's tessellation, which processes viewer chrome and no scene data.
2. **Binding by name.** Every shader parameter is bound through reflection (`cursor["name"].setBinding(...)`, `ComputeKernel::dispatch`) and never by slot, which removes the class of errors in which a slot count drifts from the shader's declaration.
3. **Errors as values.** Errors are reported as `Result<T>` with `ATHENEA_TRY`, and nothing throws. OS calls live only in `modules/core/Platform`, which is also where a Windows port would start.
4. **One Slang, one slang-rhi, one TBB per process.** OpenUSD, OIDN and the engine share one TBB, and the `single_tbb` test checks the count. For the same reason OIDN is built from source: its release binaries ship a second `libtbb.12`.
5. **GPU devices only, everywhere.** OIDN is built for GPU devices only, because a denoiser that falls back to the CPU is a CPU fallback. A test whose capability is missing on a device skips and names the missing capability.

The backends are those of slang-rhi: Metal (macOS, iOS), Vulkan and CUDA (Linux, NVIDIA L4). All kernels are written in Slang.

2.2 Module architecture

Each module under `modules/<name>` is a static library, `athenea::<name>`. Modules are listed in dependency order and each links only the ones above it (`modules/CMakeLists.txt`) (Figure 2).

Module	Contents
core	`Result`, logging, `Platform` (all OS calls)
ui	viewer panels as a description (`Controls.h`, `GaussianPanel.h`), walked by `athenea view` and the iOS app

Module	Contents
sched	`FrameClock`: genlock PTP and ST 2059-1 alignment, timecode
image	`Image`: float32 RGBA premultiplied, bottom-left origin; the type gpe and AOFX exchange
io	CPU file readers (PLY header, SPZ, SOG zip/WebP, IES), EXR with attributes
gpu	slang-rhi device, `ShaderLibrary`, `ComputeKernel`, `CommandBatch`, `Buffer`, `AsyncReadback`; `gpu/algo` (Prefix-Sum, RadixSort)
gpu_host	gpe adopting slang-rhi's device: one `MTLDevice` or CUDA context, buffers shared without copies
colour	`ColourCompiler` (OpenColorIO as a compiler into Slang), `ColourNames`
scene	`CloudLoader`: raw records uploaded and decoded on the GPU into `GpuSplats` / `GpuPoints`
render	`TileRasterizer`, `GaussianRayTracer`, `PointRasterizer`, `ReferenceRenderer`, `Camera/Projection`, `SplatEdit`
geom	`MeshBuilder`: Hydra meshes triangulated, smooth-normalled and primvar-expanded on the GPU
material	`TextureStore`, `MaterialCompiler` (MaterialX → Slang), the lobe library
light	UsdLux lights on the device: one record per light, sampling, densities, light BVH
world	`GpuScene`, `Instancing`, `RayTracingScene` (BLAS/TLAS), `BvhScene` (two-level compute LBVH)
technique	`VisibilityRaster/Trace/Bvh`, shading, `AovShading`, path tracer, `Denoiser`, `DisplayTransform`, `SplatVisibility`
lod	`LodBuilder`, `CutSelector`, the `.athc` reader/writer and `StreamingPool` (Section 6)
usd	`Engine`, `StageRenderer`, `Export`, the `hdAthenea` plugin, codeless schemas
mcp	the engine as an MCP server (JSON-RPC 2.0)
aofx	this engine's side of the AOFX host (aopenfx `host/`), `RenderEffect`, the bundles
view	`athenea view`: GLFW window, Dear ImGui on the engine's device

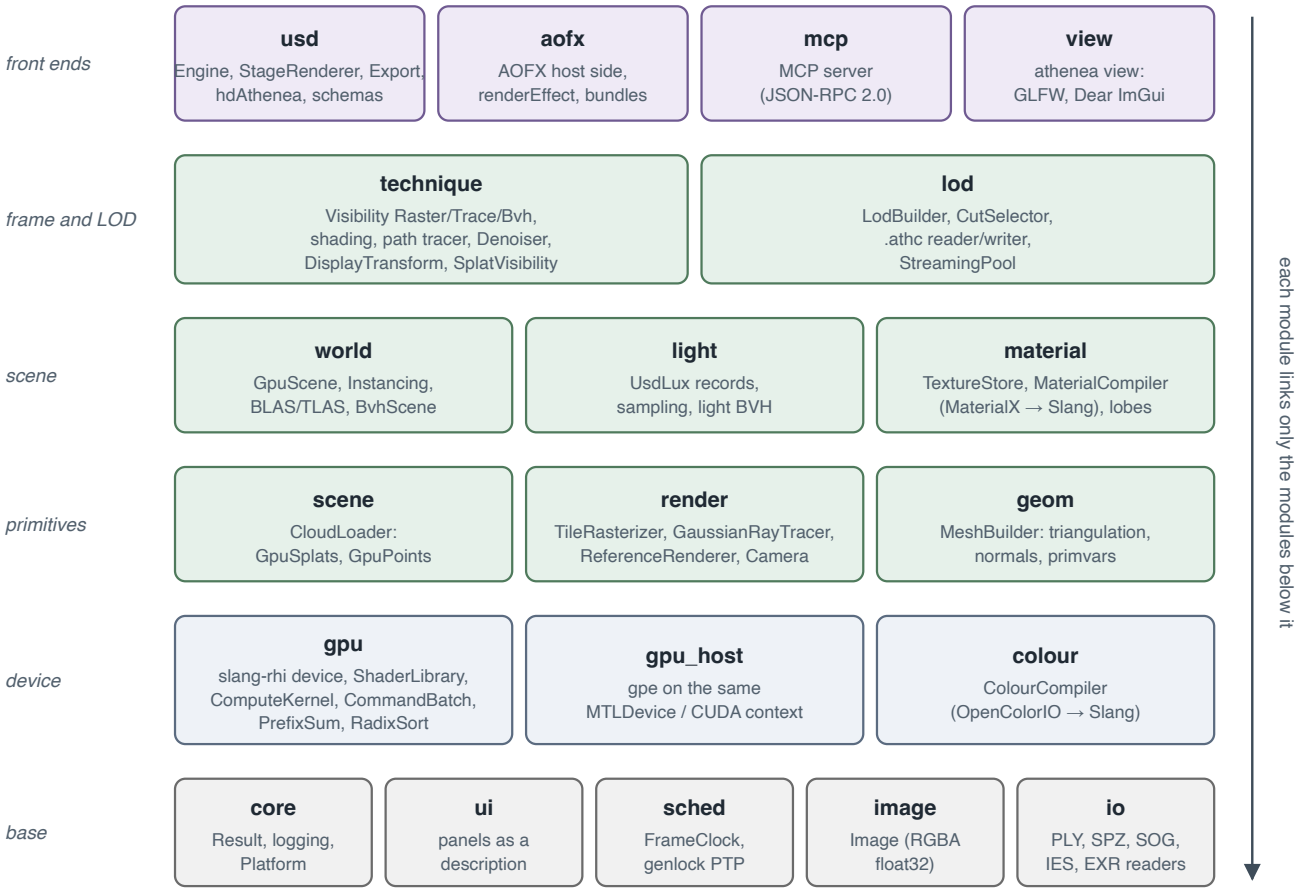


Figure 2. Engine modules in dependency order, with their main classes. Each static library links only the modules below it.

The shader tree `shaders/athenea/` mirrors the modules (common, algo, scene, splat, rt, points, reference, lod, geom, world, material, light, technique, usd, volume, view). Shaders are copied into the build tree and compiled at run time.

2.3 GPU foundations

Device and kernels. `gpu::Device` wraps a slang-rhi device. `ComputeKernel` is one compute entry point compiled as a pipeline. It is dispatched in threads, not groups: the group count is rounded up on every backend and each kernel does its own bounds check. `RasterKernel` is a vertex and fragment pipeline that pulls its data from `StructuredBuffer`s, with no vertex buffers or input layouts. `RayTracingKernel` covers OptiX and Vulkan RT pipelines. On Metal, slang-rhi offers no ray tracing pipelines, so tracing uses inline `RayQuery` inside compute kernels.

ShaderLibrary. The library loads Slang modules. A module may declare link-time constants (`extern static const uint kName;`) and is then linked against a generated exports module, which yields one program per distinct set of values. It also compiles modules that exist only as generated source, such as materials and the colour functions; loading an existing name with different source is refused. A persistent `DiskShaderCache` writes one file per compiled program, keyed by slang-rhi's hash of the linked program. With a warm cache the whole test suite went from 89 s to 26 s.

Parallel primitives. `PrefixSum` and `RadixSort` are the only sorting and scanning in the engine. The tiled radix route (`radix_block.slang`) is a reduce-then-scan form of the stable LSD pass. A group of 256 threads counts a tile of 1024 keys in group memory, a digit-major prefix sum places each tile's share of each digit, and the group sorts its tile with eight one-bit splits. It sorts 10 M 32-bit pairs in **12.4 ms**, against 39.9 ms for the chunked passes it replaced from 65 536 elements up. It uses no wave operations and no device atomics.

Mips. slang-rhi does not generate mip chains, so `MipGenerator` is a kernel. Each texel of a level is the area-weighted mean of the texels above it. An odd edge $2n+1$ folds into n with weights $(n-x,\ n,\ x+1)/(2n+1)$, so every source texel gives exactly $n/(2n+1)$ of itself to the level below and the chain preserves level 0's mean. On sizes 64^2, 37×23, 1×9 and 128×5, the means agree to $2\cdot10^{-6}$.

Memory as a first-class error. A Metal command buffer that runs out of memory used to assert inside slang-rhi on Metal's completion thread, which ended the process. A patch now records the error on the queue (`SLANG_E_OUT_OF_MEMORY`) and returns it from the next `submit()` or `waitOnHost()`. `Device::admit` refuses any allocation that would exceed the budget: `ATHENEA_GPU_BUDGET`, or else Metal's `recommendedMaxWorkingSetSize`, which is 18 186 MiB on the M5 Pro. An allocation must also fit the free physical memory less 1.5 GiB. On unified memory, several processes each within Metal's figure can together exhaust physical memory and force the system to swap. When a frame runs out of memory, `Engine::relieveMemory` steps down: it drops splat shadows first, then moves up to four LOD levels coarser, and finally reopens streams at half their budget.

2.4 The Gaussian representation

Every renderer, the loaders and the LOD share one GPU layout, `GpuSplats`. Each splat takes **32 bytes before its harmonics**, defined in `common/packing.slang`:

Field	Encoding	Reason
position, opacity	4 × float32	fp16 positions would step two pixels at 4K close up
rotation	smallest-three quaternion, 3 × 10 bits + dropped index	within a tenth of a degree
scale	f16 of log*s* per axis	0.1 % relative precision at every size
base colour	3 × f16	may leave [0, 1]
SH rest	`shWords` × u32, f16 per coefficient	signed
normal (optional)	octahedral 2 × 16 bits	shading normal kept apart from the frame
emission (optional)	RGB9E5	radiance the surface emits

`CloudLoader` reads PLY, SPZ, SOG, `.splat` and USD `ParticleField` prims. It uploads the raw records and decodes them on the GPU. For SOG, the host's only work is to unzip the archive and decode the WebP images.

Linear light. Every Gaussian is blended in linear Rec.709, and sRGB appears only where an image is displayed. A capture keeps the sRGB harmonics it was trained with: each projection evaluates them, clamps at zero and converts to linear (`cloudLight`) before blending. A cloud written by athenea declares that it holds light (`primvars:athenea:splat:linear`) and is used as stored. On two half-opaque overlapping discs, the rasteriser, the reference and the ray tracer each match the analytic over-operator at p99 0. Under the previous encoded blend, all three read p99 26.

2.5 The tile rasteriser

`render::TileRasterizer` is the interactive route for splats. It is a sort-based, deterministic tile rasteriser in the 3DGS family. Seven stages run on the device (Figure 3):

Stage	Work
project	per splat: covariance, conic, colour, 24-bit depth key, footprint, exact count of tiles touched
sort	splats by depth (RadixSort)
gather	tiles-touched counts in depth order, prefix sum
emit	one (tile, splat) pair per touched tile, in depth order
sort	pairs by tile, stably, on tile bits only
ranges	start and end of each tile's pairs
blend	per pixel: its tile's pairs, front to back

The engine sorts splats by depth once (N 32-bit keys) and then sorts pairs stably by tile (P keys of about 15 bits). This gives every tile its splats in depth order, with ties broken by splat index, exactly as a single 64-bit (tile, depth) sort would. The construction is deterministic. Tiles are 16×16 pixels.

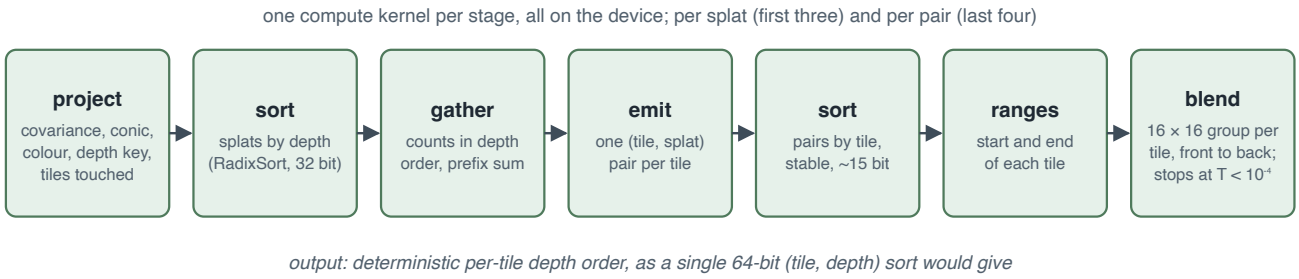


Figure 3. The tile rasterizer: seven compute stages, three per splat and four per (tile, splat) pair.

2.5.1 Projection

A splat with rotation R and scales $S = \mathrm{diag}(s)$ has the 3D covariance $\Sigma = RSS^{\top}R^{\top}$, built in the cloud's own space. With W the linear part of object-to-view and J the Jacobian of the perspective map at the view-space centre t, the screen covariance is the EWA local affine approximation

$$\Sigma_{2D} = (JWM)(JWM)^{\top}, \qquad M = RS, \qquad J = \begin{pmatrix} f_x/t_z & 0 & -f_x t_x/t_z^2 \\ 0 & f_y/t_z & -f_y t_y/t_z^2 \end{pmatrix}.$$

Before J is built, t_x/t_z and t_y/t_z are clamped to the frustum widened to $[-0.15, 1.15]$ of the image. A splat far off screen would otherwise smear across the whole frame, and 3DGS clamps in the same way. Orthographic cameras use $J = \mathrm{diag}(f_x, f_y)$.

Shutter and lens, as convolutions. A Gaussian smeared along a straight screen path d under a uniform shutter is that Gaussian convolved with a segment. Matching second moments turns this into a rank-one update. A thin lens convolves with the disc of the circle of confusion:

$$\Sigma' = \Sigma + \frac{1}{12} d\, d^{\top}, \qquad \Sigma' = \Sigma + \frac{R^2}{4} I, \quad R = f A \left| \frac{1}{z} - \frac{1}{z_f} \right|.$$

The displacement d has two sources: what a skeleton moved the splat over the shutter (a per-splat buffer), and a view-space step that combines camera and prim motion (`viewStep`). Both are projected by the same rows JW as the covariance. The update is checked against the same rasteriser run at N instants and averaged on the device (§2.5.3).

Antialiasing filter and the opacity it pays back. The covariance is then dilated by 0.3 px², as in Mip-Splatting's 2D filter. Each convolution spreads the splat, so its peak must drop by $k = \sqrt{\det\Sigma_{\text{still}}/\det\Sigma'}$ to keep its energy. One square root covers the filter, the shutter and the lens together. Scaling alone loses mass, however, because a splat is drawn only out to where it falls below 1/255, and the spread enlarged that area by $1/k$. A converted plane meant to cover 0.25 covered 0.18 at a twentieth of the screen. The opacity is therefore paid back as

$$\alpha' = \begin{cases} k\alpha + (1-k)/255 & k\alpha \geq 1/255 \\ k\alpha & \text{otherwise.} \end{cases}$$

This keeps the sharp splat's covered area (`paidOpacity`). Paying back every splat would also draw those the filter had made too faint to keep. That raised a distant converted pawn from 8.7 to 29.8 ms (debug build), and the rule above costs 11.0 ms there.

Footprint and tiles. A splat is cut at the squared Mahalanobis radius

$$r^2_{\max}(\alpha) = \min\big(2\ln(255\,\alpha),\ 9\big),$$

which is where its own contribution falls below 1/255, and never beyond 3σ. The tile count is exact rather than taken from a bounding box. For each tile in the axis-aligned extent, the minimum of the quadratic form $q(\delta) = \delta^\top \Sigma'^{-1}\delta$ over the box of the tile's pixel centres is computed in closed form: since q is convex, it is either 0 or lies on an edge, where it is a clamped 1D quadratic. The emit stage repeats the same test. Because the blend cuts at the same radius, no pixel outside a counted tile could have been touched, so tile edges are seamless. A culled slot writes the reason for culling into its unused depth-key word (edit, depth, no area, faint, off-screen, no tile), and the viewer's Gaussians panel counts these reasons without adding a counter to the projection.

Depth key. The key is `bit_cast<uint>(z) >> 7`, which is order-preserving for positive floats and 24 bits wide.

2.5.2 Blending

One 16 × 16 thread group per tile loads the tile's records in batches of 256 into group memory, and every pixel walks them from there. For pixel p and splat i with conic $Q_i = \Sigma_i'^{-1}$, centre μ_i and colour c_i:

$$\alpha_i(p) = \min\Big(0.99,\ \alpha_i' \exp\big(-\tfrac{1}{2}(p-\mu_i)^\top Q_i (p-\mu_i)\big)\Big), \qquad C = \sum_i c_i\,\alpha_i \prod_{j<i}(1-\alpha_j) + T\,C_{\text{behind}}.$$

Contributions below 1/255 are skipped. A pixel stops once $T = \prod(1-\alpha_j) < 10^{-4}$, and the group stops loading once all 256 of its pixels are done. C_{behind} is either the background or the opaque *under* layer of meshes and points, composited by view z: the walk ends at the first record behind the layer's depth. Depth is written either as the alpha-weighted mean or as the first crossing of a coverage threshold. The same walk also accumulates a six-rank Cryptomatte, by id coverage, and for pruning the per-splat contribution $T\alpha$ or the squared error the splat's removal would leave (the GaussianPOP score). Specialised entry points compile the composite and matte branches out, because leaving them live in a frame that has neither cost the blend half again.

Surfels and harder kernels (athenea-pieces branch). A Gaussian with one scale exactly 0 is a *surfel*. The rasteriser draws it by intersecting each pixel's ray with the disc's plane (Huang et al. 2024): the disc's axes are stored in homogeneous screen coordinates and pulled back through the pixel's two planes. The screen filter is a max, not a convolution: the larger of the disc and a round Gaussian of variance ½ px². Its opacity is paid back by area,

$$\alpha' = \alpha \cdot \min\left(1,\ \frac{\sqrt{\lambda_1\lambda_2}}{\sqrt{\lambda_1\lambda_2}\ + \frac{1}{2} - \sqrt{\min(\lambda_1, \frac{1}{2})\min(\lambda_2, \frac{1}{2})}}\right),$$

where $\lambda_{1,2}$ are the eigenvalues of the undilated screen covariance. A merged Gaussian may carry an opacity $D > 1$. That value selects Spark's harder kernel, $1 - (1 - g)^A$ with $A = \exp((D^2 - 1)/e)$ and D capped at 5. This is the composite of A coincident Gaussians, and its cut is widened by $0.7(D - 1)\sigma$. A *hard surfel* uses $\exp(-(|u|^p + |v|^p)/2)$, with p fitted to its members' kurtosis. Section 6 covers both, together with the merges that produce them.

2.5.3 Measured

Coverage of converted surfaces (`athenea_coverage_tests`, 256 × 256, alpha of planes converted by mesh2splat, against the mesh's alpha; sizes are fractions of full screen):

Plane	Mesh	Raster 1× / 0.25× / 0.05×	Traced 1× / 0.25× / 0.05×
constant 0.25	0.250	0.251 / 0.250 / 0.248	0.251 / 0.251 / 0.251
constant 0.5	0.500	0.501 / 0.499 / 0.493	0.501 / 0.501 / 0.500
constant 0.75	0.750	0.752 / 0.749 / 0.736	0.752 / 0.752 / 0.752
opaque	1.000	1.000 / 1.000 / 0.999	1.000 / 1.000 / 1.000

Without `paidOpacity`, the raster read 0.18 / 0.455 / 0.73 at 0.05×.

Motion blur and lens (`[motion]`, `[lens]`; truth = the same rasteriser at N instants, or 64 lens positions, averaged on the device):

Case	Result
shutter step 0.10 (many splat widths)	p99 8, max 24
shutter step 0.03	p99 2, max 9; energy ratio 1.0000
shutter step 0.9	energy ratio 0.9030 (tails below the 1/255 cut)
zero shutter vs still frame	max 0, bit for bit
f/2, 12.7 px confusion	p99 17 (not blurring: p99 113)
f/11, 2.3 px confusion	p99 9 (not blurring: p99 33)
at the focus distance	max 0, bit for bit

Frame cost by stage (`ATHENEA_STAGES=1`, every stage waited for; the sparrow's clips, 5 887 323 Gaussians at 1600 × 1000, playing):

Stage	Before	After	Change
depth sort (24 bits)	11.3 ms	6.6 ms	tiled radix
tile sort (8.8 M pairs)	8.4 ms	6.0 ms	tiled radix
blend	13–25 ms	8–13 ms	tile records shared through group memory

Stage	Before	After	Change
project, emit, counts, commit	13.6 ms	13.4 ms	–
frame, playing	102 ms	49 ms	cloud no longer re-uploaded each step

A frame of six million Gaussians spends its time on the Gaussians rather than on pixels: changing the render scale from 800 × 500 to 3200 × 2000 leaves the time unchanged.

2.6 The Gaussian ray tracer

`render::GaussianRayTracer` evaluates each particle in 3D at its peak response along the ray, with no EWA approximation and no 0.3 px dilation. Particles are blended front to back in order of their peak along the ray. In the particle's whitened frame (origin o, direction d, with $\Sigma^{-1/2}$ applied), the peak and the entry into the cut ellipsoid are

$$t^* = -\frac{o \cdot d}{d \cdot d}, \qquad \rho = \left\| o + t^* d \right\|^2, \qquad t_{\text{entry}} = t^* - \sqrt{(r_{\max}^2 - \rho)/(d \cdot d)}\,.$$

The textbook discriminant $b^2 - ac$ cancels catastrophically for flat particles and lost 176 of 400 thin layers, which is why the peak form is used. Colour is the rasteriser's: the harmonics evaluated once per frame in the direction from the eye to the centre.

Two routes, one integrator. Both routes feed the same `Integrator` (`rt_integrate.slang`). The *Hardware* route gives every particle a stretched-icosahedron proxy (3DGRT's) in a device BVH queried with inline `RayQuery`. The *ComputeBvh* route is a Karras LBVH built on the GPU (Morton codes, radix sort, hierarchy, refit until a pass changes nothing) and traversed in compute. Each traversal records up to 256 unsorted entries, which are evaluated and ordered after traversal by a 128-bucket counting sort and then an insertion sort over an index permutation. When a ray meets more proxies than that, the segment closes at the nearest dropped entry and the next traversal resumes a relative 10^{-5} before it. Particles peaking beyond the boundary are carried into the next segment; only when more than 64 are carried are the nearest blended early, which is the route's single approximation.

train_7k (742 k splats), 1080p	Frame	Build
Hardware (Metal)	435 ms	731 ms
ComputeBvh (Metal)	343 ms	168 ms
tile rasteriser, same frame	13 ms	–

On Metal, each non-opaque candidate costs a round trip out of hardware traversal. An opaque closest-hit query over the same 15 M proxy triangles takes 6 ms, but enumerating every candidate takes 255 ms, so `Auto` picks ComputeBvh on Metal. The optimisation history on this frame went from 2.52 s (a 3DGRT-style 16-entry k-buffer) to 343 ms. The largest steps were recording unsorted entries and evaluating after traversal (1.94 s → 658 ms) and the bucket sort (−85 ms); a record size of 256 was chosen from a sweep of 64 to 512.

Shadow and secondary queries. Transmittance is a product, so `rt_shadow.slang` takes proxies in whatever order they arrive, with no k-buffer, and stops when the product falls below its cut. It does not cull back faces, because a shadow ray born inside a proxy must still see the particle it starts in. Three closed forms check it: $1 - \alpha$ through one centre, measured 0.50000; $(1 - \alpha)^4$ through four, measured 0.06250; and early

termination after three of four. The integrator answers a query `rtTraceSegment(ray, [a, b))` rather than writing a pixel, and `segmentOver` composes two segments: a sparse cloud split at any depth is bit-exact with the single query.

2.7 Points

`PointRasterizer` draws point clouds through a z-buffered raster pipeline with no sort, optionally followed by surface splatting and eye-dome lighting. On devices without rasterisation (CUDA), points are drawn as discs in the splat pipeline. In that case a record's `colour.w > 0.5` marks it as an opaque disc, which ends the walk exactly as the under layer does.

2.8 Meshes: geometry, scene and visibility

On the device, in Hydra's order. `geom::MeshBuilder` triangulates Hydra topology in `HdMeshUtil`'s fan order, including holes and left-handed orientation. It computes smooth normals with `Hd_SmoothNormals`' formula, as per-corner cross products scattered to points through a radix sort, and expands primvars of every interpolation per corner, resolving indexed primvars and decoding doubles on the device. A height field's normals come within $1.1{\cdot}10^{-7}$ rad of a brute-force sum. `world::GpuScene` pools positions, indices, corners, faces and primvar values, with a 64-byte record per mesh and a 176-byte `InstanceRecord` per drawn copy. `world::Instancing` composes instancer chains exactly as Storm does, instancer $\cdot\, T \cdot R \cdot S \cdot$ instanceTransform, nested as parent$[i] \cdot$ level$[j]$, with one dispatch for all instanced sets. For Kitchen_set_instanced's 1462 sets this took command recording from 32 ms to 0.4 ms.

Three routes, one visibility buffer. Each route writes (instance + 1, triangle) per pixel. Shading and AOVs rebuild the hit from those two numbers with Möller–Trumbore in view space (`technique/surface.slang`), so all routes shade alike.

- `VisibilityRaster` uses reversed infinite Z (depth $=$ near$/z$, D32Float), one draw per mesh with `instanceCount`, and a single shared root object.
- `VisibilityTrace` builds one BLAS per mesh and a TLAS per frame. A kernel writes the instance descriptors in each backend's layout (64 B generic, 80 B OptiX, 68 B Metal).
- `VisibilityBvh` builds a Karras LBVH per mesh and one over instances, for devices without RT hardware.

Single-sided culling matches Storm in all three routes.

Storm as the oracle (Kitchen_set, 480 × 270; ids compared as segmentation, because Storm numbers prims differently):

Route	Coverage	Segmentation	Depth, worst	Neye > 6/255
raster	0 differ	18 of 57 219	$1.2{\cdot}10^{-6}$	17
rays	4 differ	41 of 57 236	$6.3{\cdot}10^{-6}$	17
bvh	4 differ	36 of 57 233	$6.3{\cdot}10^{-6}$	17

Visibility timings (`athenea stage --frames 40`, median frame after the first, headlight shading, readback included):

Scene	Size	raster	rays	bvh
Kitchen_set (1788 meshes)	480 × 270	18.0 ms	4.8 ms	8.9 ms
Kitchen_set	1920 × 1080	31.7 ms	19.2 ms	29.2 ms
Kitchen_set_instanced (1462 sets)	1920 × 1080	29.1 ms	19.2 ms	28.6 ms

Raster loses mainly on the host side: recording about 1800 draws costs about 8 ms, because slang-rhi writes render state per draw. The automatic setting therefore prefers rays wherever the device has ray queries.

2.9 Materials

Textures. The `TextureStore` uses Hio only to read a file's bytes. The bytes are uploaded raw and a kernel decodes them into the working space. Mips are a kernel that decodes sRGB, filters in light and re-encodes. UDIM is an indirection table. All textures sit in one `ParameterBlock` of 1024 slots, which Metal takes as an argument buffer. Filtering is chosen per target with `__target_switch`: `SampleGrad` where compute allows it, and an explicit level from the footprint's wider side elsewhere (CUDA). On Metal, where both are available, they pick the same levels. A decoded texture is the file to the last bit (0 of 3404 components differ). Input colour spaces resolve through `colour::ColourNames`, and transforms are generated by OpenColorIO (§2.12).

MaterialX into Slang. The generator derives from MaterialX 1.39.5's `SlangShaderGenerator`. Every node keeps its genglsl or genslang implementation except four kinds whose rasteriser meaning cannot hold here:

- the surface node, which has no light loop here: its BSDF graph runs once, pushing lobes;
- BSDF and EDF nodes, which push lobes instead of answering a light;
- image nodes, which sample the texture table;
- `heighttonormal`, which needs screen derivatives.

A BSDF value is a *weight per built lobe*, so `mix`, `layer`, `add` and `multiply` combine weights. Uniforms are not baked into the code: each input is read from a float blob, and the module is named by a hash of its source, so materials that differ only in values share one compiled module. Generated sizes include UsdPreviewSurface at 580 lines and 23 blob words, standard_surface at 821 and 59, and OpenPBR at 945 and 55. `technique::MaterialPrograms` generates one module that imports every compiled material and dispatches on a material row. A kernel that evaluates materials walks its pixels in quad order (`athenea-QuadPixel`), because on Metal the lanes run along rows, and in row order every vertical derivative is wrong (15 939 of 15 939 threads).

The lobe library. It contains Oren–Nayar and energy-compensated EON, Burley, translucent, dielectric (reflection, transmission, both), conductor, generalised Schlick with an F82 tint, and sheen in both its Imageworks and Zeltner forms. Each lobe has `eval`, `sample` and `pdf`. Microfacet lobes sample the visible normal distribution and transmission follows Walter. A material's lobes form a `LobeStack`, which is sampled with one-sample MIS by the lobes' weighted directional albedos.

Checks against MaterialX itself. The same graphs are compiled a second time in a reference variant whose closures are MaterialX's own genglsl responses, and a kernel evaluates both for 65 536 light directions. The worst component difference over eleven graphs is $\mathbf{3.2{\cdot}10^{-5}}$. On the device, the lobes pass a χ^2

test of sampled directions against the pdf (393.2 on 399 degrees of freedom for the diffuse lobes) and a white furnace that agrees within 3 %.

Cutouts and bump. Visibility evaluates `opacityThreshold`: the raster uses a generated discarding fragment shader, and the ray routes carry the ray past a cut sample up to sixteen times. Only rows flagged as cutouts pay for this. A cutout square shows what lies behind it exactly (8464 pixels, 0 mismatches) on all three routes.

2.10 Lights

Records and units. Each UsdLux light becomes one record in world space, in the units the stage authored. Exposure, the blackbody of a colour temperature (Krystek's fit, normalised to luminance 1) and `normalize` are derived in the shader. Supported kinds are sphere, disk, rect, distant with an angle, dome and cylinder, plus shaping cones and IES profiles. An IES profile is read as authored, sampled bilinearly with its declared symmetry, and modulates radiance but never density. Lights under instancers are composed on the device as meshes are.

Sampling where the light stands. A sphere or sun is sampled uniformly over the cone it subtends. A disk, rect or cylinder is sampled uniformly by area and converted to solid angle by $p_\omega = d^2/(\cos\theta_l A)$. A dome is sampled either cosine-weighted about the shading normal or by warping down its image's mip chain by luminance. The warp's density needs no tables:

$$p(\omega) = \frac{L(u,v)}{\bar{L}\, 2\pi^2 \sin\theta},$$

where $\bar{L}$ is the 1×1 mip level's mean. The image's variation chooses between the two strategies: on a flat sky the warp is 20 % out where the cosine is 0.08 %. `lightPdf` returns the density of any direction and `lightHit` answers the converse question, which light a ray meets and with what radiance. Together they are what MIS needs. With `athenea:chooseLights`, a sample takes one light chosen through a **light BVH** (`light_bvh.slang`). Each node is a box, an orientation cone (Conty and Kulla 2018) and the power beneath it, and the BVH is built by the same compute LBVH kernels as the geometry.

Checks. Every light is held to a closed form that shares no code with the renderer: a Lambert plane under a sphere, a disk, a rect (Lambert's edge formula), a sun and a dome. In each case 0 of 8281 pixels are beyond 2 %; the worst is 1.6 % for the disk. Each light also runs a χ^2 test with 10^6 samples, binned in the light's own support, against `lightPdf` integrated per bin: sphere $z = -0.23$, disk 0.14, rect -0.27, sun -0.29, dome -1.61, every pdf integrating to 1.0000. The dome warp needed two fixes, both found by χ^2 in image space. The bug reached $z = 61\,543$ and later $1\,164\,589$, against 0.84 after the fixes. The per-sample consistency check had passed throughout, because it compares a density with itself.

Measured (Kitchen_set with four authored lights, 1600×900, `athenea view --frames 200`, draw median):

Configuration	Draw
headlight only, rays	10.61 ms
four lights, 1 sample, rays	27.77 ms
the same, compute BVH	35.14 ms
the same, raster	40.94 ms
four lights, 16 samples, rays	275.40 ms

2.11 The path tracer

The path tracer is the engine's ground truth. Its paths start from the same visibility buffer the raster shading reads, so the two techniques differ by exactly one thing: the bounces. With a lens, the tracer additionally casts its own primary rays. A hit is rebuilt by `material_surface.slang`, its material is evaluated into the same lobe stack, and direct light is gathered by next-event estimation (NEE). The bounce samples the stack, and the path throughput is multiplied by `LobeSample.weight` $= f|\cos\theta|/p$ (Figure 4).

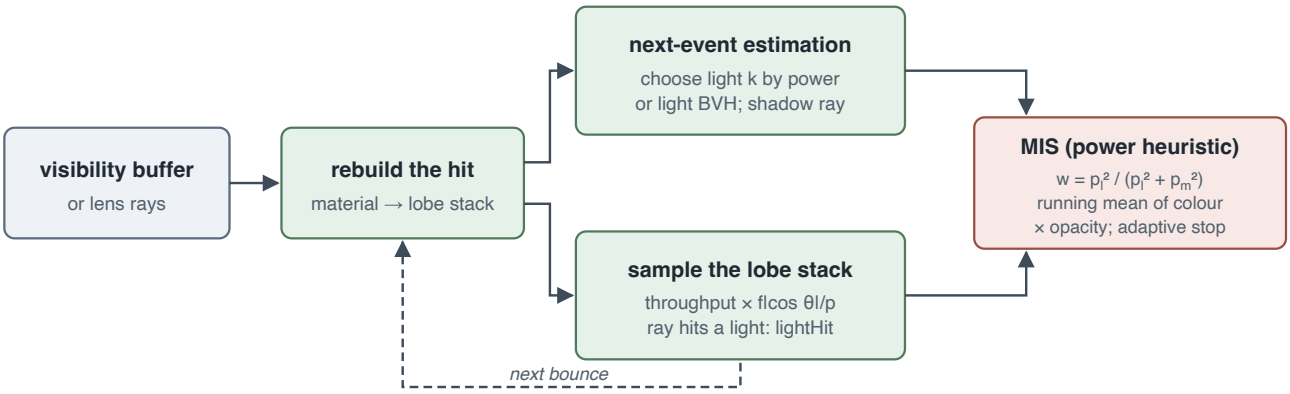


Figure 4. One vertex of the path tracer: the hit is rebuilt from the visibility buffer, direct light is gathered by next-event estimation, the lobe stack is sampled for the next bounce, and both estimates are combined by multiple importance sampling.

MIS. At each vertex that continues the path by sampling its material, NEE's sample of light k and the material sample's hits on lights (`lightHit`, including escaping rays meeting domes and suns) are weighed by the power heuristic,

$$w_{\text{light}} = \frac{p_l^2}{p_l^2 + p_m^2}, \qquad p_l = P(k \mid x)\; p_k(\omega),$$

where $P(k \mid x)$ is the light-choice probability (by power, or through the light BVH) and p_m is `stackPdf`. Delta lights, delta lobes and the last vertex keep a weight of 1. **Emissive triangles** form a light table (`technique::EmissiveTable`): each material row's emission is probed once, triangles are weighted by world area times that luminance, and the choice between lights and triangles is made by power, with each side's density carrying the other's share. On a floor of roughness 0.2 at 32 paths, MIS reduced error against a deep frame by 4.5× under a rect, 57.7× under an imageless dome and 2.4× under a sphere. An emissive quad and an equivalent UsdLux rect agree to relMSE $4.1{\cdot}10^{-7}$ at 8192 paths.

Accumulation, sampler, adaptivity. A frame is a running mean of colour × opacity. An earlier implementation took the product of means, which is unbiased only while every sample is opaque. The sampler folds pixel, stream, absolute path index, bounce and dimension through a PCG hash, so 256 paths in one pass and 4 × 64 differ by relMSE $3.9{\cdot}10^{-15}$. Between passes the camera is jittered by a Halton (2, 3) offset inside the pixel; the first pass takes no offset. This is antialiasing at no cost. Adaptive sampling stops a pixel once

$$\frac{\sqrt{(\mathbb{E}[l^2] - \mathbb{E}[l]^2)/N}}{\mathbb{E}[l]} < \varepsilon$$

after a minimum count. The variance used is the larger of the pixel's own and the mean of its 3 × 3 neighbours', which reduced the stopped pixels beyond three of their own σ from 526 to 100 of 4212. **Thin lens and radial distortion** come from `UsdGeomCamera`: the lens radius is $f/(2N)$, and rays are scaled by

$1 + k_1 r^2 + k_2 r^4$.

Verification. Every check is a GPU kernel against a closed form or against a second, independent estimator:

Check	Result
closed emissive sphere, $E(1 + \rho + ... + \rho^N)$, N = 0, 1, 2, 3, 6	0 of 3072 pixels beyond 10^{-4}, worst $1.7{\cdot}10^{-7}$
white furnace under an imageless dome vs unweighted NEE	p99 relative 0.0000, max 0.0003
one bounce vs raster direct light (nothing to bounce off)	p99 1, max 1 (8-bit codes), 4096 paths
error vs N, pairs of independent estimates, 64–1024 paths	exponent -0.499 (window -0.42 to -0.58)
in focus vs pinhole (lens radius 0.2)	relMSE 0, bit for bit
out of focus vs circular-segment profile	0 of 4235 pixels beyond 4 %
OIDN, 16 paths vs 4096-path reference	relMSE $3.65{\cdot}10^{-4} \rightarrow 7.17{\cdot}10^{-5}$

The furnace exposed a one-sided MIS weight. On a dome against a Lambert lobe the two densities are the same function, so the weight was exactly ½, and the furnace read 0.5453 instead of 1. The shell exposed a double division, $\rho\pi/\cos$ instead of ρ, whose infinite second moment had held the convergence exponent at -0.43 to -0.45.

Denoiser. `technique::Denoiser` runs OIDN 2.5.1's `RT` filter on the engine's own queue, guided by the first hit's albedo and normal. On Metal, OIDN shares only hazard-tracked buffers, which slang-rhi forbids. The engine therefore copies through tracked staging buffers using a kernel, since slang-rhi's blit silently did nothing on a wrapped buffer (4095 of 4096 words untouched). On CUDA the engine's buffers are shared directly. On Vulkan, OIDN runs through CUDA on imported memory (`oidnNewSharedBufferFromFD`).

Measured (Kitchen_set, 1920 × 1080, `athenea stage --frames 11`, median, release):

Stage	raster	rt (1 path, 1 bounce)
no lights (headlight)	43.1 ms	333.3 ms
lit (dome, rect, two spheres)	96.0 ms	358.9 ms

Through MCP, the OpenChessSet path traced at 640 × 360 with 64 paths, one bounce and ACES 2.0 takes 25.7 s.

2.12 Display and colour

`technique::DisplayTransform` writes the window's surface texture directly. It supports Standard; AgX, in Wrensch's analytic fit (log2 over [−12.47, 4.03] stops, a sixth-order sigmoid); and ACES 2.0. The ACES 2.0 output transform (`aces2.slang`) is ported function for function from the Academy's CTL, including Hellwig 2022's appearance model, tonescale and gamut compression. The reference's init-time tables (362 entries per hue) are built on the device by three kernels. Checks: 256 greys over sixteen stops stay neutral to $5{\cdot}10^{-6}$; 18 % grey reads 0.10000 of reference white at 100 nits; inverse-then-forward over 8192 colours returns within $6{\cdot}10^{-6}$ of peak. The display formulas, re-derived in a second kernel, agree to $3.5{\cdot}10^{-6}$. OpenColorIO is used **as a compiler**: `colour::ColourCompiler` asks OCIO for HLSL, loads it as a Slang module `athenea_cs_<hash>`, and fills its LUTs on the device. The host computes shader text and LUT values but nothing per pixel.

2.13 USD and Hydra

The `hdAthenea` plugin implements `HdRenderDelegate` with rprims for meshes, basis curves, points, `ParticleField` (Gaussian splats) and volumes; sprims for cameras, materials, the six UsdLux light types, ext computations and coordinate systems; and bprims for render buffers and render settings. Implicit surfaces (sphere, cube, cone, cylinder, capsule, plane) reach the delegate as meshes, converted by hdsi's scene indices. **Threading:** `Sync`, which Hydra may run on any thread, only hands CPU records to `Engine` under a lock. The render pass thread commits uploads, opens `.athc` files and compiles materials, then renders, so the device has a single caller. A skinned cloud whose arrays carry no time samples is held across steps (`_held`) and not re-uploaded; this took the sparrow's playing frame from 102.4 to 71.7 ms before the sort and blend work of §2.5.3. **Outputs:** colour, depth, primId, instanceId, elementId, Neye, normal, `primvars:*`, Cryptomatte and light groups. They are converted on the device (`usd/aov_convert.slang`) only when a host maps a buffer, and are stored bottom row first, as Storm stores them. `StageRenderer` is the programmatic entry point and draws until a path-traced frame holds its total.

The engine reads codeless schemas of its own (`AtheneaSplatLightingAPI`, `AtheneaStreamedAssetAPI` and others). Render settings arrive as `athenea:*` settings: technique, visibility route, path samples, bounces, totals, MIS, denoise, antialias and lens.

2.14 Hosts and front ends

- **`athenea view`** keeps frames on the device: `StageRenderer::draw`, then `DisplayTransform` writes the surface texture and Dear ImGui draws on the same device. Measured with materials at 1600 × 900, Kitchen_set draws in 9.27 ms on rays. The panels are described once in `modules/ui`, so the iOS app draws the same ones.
- **AOFX host.** The AOFX SDK and host are aopenfx's (`third_party/aopenfx`), and the engine declares only what it provides (`Host.cpp`). They change only additively under the ABI's rule, so the compositor's bundles load unchanged. `aofx_sdk_manifest` fails on any header change. Data-processing steps such as mesh2splat (Section 3) and `athenea compare` are AOFX bundles.
- **MCP server** (`athenea-mcp`): JSON-RPC 2.0 over stdio, with a warm device and stage. Tools are `open_stage`, `render` (returning a PNG image block), `pick`, `bounds`, `convert`, `timings` and `settings`. A protocol test suite checks handshake, notifications and error codes.

2.15 Verification without a CPU oracle

Every image comparison is a kernel. `ReferenceRenderer` is the ground truth for splats: every pixel considers every splat, sorts exactly what reaches it (depth, then index, the rasteriser's tie rule) and blends front to back, with no tiles, ranges or pairs. A comparison is valid only when no pixel overflows its 1024 contributors. `renderPeaks` is the ray tracer's reference, sorting exactly by peak. The metrics read back to the host are `compareImages` (p99, max and the number of pixels over 2, in 8-bit sRGB codes), `compareHdr` (relMSE $= (a-b)^2/(\bar{b}^2 + 10^{-2})$ and the relative p99) and `countDifferent` for id buffers. The ray tracer is held to its reference at p99 ≤ 1 with at most 0.05 % of pixels over 2. Against the rasteriser, the tolerance holds only where the two definitions agree: p99 2 for orthographic sparse scenes, where EWA is exact, and 4 for perspective scenes with particles a few pixels across. Dense scenes differ by tens of codes, which reflects ordering rather than error.

Three devices (each machine computes the same GPU metric itself, and no pixels travel between machines):

	Metal (M5 Pro)	Vulkan (L4)	CUDA (L4, OptiX 9, no inline rays)
Cases	188	188	188
Passed	188	180	96
Failed	0	0	1
Skipped (capability named)	0	8	91

2.16 Limitations

- Splats are not yet inside the path tracer's secondary rays. A splats-only stage is traced whole by `GaussianRayTracer`. With meshes, splats are composited over the path-traced surfaces by depth; they cast shadows onto meshes, but a bounce ray does not see them.
- The rasteriser's motion and lens blur are moment-matched Gaussians, not plateaus or bokeh discs: up to 38 % too peaked for long streaks.
- Clouds whose Gaussians are a few hundredths of a pixel across vanish from the raster ($5{\cdot}10^{-6}$ of the frame against $3{\cdot}10^{-4}$ for the mesh and the tracer); levels of detail are the remedy (Section 6).
- Raster mesh visibility pays about 8 ms of host time per 1800 draws.
- Material compilation is synchronous.
- The TLAS is rebuilt every frame.
- Lens distortion beyond k_1, k_2 is not applied.

3. From USD meshes to Gaussians (mesh2splat)

The conversion is the relight baker's first stage: it turns a USD stage of polygonal meshes and their materials into a cloud of anisotropic 3D Gaussians, and that cloud is what the transfer bake of Section 4 fills with light. It is invoked as `athenea mesh2splat STAGE`; every Gaussian the rest of the system draws, bakes, decimates and streams was a triangle of a mesh first. This section describes how the stage is read, how large a Gaussian's cell is, how each Gaussian is placed, shaped and coloured, how the conversion handles the places where a surface stops or folds, how it captures a layered physically based material, how it treats glass, and what was measured on the Chevrolet Corvette C7 asset that drives most of the project. The path-traced bake that follows emission (the *transfer*) is the subject of Section 4 and is touched on here only where it constrains the conversion.

Numbers in this section come from `docs/decisions.md` and `docs/pieces.md` of the repository and from the source (`plugins/mesh2splat/mesh2splat.slang`, `apps/athenea/src/CmdMesh2Splat.cpp`, `shaders/athenea/usd/mesh2splat_*.slang`). All measurements were taken on an Apple M5 Pro (Metal) unless said otherwise.

3.1 Origin, and where the conversion runs

The emission algorithm is that of Electronic Arts' *mesh2splat* (BSD-3): every triangle is projected onto the axis plane its normal points along most, the projection is rasterised at a fixed resolution, and each covered texel becomes one Gaussian lying in the triangle's plane, as wide as a texel measured in the world. The original is an OpenGL geometry and fragment shader pair; here it is one compute kernel in which each thread owns a triangle and walks the cells of its projected bounding box.

The kernel is not a module of the engine but an AOFX effect, the bundle `plugins/mesh2splat`, loaded through the same host registry every program shares (the AOFX ABI, Section 2.14). The conversion is a stateless kernel over triangles, which is exactly the shape of an AOFX effect, and making it one keeps a converter out of the renderer's modules and lets the same bundle run in other AOFX hosts. The cost is a boundary: an effect is handed pictures, so a mesh travels as a *picture of numbers*, six `float4` per triangle — $(\boldsymbol{p}, u)$ and $(\boldsymbol{n}, v)$ for each corner, in world space (`mesh_pack.slang`) — and every texture a material names travels as rows of linear `float4` sampled out of the engine's texture table (`texture_rows.slang`). Neither is a copy: the pictures live in the gpe image storage that adopts the engine's Metal device, and `Context::renderView` gives a view of the same bytes, so the packing kernel writes where the effect reads. The records the effect writes are gathered into the cloud's device buffers (`mesh2splat_gather`), the cloud's bounds are folded on the device (`mesh2splat_span`), and the bake reads them in place. Only counters and the final USD arrays cross to the processor.

The command is therefore a pipeline of device passes orchestrated by the host (Figure 5):

1. read the stage (`usd::MeshStage`) and split meshes into *pieces*;
2. pack each piece into a mesh picture, fold boxes, choose each piece's cell (`mesh2splat_cells`);
3. mark hard edges and closed shells per mesh (`mesh2splat_edges`), classify glass (`mesh2splat_sheet`);
4. run the Mesh2Splat effect per piece — count, scan, emit — into records;
5. optionally bake (radiance, or a transfer) at every Gaussian with the path tracer;
6. write a `ParticleField3DGaussianSplat` stage (or a `.athc`) atomically.

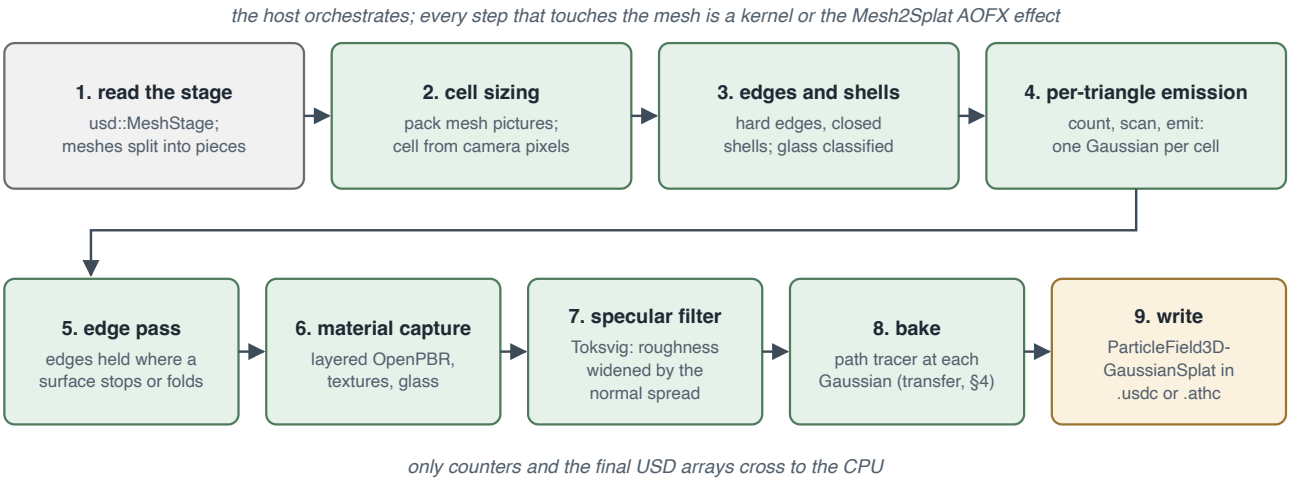


Figure 5. The conversion as a sequence of device passes. Steps 2–7 are kernels or the Mesh2Splat AOFX effect; step 8 is the transfer bake of Section 4.

3.2 Reading the stage

`usd::MeshStage` walks the stage with `UsdShade` directly rather than through Hydra: the conversion wants no render index, and the material side is built on the stage's own bindings.

Selection. `--prim P` restricts conversion to meshes at or under `P`; `--hide P` (repeatable) authors invisibility on `P` and everything below it, in a non-persistent layer over the stage, before the read. A mesh that computes invisible at the read's time code is skipped, so the conversion converts what the renderer draws. `--hide` affects the conversion only: the bake traces the whole stage, so a part converted with every other part hidden still receives the others' shadows and inter-reflections. This is what allows the Corvette to be converted in groups (Section 3.11).

Instancing. A `PointInstancer` prototype's mesh is emitted once per instance of the innermost instancer that holds it, at (mesh relative to its prototype root) × (instance transform, prototype root included) × (instancer to world); the built mesh buffers are shared between copies. Without this expansion the chess set's sixteen pawns convert to two. Nested instancers are expanded at the innermost level only.

Time and pose. `--time T` sets the time code at which positions, transforms and primvars (including their indices) are read and at which the bake traces. A skinned mesh's `points` do not animate in USD, so the stage is first posed with `UsdSkelBakeSkinning` into a non-persistent layer over $[T, T]$, which costs one pose and leaves the file untouched; a stage without a `SkelRoot` is unchanged and converts byte-identically. With `--skinned` the Gaussians are built in the bind pose and each keeps four joints and their weight gradients; that path is described in Section 5.

Pieces. A mesh whose `materialBind` GeomSubsets bind different materials is converted as several *pieces*: each subset's triangles, and those no subset claims. A piece's triangles are listed on the device in the mesh's order (flag, prefix sum, scatter) and packed as if they were a mesh, so the effect is untouched. A test mesh with two faces bound red and blue by subsets and the mesh bound green converts to 272 red and 272 blue Gaussians and none green; reading the mesh's own binding alone makes it all green.

Materials in their own words. The bound surface — MaterialX `standard_surface`, OpenPBR `open_pbr_surface`, `gltf_pbr` or `UsdPreviewSurface` — is read with its own input names and its own defaults (standard_surface 0.8 grey at roughness 0.2, OpenPBR roughness 0.3, glTF a fully rough metal, UsdPreviewSurface 0.18 at 0.5), following connections through node graphs and interface inputs to the im-

age node that produces a value. A value computed by a graph (a mix, a noise) has no answer without shading a point, so the default stands and a warning is printed; a TX bake later reads such fields back from the compiled material on the device. A texture's colour space is taken from the file input's `colorSpace` metadata (or `sourceColorSpace`), and maps are capped at `--texture-size` (1024 by default): a map is a `float4` picture on the device, so a 4k map would occupy 268 MB.

3.3 The size of a cell

Everything downstream scales with the cell h, the world length one Gaussian stands for. The conversion offers four ways to choose it, in increasing order of preference.

Per model (`--density per-model`, the default when no camera is given). $h = R/N$, where N is `--resolution` (512 by default) and R the wider of the projection plane's two ranges of the model's box — the original's `orthogonalUvs`. Two defects followed: a 16-unit floor diluted the car standing on it, and a panel facing forward walked a cell 2.5 times finer than the hood, because its projection plane is the car's short side.

Per mesh (`--density per-mesh`). $h = L_m/N$ with L_m the longest side of the mesh's own box, the same for every triangle of it (`cellByLongest`), clamped to $[h_{\min}, h_{\max}]$ given by `--cell-min`/`--cell-max`; left at 0 they derive from the model's cell h_M: $h_{\max} = h_M$, $h_{\min} = h_M/8$, a density ratio of 64 between the finest and coarsest part. Measured on `MustangHero.usda` (car on a 16-unit floor) at `--resolution 512`:

mode	total	bumper `Object_199`	hood `Kapoot`	floor
per-model	375 199	762 (cell 0.0313)	2 326	262 144
per-mesh, defaults	934 338	3 082 (cell 0.0039)	20 279	262 144
per-mesh, `--cell-min` = `--cell-max` = 0.03125	375 157	764	2 339	262 144

The control row reproduces per-model to 0.01 %; the remaining difference is the orientation effect alone.

From a camera (`--cell-from-camera CAM --camera-pixels P`, after Mesh2GS, arXiv 2606.21898). Each piece's cell is what one pixel of camera `CAM`, `P` pixels across, covers where the piece's mesh box is nearest to it:

$$h = \frac{z\,(A/P)}{f}, \qquad z = \max\Big(\,\big\|\mathrm{clamp}(\boldsymbol{e}, \boldsymbol{b}_{\min}, \boldsymbol{b}_{\max}) - \boldsymbol{e}\big\|,\; z_{\text{near}}\Big),$$

with A the camera's horizontal aperture, f its focal length, $\boldsymbol{e}$ the eye, and the result bounded only by explicit `--cell-min`/`--cell-max`. Mesh2GS takes $z_{\min}$ over the box inside the frustum and over the camera's time samples; the Euclidean distance at `--time` is the same for a box in view and finer, never coarser, for one beside the frustum. On two equal cards three and seven units from a 64-pixel camera, the near card received 930 Gaussians and the far one 169: a ratio of 5.5 against $(7/3)^2 = 5.4$ expected. This is the mode used for every production bake of the Corvette (`--cell-from-camera /World/Camera --camera-pixels` 640, 1280 or up to 5120).

The budget. `--max-splats` (2 M by default) is shared over the whole stage. Every piece is first counted (a run of the effect with room for one Gaussian, whose count and scan cover everything a run would write); where the total W exceeds the budget B, piece i wanting w_i gets $s_i = \max(1, Bw_i/W)$ and walks a cell $\sqrt{w_i/s_i}$ times coarser. A cell k times coarser covers the surface with k^2 fewer Gaussians, so proportional shares and uniform coarsening are the same thing. On two cards wanting about 1 300 under a budget of 800 the shares came out 356 and 361; spent in mesh order, the second card held about 145.

Per-triangle and per-run limits. A triangle walks at most `--max-cells` cells of its projected bounding box (2^{18} by default); cells beyond are counted and reported as a warning. At 4736 px the Corvette's `Infinite_Black` lost 359 628 cells this way (visible holes) and was re-baked with `--max-cells 1048576`. A run's output picture is capped at 2 M Gaussians (`kRunCeiling`, 192 MB); a piece wanting more is converted in slices — the effect reports the first triangle the ceiling cut into, and the host reruns from there with `firstTriangle` — giving the same array a single run would have written.

3.4 One Gaussian per cell

The projection. For a triangle $(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c})$ with geometric normal $\widehat{\boldsymbol{n}}_g$, the two axes (i, j) along which $\widehat{\boldsymbol{n}}_g$ points least define the projection; a point maps to $\boldsymbol{o}(\boldsymbol{x}) = (x_i - b_{\min,i},\ x_j - b_{\min,j})/R$, with R the range in use (the model's, the mesh's longest side, or the camera-derived box). A cell is $1/N$ of that unit square, i.e. $h = R/N$ in the world, bounded as in Section 3.3. A Gaussian is emitted at every cell whose centre falls inside the projected triangle, located by its barycentric coordinates $\boldsymbol{\lambda}$, at the world point $\boldsymbol{\mu} = \lambda_0\boldsymbol{a} + \lambda_1\boldsymbol{b} + \lambda_2\boldsymbol{c}$.

The sizes. With $E = [\boldsymbol{b} - \boldsymbol{a},\ \boldsymbol{c} - \boldsymbol{a}]$ and $O = [\boldsymbol{o}_b - \boldsymbol{o}_a,\ \boldsymbol{o}_c - \boldsymbol{o}_a]$, the Jacobian of the projection's inverse is $J = E\,O^{-1} = [\boldsymbol{j}_u, \boldsymbol{j}_v]$ (guarded at $|\det O| > 10^{-24}$). The two in-plane standard deviations and the third are

$$s_u = \min\Big(\big\|\boldsymbol{j}_u\big\|\sigma\tfrac{1}{N},\ \ell_{\max}\Big), \qquad s_v = \min\Big(\big\|\boldsymbol{j}_v\big\|\sigma\tfrac{1}{N},\ \ell_{\max}\Big), \qquad s_n = \varphi\min(s_u, s_v),$$

where σ is `--sigma`, $\ell_{\max}$ the triangle's longest edge and φ `--flatness`. Three details follow the original source: the Jacobian is over the triplanar coordinates, not over texture coordinates; the $1/N$ factor is applied by the original's exporter, not by a shader (without it every Gaussian is the size of the model); and the frame is not the Jacobian's columns but the triangle's longest edge $\widehat{\boldsymbol{t}}$, the interpolated shading normal $\widehat{\boldsymbol{n}} = \text{normalize}(\sum_k \lambda_k \boldsymbol{n}_k)$, and their cross product. The covariance is therefore

$$\Sigma = R_{\text{frame}}\,\text{diag}(s_u^2, s_v^2, s_n^2)\,R_{\text{frame}}^{\top}, \qquad R_{\text{frame}} = [\widehat{\boldsymbol{t}}_\perp,\ \widehat{\boldsymbol{n}}\times\widehat{\boldsymbol{t}}_\perp,\ \widehat{\boldsymbol{n}}],$$

with $\widehat{\boldsymbol{t}}_\perp$ the edge made orthogonal to $\widehat{\boldsymbol{n}}$. Without the bound $\ell_{\max}$, 1 859 slivers of a BMW, seen almost edge-on by their own projection, produce Jacobian columns twenty orders of magnitude too long and draw white spikes metres long.

Width. The original's $\sigma = 0.65$ leaves a converted surface about 30 % transparent to a ray tracer. Measured on a unit quad lit head-on (mesh pixel 0.5032):

σ	coverage	pixel
0.65	0.702	0.351
0.85	0.848	0.425
1.0	0.911	0.456
1.2	0.957	0.479
1.5	0.986	0.493

The default is therefore $\sigma = 1.0$; `--sigma 0.65` restores the original.

Flatness and surfels. The original wrote 10^{-7} model units as the third size, which made a Gaussian's thinness depend on the model's scale: a 66 mm pawn traced correctly while a 100-unit fox became a ghost under the ray tracer. The third size is now the fraction φ (default 0.1) of the smaller in-plane size. `--flatness 0` writes surfels — Gaussians with one scale exactly zero, the 2D Gaussian-splatting disc — which the rasteriser draws by a ray–plane intersection per pixel and the path tracer clamps to 10^{-4} of the widest axis.

Opacity as coverage. A point of a converted surface lies under several Gaussians at once, so an opacity written straight into each over-covers: a mask of 0.5 covered 96 % of what stood behind it. Every opacity the conversion reads (`--opacity`, the material's constant, a map's value, what a glass keeps) is treated as the surface's *coverage* F and converted into the per-Gaussian α by expanding the transmittance of the stack to second order: on the cell lattice the footprints sum to $2\pi\sigma^2$ and their squares to $\pi\sigma^2$, so

$$\tfrac{\pi\sigma^2}{2}\,\alpha^2 + 2\pi\sigma^2\,\alpha = -\ln(1-F) \;\Rightarrow\; \alpha = \frac{\sqrt{4S^2 + 2SL}\; - 2S}{S} + \tfrac{1}{255}, \quad S = \pi\sigma^2,\; L = -\ln(1-F),$$

the 1/255 paying back the area a Gaussian loses where it is cut at 1/255; $F \geq 0.999$ maps to 1 and 0 to 0. Measured on planes of constant coverage 0.25 / 0.5 / 0.75, the raster gives 0.251 / 0.501 / 0.752 and the ray tracer the same at full size, against 0.253 / 0.510 / 0.770 with the first-order term alone. A cut-out map below `--opacity-cut` (0.5) writes no Gaussian at all, so the budget goes where the surface is.

A triangle that catches no cell. A triangle smaller than a cell need contain no cell centre; rasterised, it vanishes. Without a fallback, that removes the Buick's grille and window rubbers (80 137 triangles become 34 011 Gaussians). Such a triangle therefore gets one Gaussian at its centroid, sized by its own second moments (`m2sOwnSize`): the uniform triangle's covariance $\Sigma_\triangle = \frac{1}{12}\sum_k \boldsymbol{d}_k \boldsymbol{d}_k^\top$ (corners about the centroid, in the tangent plane), widened as a cell is, $\Sigma = 12\sigma^2\Sigma_\triangle$, with the wide axis clamped to $[0.05\,s_{\max}, s_{\max}]$ of the cell's size and the narrow axis to at least 0.02 of the wide. Without it, a panel's rolled lip — a strip of sub-cell triangles — draws a band of cell-wide beads three pixels thick along every Corvette edge where the path tracer draws a hairline. In the Corvette bakes, glass and tyres come out at about one Gaussian per triangle at both 640 and 1280 px.

Deterministic order. The original appended with an atomic counter, so two conversions of the pawn produced the same 729 073 Gaussians in different orders. The emit is now three passes — count each triangle's covered cells, an exclusive scan in one 256-thread workgroup, write each triangle's Gaussians at its offset — sharing one function (`m2sTriangleOf`) so that counting and writing cannot disagree. Two conversions are now byte-identical, a budget keeps the first Gaussians in mesh order, and Gaussian i is the same Gaussian from one pose to the next. A record that decodes to nothing is written as an empty slot rather than dropped, so array indices stay aligned.

3.5 Where a surface stops or folds

A Gaussian is a cell wide, and on the Corvette a cell is a pixel of the conversion camera. A part narrower than that — the flange a hood folds down into its gap — still received cell-wide Gaussians in its own plane; they reached out of the gap over the paint beside it, carrying the gap's baked darkness, and every panel edge became a thick black wavy line where the path tracer draws a thin one.

Hard edges (`mesh2splat_edges.slang`). Per mesh, over all its pieces, every edge and corner is keyed by where its points stand and radix-sorted on the device; each run of equal edge keys is one edge. An edge is *hard* when it is open (one triangle uses it: the part stops or another mesh begins), when the two triangles sharing it fold by more than 40° (`kCreaseDegrees`), or when more than two triangles share it (non-manifold). Because keys are positions over the whole mesh, a boundary between two materials on one smooth surface is not an edge. Each hard edge becomes a line with an inside, $(\widehat{\boldsymbol{m}}, c)$ with inside $\widehat{\boldsymbol{m}}\cdot\boldsymbol{x} \geq c$ — the plane through the edge, orthogonal to its triangle, facing the third corner — known to its own triangle as a bit and told to every triangle touching either end (up to five lines each), since a Gaussian near a corner can stand beside the next triangle's hard edge.

Holding the reach (`m2sHoldEdges`, `--edge-reach` ρ, default 1.5). For each line in the Gaussian's plane, let $\boldsymbol{m}$ be its normal in the tangent basis, C the in-plane covariance, $r^2 = \boldsymbol{m}^\top C\,\boldsymbol{m}$ the variance across the line and d the centre's signed distance to it. The size allowed across the line is $a = \max(d/\rho,\ 0.05\,r)$, and where $r > a$

$$C \leftarrow C - \kappa\,(C\boldsymbol{m})(C\boldsymbol{m})^\top, \qquad \kappa = \frac{1 - a^2/r^2}{r^2},$$

which makes $\boldsymbol{m}^\top C\,\boldsymbol{m} = a^2$ exactly while keeping the variance along the line's conjugate direction, so a strip of Gaussians still covers the strip. Lines not lying in the Gaussian's plane (the flange seen from the paint it folds off), and lines the centre stands more than a quarter size behind, are ignored; lines are applied in a canonical order so the result does not depend on discovery order. The new covariance is re-diagonalised into a tangent and two sizes. The count of Gaussians is unchanged — only their shapes are.

Closed shells. The same pass counts edges not shared by exactly two triangles, one counter read back per mesh. A mesh is marked closed when

$$n_{\text{open}} < 2\sqrt{1.5\,T}\ ,$$

T its triangles ($1.5T$ approximates its welded edges), and its Gaussians carry `primvars:athenea:splat:closed = 1`, which allows a player or a cut to drop nodes facing away. A fixed share of edges (2 %) is not sufficient: an open sheet's border grows as the square root of its area, so a fine enough tube passes as closed — with that rule the rims' barrels (1 %) are culled and light shows between the spokes. Against the square root, the doors (no open edge), logos (1.4) and calipers (1.5) stay closed and rims (4.0), tyres (3.1), hood (3.1) and bumper (4.4) do not: 102 of the Corvette's 192 meshes. A car's door is two skins; drawn without culling, the inner skin sorts in front of the outer in patches, a net of dark cracks at small screen sizes.

Measured. The Corvette's TX base at 640 px of `/World/Camera`, rebaked with edges held (default 1.5) and as a control with `--edge-reach 0`, both 5 380 804 Gaussians; day frame at 1920 × 1080 against the mesh path trace:

bake	relMSE (HDR)	PSNR (display)	SSIM
previous base (earlier build)	0.0900	21.83 dB	0.8555
control, edges not held	0.0902	21.86 dB	0.8554
edges held, $\rho = 1.5$	0.0887	21.93 dB	0.8567

The global gain is small because the edges are a small fraction of the frame; the visible effect is on the panel gaps.

3.6 Material capture

Each Gaussian record carries position, coverage-derived opacity, three sizes, a rotation and a colour, followed — depending on what the cloud needs — by the shading normal and metallic, roughness, transmission and texture coordinates; four joints and their gradients; three relief entries for displacement; an emission entry; four layer entries; and the shape operator.

Base. The colour is the material's base colour times its base weight, times the albedo map where there is one (decoded to linear). Metallic and roughness are the material's constants, multiplied by the packed map's channels where there is one (blue metallic, green roughness), as glTF multiplies its factors; a missing

channel is a factor of one. The original defaults to $(0.1, 0.5)$ from the map alone, so every material without a map becomes the same plastic; with the material's own values passed, a converted quad under a head-on light reads 0.5009 against the mesh's 0.5032.

Shading normal. A normal map is read in the frame its UVs give — $\partial \boldsymbol{p}/\partial u$ from the triangle's positions and UVs, orthonormalised against the normal, as the mesh shader does — and the result is kept as the Gaussian's shading normal (`primvars:athenea:splat:normal`) beside its geometric frame. Read against the longest edge instead, every facet of the pawn's marble turns its veins by its own angle. The disc itself is not turned unless `--normal-map-turns` is given, which would open the surface where the relief is steep.

Layers. A Gaussian carries the specular weight, colour and index; the coat's weight, roughness and index; the sheen's colour times its weight and its roughness; and the coat darkening switch — OpenPBR's units, three words per splat on the device. Up to three maps on these inputs are sampled per Gaussian (`m2sLayersAt`). This is required because the Corvette's paint (`Car_Paint_Main`) is OpenPBR metalness 1 over a base of about 0.05 under a clear coat of weight 1: everything the eye sees of it is the lacquer, and without the coat a converted car comes out nearly black.

Schlick metals. OpenPBR and glTF metals are MaterialX `generalized_schlick_bsdf` (colour head-on, specular colour at 82°), not a conductor of the artistic index; for a dark metal the two differ by a factor of two at grazing angles (at 60°: 0.172 for the conductor, 0.077 for the Schlick from $F_0 = 0.047$). The material's kind is written per Gaussian (`primvars:athenea:splat:schlickMetal`) and the renderer reflects it with the DFG fit of its two ends. Treating every OpenPBR metal as dielectric polish in the bake has a measured cost: a Mustang's front bumper stores the constant that decodes to black in all 27 016 Gaussians; with Schlick lobes of $F_0 > 0.2$ kept as body, the bumper's mean colour is (0.45, 0.51, 0.61).

Emission and transmission. Emission is the material's colour times its map, linear and unbounded, in an entry of its own. Transmission T and transmission colour $\boldsymbol{c}_T$ are a channel the original did not have (glass comes out opaque white). The colour is taken towards the transmission colour,

$$\boldsymbol{c} \leftarrow \mathrm{lerp}\big(\boldsymbol{c},\ \mathrm{lerp}(\mathbf{1}, \boldsymbol{c}, w_T)\,\boldsymbol{c}_T,\ T\big),$$

with $w_T = 1$ for glTF (transmission tinted by the base colour) and 0 for OpenPBR and standard_surface. The opacity is the coverage mapping of Section 3.4 applied to $o \cdot o_{\mathrm{mat}} \cdot \mathrm{mask} \cdot \mathrm{lerp}(1, g, T)$, g being what a fully transmitting surface still covers (Section 3.8).

The material roughness mark. The stored roughness may be widened by the specular filter (Section 3.7). The material's own roughness before widening is written separately, `primvars:athenea:splat:materialRoughness` (-1 under a roughness map, read as sharp). The player uses it to decide whether the polish lobe answers its occluded part from the baked reflected field (material roughness at least 0.3: paint, 0.34) or keeps it dark (chrome 0, rough metal 0.22); in the decimated tree it is packed into the `pbr` word (bit 26 present, bits 27–31 the value in five bits).

3.7 The specular filter (Toksvig widening)

A Gaussian has one normal. Where a strip of chrome curves under it, the path-traced pixel averages a highlight that moves across the footprint and shows a thin line; the Gaussian shows the light its single normal meets, at full brightness, over its whole footprint. On the Corvette, without filtering, the grille trim — four pixels high — comes out as a row of white beads, and the chrome reads 1.6 times the path trace while a chrome ball validates within 1 %.

The conversion therefore widens the roughness by the spread of normals under the Gaussian (`m2sSpread`, `m2sWidened`). The turn rate is the largest angle between corner normals per unit corner distance, over the triangle's three edges, and the spread is that rate times the Gaussian's wider size:

$$s = k \max_e \frac{\angle(\boldsymbol{n}_{e_0}, \boldsymbol{n}_{e_1})}{\left\| \boldsymbol{p}_{e_1} - \boldsymbol{p}_{e_0} \right\|} \max(s_u, s_v),$$

with k = `--specular-filter` (1 with `--transfer`, 0 otherwise). With perceptual roughness r and GGX $\alpha = r^2$, the spread is added to the slope variance (Toksvig):

$$\alpha'^2 = \alpha^2 + 2s^2, \qquad r' = \text{saturate}\Big((r^4 + 2s^2)^{1/4} \Big).$$

A flat triangle, or one whose corners share a normal, is unchanged. The coat's roughness is widened the same way unless the cloud carries the shape operator.

The shape operator. For a transfer the conversion also writes, in the Gaussian's own axes $(\boldsymbol{u}, \boldsymbol{v})$, the symmetric shape operator S with $\mathrm{d}\boldsymbol{n} = S\,\mathrm{d}\boldsymbol{x}$, solved from the corners' normal differences against their position differences (`m2sShape`, stored as (S_{uu}, S_{uv}, S_{vv})). The renderer uses it to turn the reflection across the footprint pixel by pixel (the per-pixel "sharp lobe"). Widening the coat as well would blur twice, so where S is written the coat is not widened, and the player reads the sharp lobes at the unwidened roughness $r_0 = \max(r'^4 - 2s^2, 0)^{1/4}$ (`unwidened`), with s carried per Gaussian.

Measured. At 640 px the coat came out at roughness 0.17–0.24 with 5–14 % high-pass on folds; at 1280 px the spread under each Gaussian is smaller and the coat's roughness comes out at exactly 0, so the widening matters less as the bake gets finer. Turning it off is nevertheless worse (Section 3.11): with `--specular-filter 0` a dashed dark line appears on the sill's fold and the chrome strip sparkles.

3.8 Glass

A Gaussian cannot refract, but it can let what is behind it through by its opacity, reflect through its specular lobe, and carry an index of refraction that the renderer bends by (`primvars:athenea:splat:ior`, one per cloud; with layers each Gaussian bends by its own specular index). The conversion's job is to decide, per transmitting piece, whether the glass is a *solid* (a body: a ball, a lens) or a *thin wall* (a sheet or slab whose two faces bend nothing), because the two are drawn differently.

Classification (`classifySheets`, `mesh2splat_sheet.slang`), for every transmitting piece whose material does not already declare itself thin-walled. On the device, the piece's edges are keyed by position, sorted, and their runs counted; the signed volume V (tetrahedra to the origin) and the area A are summed. Then

- *open*: more than 1/200 of the edges have one triangle, or any edge has more than two;
- *slab*: closed, and the thickness $2V/A$ is below four of the model's cells or below $0.02\sqrt{A}$ ($2V/A$ is a slab's gap and two thirds of a ball's radius).

An open or slab piece is read thin-walled — unless it is *tinted* (transmission colour luminance below 0.9), which stays solid for now, since a thin wall lets the background through grey. The Corvette's glass consists of closed slabs 3–4 mm thick, which the edge rule alone calls solid; the thickness rule reads them. Each decision is printed, e.g. for the rear light lens: "tinted, kept solid glass: 200 of its 17284 edges open, 0 shared by more than two; 1.199 thick (2V/A) against a cell of 0.007921". `--thin-glass MAT` and `--solid-glass MAT` override per material.

Coverage. A solid glass covers `--glass-opacity` of what is behind it (default 0.6; a TX lens answers its background from the domes alone, so a glass that covered fully hid every other light shining through it). A thin wall covers exactly what the sheet reflects head-on, both faces with inter-reflection:

$$g_{\text{thin}} = \min\left(\frac{2R}{1+R},\ 0.998\right), \qquad R = \left(\frac{n-1}{n+1}\right)^2,$$

and the frame scales its reflection back by $1/R$. At the windscreen's index of 1.16 that is an opacity of 0.0055, and the rasteriser draws such sheets at an opacity of at least 0.1 with colour scaled down by the same factor, which keeps the product $\alpha \boldsymbol{c}$ — the reflection — intact (windshield relMSE 2.26 → 0.89 under the workshop sky).

3.9 Shadow and light catchers

The ground is not converted as a surface; its interaction with the object is converted instead, so the playback is Gaussians only.

Shadow catcher (`--shadow-catcher --prim OBJECT`). The ground is found automatically (the largest flat mesh outside the object whose top is at the object's bottom; `--catcher-ground` names one). A patch of quads — the object's footprint widened by `--catcher-margin` heights (1.5), split into a grid so that no triangle exceeds `--max-cells` — is added to an in-memory copy of the stage and converted alone at a cell of a hundredth of the object's height, as a TX transfer whose rays meet the object and the ground. At draw time the gaussian's opacity is $-\ln(\text{ratio})/2\pi$, where the ratio is light reaching it with the object against without, since a lattice of $\sigma = 1$-cell Gaussians sums to 2π and a layer of opacity a leaves $e^{-2\pi a}$. The Corvette's: a 5.9 × 8.3 m patch, 11 × 11 quads, 309 k Gaussians at a 1.26 cm cell, baked in 79 s. Against the path trace:

	under the car (GT 0.182)	**bumper contact (GT 0.043)**	**open ground (GT 0.217)**
mesh ground's own shadow map	0.182	0.041	0.215
catcher, depth 2.4, two triangles	0.144	0.002	0.206
catcher, depth 2π, grid	0.192	0.030	0.220

Light catcher (`--light-catcher`). The same patch, reaching `--light-reach` object lengths (3) ahead and behind, at a cell of a thirtieth of the height, converted as a radiance bake per light group with the dome black; drawn additively, it carries the low beam's pool and the tail lamps' glow.

3.10 Validation against the path trace

`--validate DIR` measures a conversion material by material against the stage path-traced, for any stage. The groups are the stage's material bindings (a mesh of two materials is in both); the conversion is the command's own, run once per group with every mesh outside it hidden, so every option is what is measured. The ground truth is path traced once (`--validate-paths` 512, `--validate-bounces` 6) and kept; the stage is rasterised once as meshes with its Cryptomatte, and each group's mask is the coverage of every rank whose id hashes one of its meshes. Within the mask's box the Measure effect gives relMSE, p99 and the means of the cloud and of the mesh raster beside it, written to `validate.json`. `--validate-sky white` (or an image) draws everything under another sky with the stage's other lights off, so a transfer's er-

ror is its occlusion's and reflection's alone. Among the defects it measures: a solid glass ball traced at half the path-traced brightness (0.096 against 0.195) when rays are not bent into it, and a paint ball under a white dome reading 13 % under because of the coat's Fresnel.

3.11 Measured results on the Corvette

All bakes below are TX transfers, `--cell-from-camera /World/Camera`, 64 paths per Gaussian, 3 bounces, no adaptive extra paths, with the specular filter and edge reach at their defaults (1 and 1.5). "Car crop" is display PSNR on the 900 × 420 crop of the car, "frame" on the full 1920 × 1080 frame, both against the mesh path trace, after the merge tree is cut at 0.25 px with back culling.

640 → 1280 px. The 640 px bake's leaves were 3–4 mm on the hood, and the door's stripes and the sill's hairs — absent from the path trace — were in the cloud. The car is re-baked at 1280 px in 8 groups of top-level parts balanced by count, each converted with all other parts hidden (part names taken from the Cryptomatte manifest). The eight bakes wrote 891 387, 896 496, 809 073, 790 403, 1 631 765, 1 073 954, 1 302 878 and 1 307 473 Gaussians (8 703 429 in total) in 212–304 s of transfer each, about 34 min altogether.

	640 px	**1280 px**
leaves	5.38 M	8.70 M
hood / door	50 k / 79 k	151 k / 266 k
glass, tyres	about one per triangle	about one per triangle
car crop / frame (PSNR)	16.55 / 22.18 dB	17.88 / 23.90 dB



Figure 6. Car crop (900 × 420 of the 1920 × 1080 frame from `/World/Camera`): path trace of the mesh (top), the 640 px bake (middle) and the 1280 px bake (bottom), both drawn as the 0.25 px cut with back-face culling. Car-crop PSNR 16.55 dB and 17.88 dB.



Figure 7. The full 1920 × 1080 frame of the 1280 px bake, 0.25 px cut with back-face culling (frame PSNR 23.90 dB against the path trace).

Analysis of the 1280 px cloud against the 640 px one: hood highlights 3.2 → 2.0 px off their path-traced position, the door's open-reflection contrast 0.35/0.53 → 0.56/0.74 of the path trace's, the grille's holes in the coarse cut 2.24 → 1.04, the sill's hairs 987 → 459 px from the resolution alone (Figures 6 and 7).

Settings for 1280 px (variants A/B/C). To select the 1280 px settings, eight parts (both doors, both sill panels, front fender, chrome grille, grille rim, hood) were baked with the whole car casting, 931 017 Gaussians in each variant:

variant	settings	result
A	as the 640 px bake ($k = 1$, $\rho = 1.5$)	selected: no hairs, the door smooth
B	`--specular-filter 0`	dashed dark line on the sill's fold; the chrome strip sparkles
C	B + `--edge-reach 1.0`	elongation and hairs identical to B



Figure 8. The 1280 px settings on the sill panel: the side of the car for context (top left), variant A (top right), variant B (bottom left) and variant C (bottom right). B and C show the dashed dark line on the fold.

The variants test whether the door's streaks are the coat's Toksvig widening saturating on folds and the sill's hairs the edge pass's elongated Gaussians. Neither holds at 1280 px: the widening is needed on folds and the tighter edge reach changes nothing, so the settings of A are used (Figure 8).

The 5120 px close-up level. A close view stands far nearer than `/World/Camera`, about 0.5 mm per screen pixel against 3–4 mm leaves. A logo-and-grille conversion from a close camera grows from 108 527 to 404 594 Gaussians between 1280 and 5120 px and shows the checkered flag, the bowtie and the grille's mesh crisp. The 66 parts the 1280 px bake had already given at most 1.5 Gaussians per triangle (badges, grilles, lamps, calipers, mirrors, wheels, wipers, trims) are re-baked at 5120 px in seven groups; the panels at 5120 px or as fine as kept a part under a cap of leaves (5 M, lowered to 4.5 M after a 5.01 M tree did not fit in device memory), with `--max-cells 1048576` (Figure 9):

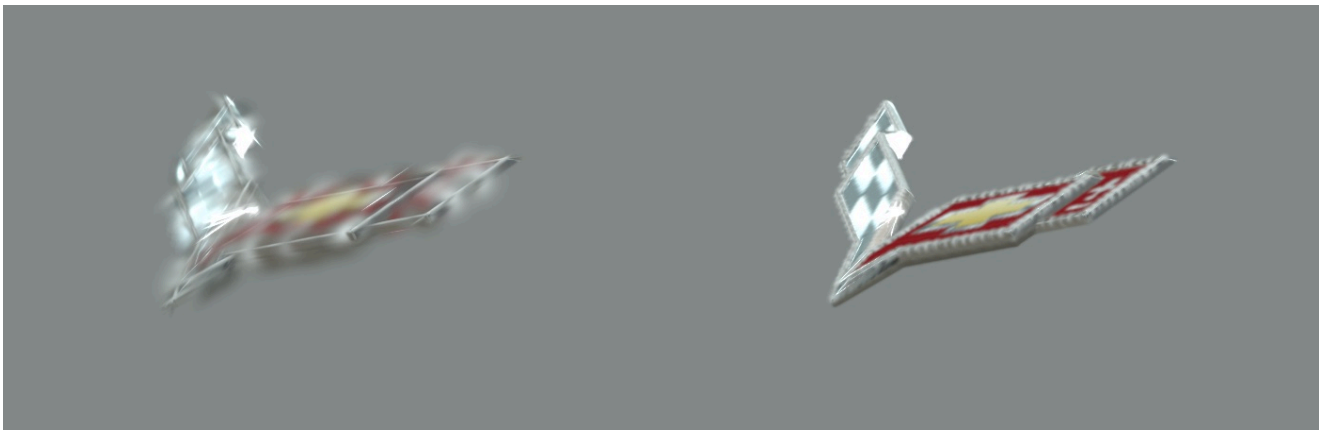


Figure 9. The logo from a close camera: leaves of the 1280 px bake (left) and of the 5120 px close-up level (right).

groups	leaves	resolution (px)
fine0–6 (66 detailed parts)	18 610 627	5120

groups	leaves	resolution (px)
door	4 084 170	5120
Bottom	4 897 428	3392
Part_3	4 370 459	4480
Infinite_Black	4 481 738	4736
six large panels and the windshield	15 994 566	5120
pan0–4 (small panels)	17 088 152	5120
Front_Bumper_001	5 008 986	4480
24 trees	**74 536 126**	

The 24 merge trees hold 74 328 031 merges, 208 095 roots and 148 864 157 nodes in 49 634 pages (87.4 GB) and share one occlusion codebook of 8 182 entries; Section 6.9 gives the details. (The conversion of Front_Bumper_001 emits 5 009 059 Gaussians; its tree holds 5 008 986 leaves, the figure used in the table.)

3.12 Limitations

- A Gaussian has one normal and one cell's footprint: a chrome tube a few pixels across is still 38 % brighter than the path trace on the front bumper, and the bumper's lower lip is ragged.
- Lamps carry no emission in the transfer bake; headlights are dark against the path trace.
- A tinted thin glass is converted solid, and a cloud keeps a single index unless it carries layers.
- The coverage mapping assumes the cell lattice's overlap $2\pi\sigma^2$; sub-cell triangles and displaced, split cells overlap otherwise and are not corrected.
- Nested point instancers are expanded only at the innermost level; motion authored as velocities is read at the sample.
- Device memory bounds a part at about 5 M leaves for the merge tree that follows; larger parts are converted at a coarser camera resolution.

4. Light transport baked into Gaussians (transfer) and relighting

A cloud converted from a mesh can carry its light in two ways. A *radiance bake* stores, per Gaussian, the light that arrived under the stage's own illumination; it is exact for that illumination and wrong under any other. A *transfer* stores instead how each Gaussian's surroundings would carry an arbitrary environment to it, and a frame combines that geometry with whatever sky and lights it has. This section describes the core of the work, the baker: the transfer that the scene's path tracer bakes into every Gaussian, and the relighting that evaluates it at draw time, in its TX form (a transfer that holds under any sky and any light), the reflection model built on top of it, the per-pixel treatment of sharp lobes, and what each piece was measured to buy against a path trace of the source mesh. Everything described runs on the GPU: the bake is the scene's own path tracer in a transfer mode, and the evaluation is the splat projection and blend kernels (`shaders/athenea/splat/splat_relight.slang`, `splat_project.slang`, `splat_blend.slang`).

4.1 Notation

A Gaussian has a position $\boldsymbol{p}$, a unit normal $\boldsymbol{n}$ (its shortest axis, or a stored shading normal, turned towards the eye), an albedo ρ, a metalness m, a perceptual roughness r with GGX parameter $\alpha = r^2$, and optionally a clear coat (weight w_c, roughness r_c, index η_c), a sheen and a transmission. $\boldsymbol{\omega}_o$ is the direction to the eye and $\boldsymbol{m} = \text{reflect}(-\boldsymbol{\omega}_o, \boldsymbol{n})$ the mirror direction. Y_k are the real spherical harmonics, indexed $k = l^2 + l + m$; degree 2 has 9 coefficients and degree 3 has 16. A sky has radiance coefficients L_k (rgb), a sun of direction $\boldsymbol{s}$ and irradiance $E_\odot$ extracted from it, and a prefiltered map $\tilde{L}(\boldsymbol{\omega}, r)$.

4.2 What a Gaussian stores

The transfer is one run of values per Gaussian, f16 on the device. There is no mode token: the count is the layout (`common/transfer_layout.slang`) (Figure 10).

values	layout	origin
9	degree 2, direct	first transfer
36	degree 2, direct + indirect (27)	first transfer
84	degree 2, both + reflected field (48)	TX at `--transfer-degree 2`
16	degree 3, direct	
64	degree 3, direct + indirect (48)	
112	degree 3, both + reflected field (48)	**TX default**
10	two zonal lobes in the Gaussian's frame	skinned clouds

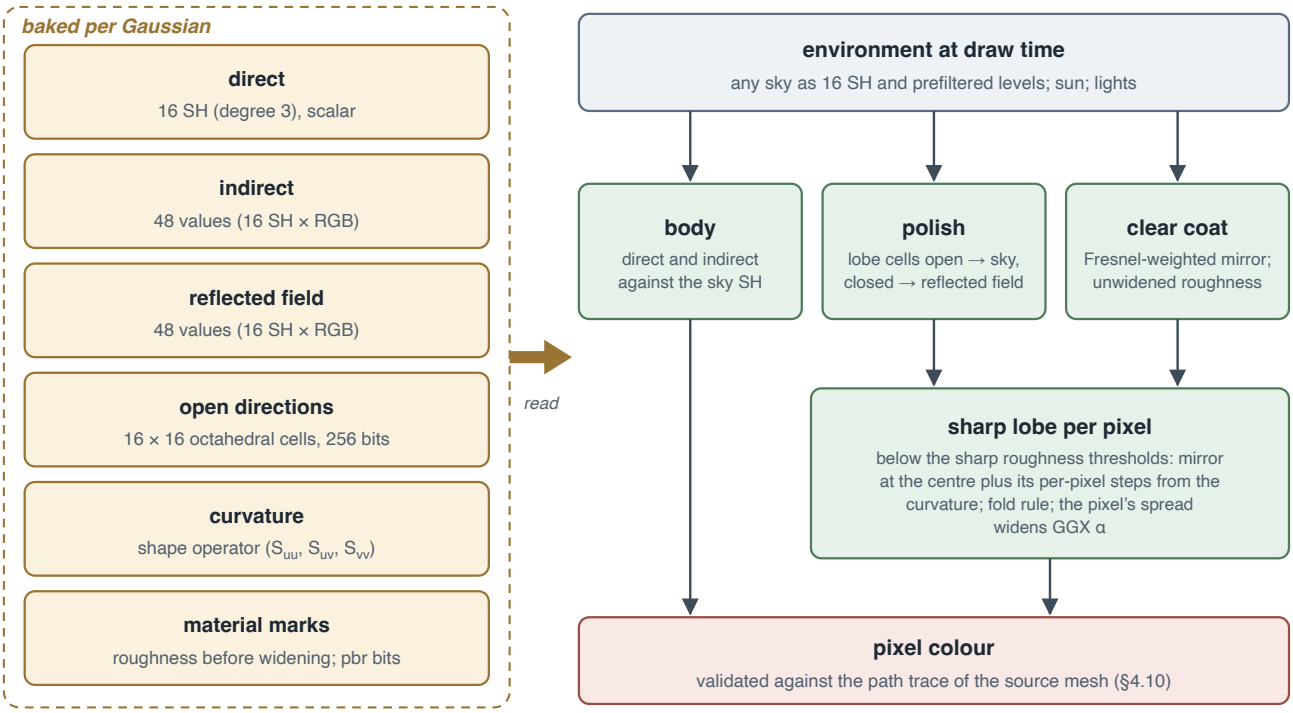


Figure 10. What the bake stores per Gaussian and how the draw combines it with the environment: the body from the direct and indirect SH, polish and clear coat from the reflected field and the open-direction cells, and the sharp lobe evaluated per pixel.

Beside it:

- **Open directions** (`primvars:athenea:splat:shadowBits`): 2 words (64 bits, an 8 x 8 octahedral grid; first transfer), 8 words (16 x 16 cells over the whole sphere; TX default) or 32 words (32 x 32). A bit is set where the bake's ray in that cell left the scene.
- **Curvature** (`primvars:athenea:splat:curvature`): the shape operator in the Gaussian's two in-plane axes, (S_{uu}, S_{uv}, S_{vv}), three floats, written with `--transfer`.
- **Material roughness** (`primvars:athenea:splat:materialRoughness`): the material's own roughness before the bake's widening (section 4.9), -1 under a texture map. On the device it rides in five bits (27-31) of the pbr word, flagged by bit 26.

The direct half is a scalar per coefficient (geometry alone); the indirect half is rgb, coloured by what light bounced off; the reflected field is rgb. The TX layout costs, per Gaussian:

	first transfer	TX
direct	9	16
indirect	27	48
reflected field	–	48
visibility	2 ints (64 bits)	8 ints (256 bits)
file, bytes	152	480
device, bytes	80	256

For the whole Corvette (about 10 M Gaussians at the time) that is 4.8 GB of file and 2.6 GB of device for the transfer alone; the codecs that reduce its storage and bandwidth (a clustered PCA of the transfer, a Hamming k-means codebook of the shadow bits) are described in Section 7, where they serve one applica-

tion, a WebGPU player.

4.3 The environment at draw time

`technique::Environment` prepares each dome in three forms, all buffers read through one module (`light/environment.slang`) by both the rasteriser and the tracer.

Spherical harmonics of radiance. `env_project` sums the texels of one level of the dome's mip chain, each weighed by the lat-long solid angle $2\pi^2 \sin\theta \, \Delta u \, \Delta v$ (the measure the dome's importance sampling uses):

$$L_k = \sum_{\text{texels}} L(\boldsymbol{\omega}_t) \, Y_k(\boldsymbol{\omega}_t) \, 2\pi^2 \sin\theta_t \, \Delta u \, \Delta v.$$

Sixteen coefficients are projected (degree 3); the irradiance of an untransferred body reads the first nine, convolved at read time with the clamped-cosine zonal weights $\{\pi, 2\pi/3, \pi/4\}$. A constant sky projects back to πL of irradiance at every normal: 6.2841 against 6.2832 (0.089 %) over 64 directions.

The sun, taken out. A disc a thousand times brighter than the sky is not a degree-2 function; projected, it smears into a lobe sixty degrees wide. `env_sun` finds the brightest region whose light is as concentrated as a disc of its solid angle (light-weighted mean square angle about its centre at most twice a disc's, $\theta_{\text{cone}}^2/2$), takes its texels out of both the harmonics and the prefilter's source, and returns a direction $\boldsymbol{s}$, an irradiance $E_\odot$ and a half angle. The irradiance is summed over exactly the texels the harmonics skip, at the same level and with the same measure, so what leaves the coefficients arrives as a light (closed form 0.1119 against 0.1113).

The prefiltered map. The sky without its sun, convolved with a GGX lobe into an octahedral mip chain of eight levels, level k standing for roughness $(k/7)^2$ (0, 0.020, 0.082, 0.184, 0.327, 0.510, 0.735, 1). The base side matches the dome's own texel, $S = \sqrt{2WH/\pi}$, rounded down to a power of two in $[256, 2048]$. The prefilter uses filtered importance sampling (Krivanek and Colbert): each GGX sample reads the source one level coarser than its own solid angle asks, never narrower than the texel it writes, with the sequence turned by a per-texel hash, 64 samples at level 1 doubling to 1024. The source's lat-long mips weigh each row by $\cos\theta_0 - \cos\theta_1$, so every level keeps the sky's solid-angle mean. Without these, a 1.5-degree sun in a rough level was a scatter of texel-sized blobs (a paint ball's 4-16 px band at 2.13 times the path trace).

4.4 The bake

4.4.1 The transfer integral and its estimator

The transfer is the projection of visibility times the clamped cosine, the albedo taken as one:

$$T_k = \frac{1}{\pi} \int_{S^2} V(\boldsymbol{\omega}) \max(\boldsymbol{n} \cdot \boldsymbol{\omega}, 0) \, Y_k(\boldsymbol{\omega}) \, d\boldsymbol{\omega},$$

and a frame recombines it with the sky as $\langle T, L \rangle = \sum_k T_k L_k$. It is gathered by a third variant of the path-traced kernel, `kTransfer`, beside the radiance bake and the camera path. At a Gaussian's first vertex the material is replaced by the sampled direction itself: directions are stratified (a jittered $\sqrt{N} \times \sqrt{N}$ grid) and uniform over the hemisphere, density $1/2\pi$, against an estimator that wants $\cos\theta/\pi$, so an escaping sample is worth $2\cos\theta \, Y_k(\boldsymbol{\omega})$ and a blocked one nothing. Uniform rather than cosine sampling, and over the hemisphere measured as a sphere with the far half empty, because the harmonics are orthonormal only over the sphere; fitted coefficient by coefficient over a hemisphere they overshoot (the pawn came back ten times too bright). The accumulation *is* the projection; no fit is solved.

The same paths give two halves at no extra ray: a path that escapes right after the first vertex contributes to the direct half (scalar); one that escapes after bouncing carries the colour of what it met and contributes to the indirect half T_k^{ind} (rgb). Checked against closed forms: an unoccluded point transfers the clamped cosine, $(0.2820, 0.3258, 0.1581)$ against the exact $(0.2821, 0.3257, 0.1577)$ for bands 0-2, and a constant sky of 0.7 reads 0.6998. A corner of floor and wall, transfer recombined against a radiance bake of the same points:

	transfer	radiance bake	worst point
no wall, 1 bounce	0.6000	0.6000	0.0 %
black wall, 1 bounce	0.3091	0.3039	2.5 %
white wall, 3 bounces	0.5095	0.5325	4.3 %

4.4.2 Open directions and the direct half by quadrature

At the first sample of each point TX traces one any-hit ray per cell of a $S_c \times S_c$ octahedral grid ($S_c = 16$ or 32) over the *whole* sphere: the far half too, starting from the far side of the surface, so a direction behind a solid meets the solid and one behind a sheet leaves. A cell's bit is set where at least half the light gets through: the occlusion ray passes up to eight surfaces whose material transmits, weighed by transmission times colour luminance less the Fresnel at that angle (a ninth closes the way). The cells are traced from four origins over the Gaussian's footprint (cell c from origin $(c + \lfloor c/S_c \rfloor) \bmod 4$: the centre and three points brought onto the surface within the Gaussian's disc), so a groove narrower than a Gaussian no longer reads as a dashed line of whole shadows.

The same rays then give the direct half by quadrature (no grain), each cell weighted by its octahedral solid angle:

$$T_k^{\text{dir}} = \sum_c \tau_c \, \frac{\max(\boldsymbol{n} \cdot \boldsymbol{\omega}_c, 0)}{\pi} \, \Delta\Omega_c \, Y_k(\boldsymbol{\omega}_c), \qquad \Delta\Omega_c = \frac{4}{S_c^2} \left\| \boldsymbol{\omega}_c \right\|_1^3,$$

with τ_c the fraction transmitted along the cell's ray. Words are written four at a time as each 128 cells are traced, so the kernel holds no array the size of the grid.

4.4.3 The reflected field

A reflection needs the bounced light *by direction*, not integrated. For every path that leaves the Gaussian, meets the scene and later escapes, the throughput past the first vertex is the radiance that arrives along its first direction $\boldsymbol{\omega}_1$ under a white sky of radiance one. It is projected onto degree 3 of $\boldsymbol{\omega}_1$, uniformly over the domain it was drawn from:

$$F_j \approx \frac{1}{N} \sum_{i:\ \text{escaped after a bounce}} \Omega_{\text{meas}} \, \frac{\beta_i}{w_i} \, Y_j(\boldsymbol{\omega}_{1,i}),$$

with β_i the throughput, w_i the first-vertex weight and $\Omega_{\text{meas}} = 2\pi$ (or 4π over the sphere). That is 16 rgb coefficients, `primvars:athenea:splat:transferReflected`. It costs no ray. Because F is zero along open directions, sixteen harmonics read a closed direction beside an open one at a fraction of what stands there (the paint's closed part was 0.0014 of 0.115). So the bake also projects the indicator of closed directions C_j over the same samples, and the conversion divides the two, a normalised convolution giving the mean radiance of the closed directions about each direction, falling back to the closed directions' mean where their share is under 0.1 (blended up to 0.4), and re-projects the result over a 16 x 16 grid by cell solid angle (`m2sFieldOverClosed`).

For a transmitting Gaussian the first direction is drawn over the whole sphere (`bakeSphereDirection`: the stratum's height stretched to $[-1, 1]$ and the lower half mirrored through the tangent plane). Front samples keep the transfer's estimator ($4\cos\theta$ at density $1/4\pi$); back samples feed only the field, after being bent into a solid dielectric by its index (`refractInto`), so the field along $-\boldsymbol{\omega}_o$ holds what the eye's ray finds through both faces. Unbent, a glass ball validated at half the path trace (0.265 against 0.541 under a white dome).

4.4.4 Slices, filtering, and the split bake

The whole TX Corvette asked for its bake's answer at once: 35 float4 per Gaussian, 7.8 GB, with 5 GB free. `Converter::transfer` therefore bakes in slices whose answer is at most 1.5 GB (never under the bake's own batch, 2^{19}), and at most 2^{20} Gaussians where the filter runs. Each slice's rays are copied on the device, baked (`StageRenderer::bakePointsOnDevice` in transfer mode), its bounced halves filtered, written by `m2sTransferInto` and read back into the file's arrays before the next. The radiance bake uses the related `StageRenderer::bakeSplitOnDevice`, which keeps the direct and indirect halves as separate sums so that the indirect one can be filtered before `combineBake` fits the two. The transfer follows the same principle: its indirect half and its field (the noisiest values, at 64 paths per Gaussian in the Corvette's bakes) go through the splat bake filter, an a-trous filter at `--bake-filter` iterations (3), whose weights are position on the tangent plane, facing and prim id; the direct half, which holds the shadows, is never filtered.

4.4.5 Zonal lobes for skinned clouds

Nine harmonics are in the world; a skeleton turns the Gaussian under them. Rotating them costs a 9 x 9 per Gaussian. A zonal lobe rotates by rotating its axis, so a skinned cloud keeps two lobes, each the octahedral (u, v) of its axis in the Gaussian's own frame and zonal coefficients z_0, z_1, z_2: ten values (`primvars:athenea:splat:transferZonal`). As harmonics a lobe about $\boldsymbol{a}$ is $g_{lm} = z_l\, k_l\, Y_{lm}(\boldsymbol{a})$ with $k_l = \sqrt{4\pi/(2l+1)}$. For a fixed axis the best coefficients are the projection $z_l = k_l \sum_m f_{lm} Y_{lm}(\boldsymbol{a})$, leaving error $\|f\|^2 - \sum_l z_l^2$, so the fit (an AOFX effect, `splattransferzonal`) searches the axis on the device (band-1 direction, the normal, 64 Fibonacci directions, then a halving walk), fits the second lobe to the residual and refits the pair twice (Sloan 2008). At draw time `splatTransferFrame` turns each axis by the skinned frame and rebuilds nine world harmonics, which everything downstream reads unchanged. The bake poses the cloud at `--time` with the engine's own skinner, so the frame the lobes are fitted against is the packed quaternion a renderer reads. The shadow bits stay two words, re-binned into the Gaussian's frame. The indirect half is not kept zonal.

4.5 The body

For a Gaussian with a transfer, the part that does not depend on the eye is

$$\boldsymbol{B} = \sum_{k<16} L_k \left(T_k^{\text{dir}} + T_k^{\text{ind}} \right) \;+\; E_\odot \frac{\max(\boldsymbol{n}\cdot\boldsymbol{s}, 0)}{\pi} O(\boldsymbol{s}) \;+\; E_\odot\, T^{\text{ind}}(\boldsymbol{s}), \qquad T^{\text{ind}}(\boldsymbol{s}) = \max\Big(\sum_k T_k^{\text{ind}} Y_k(\boldsymbol{s}), 0 \Big),$$

the last term being the sun's bounce, which the first transfer lost with the sun it took out of the harmonics. $O(\boldsymbol{s})$ is the sun's share: for TX the bilinear read of the four cells around $\boldsymbol{s}$, cells behind the normal left out; for the 64 bits the same on an 8 x 8 grid; without bits, the degree-2 ratio

$$O_{\text{SH}}(\boldsymbol{s}) = \frac{\sum_k T_k Y_k(\boldsymbol{s})}{\sum_k \widehat{A}_l\, Y_k(\boldsymbol{n})\, Y_k(\boldsymbol{s})}, \qquad \widehat{A}_l = \{1, 2/3, 1/4\},$$

numerator and denominator sharing the truncation so it reads one where nothing occludes. The diffuse body is then $U\rho(1-m)\left[(1-t)\boldsymbol{B}+t(\ldots)\right]$ with U the layers' throughput (section 4.6.4) and t the transmission. $\boldsymbol{B}$, the sun's share and the field coupling (below) are kept per Gaussian while the sky and the lights stand (`transferViewless`, the `txCache`); a frame computes only what the eye changes.

The transfer's constant term is also an ambient occlusion: over an open hemisphere $T_0 = A_0 Y_0/\pi = 0.28209$, so $o = \mathrm{sat}(T_0/0.28209)$ is the fraction of the sky that reaches the Gaussian.

Lights that are not the sky. A distant, sphere, disk or rect light is evaluated with one sample at its centre through the same lobes. With TX cells, its shadow is the cells read over the cone the light subtends ($\tan^{-1}$ of its radius over the distance, or half a distant light's angular diameter), the darker of that and any ray-measured shadow; its bounce is T^{ind} read along it, times L/p of the sample, on the body and the sheen, and it couples the reflected field as the sun does (`splatLightBounce`).

4.6 Reflection

4.6.1 Which part of a lobe sees the sky

The first transfer narrowed every reflection by Lagarde and de Rousiers' fit of the constant term,

$$O_{\mathrm{L}}(\boldsymbol{n}\cdot\boldsymbol{\omega}_o, o, r) = \mathrm{sat}\Big((\boldsymbol{n}\cdot\boldsymbol{\omega}_o + o)^{2^{-16r-1}} - 1 + o\Big),$$

one number for the hemisphere: a door reflected the open sky where the ground stands in its mirror, and the Corvette's rims read 2.4 times the path-traced mesh, `Metal_rough` 6.5 times, the brake discs a hundred. With TX cells the lobe itself is asked (`splatLobeOpen`, `splatConeOpen`): the cell read $o_c(\cdot)$ (bilinear over four cells, cells behind $\boldsymbol{n}$ excluded) at the mirror and at a ring of six directions around it at angle

$$\theta_r = \max\big(\min(1.4\tan^{-1}\alpha,\ 1.2),\ 2\cdot\tfrac{\pi}{2S_c}\big),$$

about the angle within which GGX reflects half its energy, never narrower than a cell; and

$$O(\boldsymbol{m},\alpha) = \frac{\frac{1}{3}o_c(\boldsymbol{m}) + \frac{1}{9}\sum_{k=1}^{6} o_c(\boldsymbol{d}_k)}{\frac{1}{3}+\frac{6}{9}}$$

over the reads that weighed anything. A cone inside one cell read that cell's single bit and speckled the Corvette's paint (0.393 relMSE against the path trace; 0.298 with the cone held to a cell).

4.6.2 What the closed part shows: the field coupled to the sky

The field was baked under a white sky. Under a real one its pattern stands and its brightness is scaled per channel by how much more the surroundings bounce under that sky than under white (`splatFieldCoupling`):

$$\kappa = \frac{\sum_k T_k^{\mathrm{ind}} L_k + E_\odot\, T^{\mathrm{ind}}(\boldsymbol{s})}{\sqrt{4\pi}\ \ T_0^{\mathrm{ind}}},$$

falling back to the sky's mean radiance where the Gaussian bounces nothing. The coupling is rank one: exact for a Gaussian surrounded by one surface lit by one sky, and it puts a lit ground's colour, not its shape, under a sky lit from one side. Read along a direction at a lobe's roughness, the field's bands are narrowed as a reflected GGX taken as a von Mises-Fisher of concentration $1/2\alpha^2$:

$$F(\boldsymbol{d};\alpha) = \max\Big(\sum_{j<16} F_j\, e^{-l_j(l_j+1)\alpha^2}\, Y_j(\boldsymbol{d}),\ 0\Big).$$

The two roughnesses a Gaussian needs (polish and coat) are taken from one read of the 48 values (`splatFieldAlongPair`). The alternatives weighed were a 9 x 9 transfer matrix per Gaussian (243 numbers, 5 GB for the car, still degree 2) and a per-direction index into the cloud (needs a radiance cache and sees nothing of a mesh ground).

4.6.3 Polish and coat

The base's reflection ("polish") and the coat are

$$\boldsymbol{R}_{\text{polish}} = U\,g\,P(\boldsymbol{n}\cdot\boldsymbol{\omega}_o)\left[O\,\tilde{L}(\boldsymbol{m},r) + (1-O)\,\chi\,\kappa\,F(\boldsymbol{m};\alpha)\right] + U\,g\,P\,E_\odot\,O(\boldsymbol{s})\,p_\odot(\boldsymbol{m},\boldsymbol{s}),$$

$$\boldsymbol{R}_{\text{coat}} = w_c E_c\left[O_c\,\tilde{L}(\boldsymbol{m},r_c) + (1-O_c)\,\kappa\,F(\boldsymbol{m};\alpha_c)\right] + w_c E_c\,E_\odot\,O(\boldsymbol{s})\,p_\odot(\boldsymbol{m},\boldsymbol{s}),$$

where P is the polish's environment albedo, g the thin-wall gain (section 4.7), E_c the coat's albedo and χ the polish-field gate:

- **The polish's openness and gate.** $\chi = 1$ and $O = O(\boldsymbol{m},\alpha)$ only where the material's own roughness is at least `kPolishFieldRoughness` $= 0.3$; elsewhere $\chi = 0$ and $O = O_{\text{L}}$. On the Corvette's chrome (0) and rough metal (0.22) the cells' openness read 1.64 and 0.221 relMSE against the path trace where the harmonics' read 1.05 and 0.128 for the same mean, and the field read the metals a third bright (chrome 0.261 where the path trace reads 0.197). A lobe at 0.3 or wider is wider than the field's degree resolves, so what closes it (a ball's ground) is what the field says: green and white balls at 0.3421 on a grey ground reflected the sky in their lower half without it. The gate reads the *material* roughness, not the stored one, because the bake's widening pushes a thin chrome tube's stored roughness past 0.3. The coat always uses the cells and the field.
- **The analytic sun.** The map holds no sun, so the sun is added through each lobe's distribution of reflected directions with the eye along the normal, $D(\boldsymbol{h})/4$ for the half vector between mirror and sun, the disc's half angle $\theta_\odot$ folded into the width:

  $$p_\odot(\boldsymbol{m},\boldsymbol{s}) = \frac{\tilde{\alpha}^2}{4\pi\,d^2}, \qquad d = \cos^2\!\frac{\theta}{2}(\tilde{\alpha}^2-1)+1, \qquad \tilde{\alpha}^2 = \alpha^2+\theta_\odot^2, \qquad \cos\theta = \boldsymbol{m}\cdot\boldsymbol{s}.$$

  It integrates to one, so weight times $E_\odot$ times $p_\odot$ is the sun's whole reflection. The split sum read a compact sun a quarter short at every map level (0.57-0.74 of the path trace on a floating metal ball; part of that gap came from the path tracer's own dome sampling density, since corrected).

4.6.4 Layers, Fresnel, metals, sheen

Environment albedo. P is MaterialX's `ggxDirectionalAlbedo` with multiple-scattering compensation, not Lazarov's fit, which differed by up to 76 % at grazing; a dielectric uses the exact Fresnel against F_0, a conductor its colour at both ends, mixed by m. Where the material's metal is a generalized Schlick (OpenPBR, glTF), it runs from its colour head on to the specular colour at grazing (`ggxEnvSchlickMetal`); for a dark metal a conductor of artistic index and a Schlick differ by twice at sixty degrees, and the car's paint read matte and bright as a conductor.

The coat's Fresnel. Schlick is low through the middle of a dielectric's curve (0.070 at sixty degrees where a lacquer at 1.45 reflects 0.081); a coat over black read 0.084 against the path trace's 0.099. The coat's albedo is the fit scaled by exact over Schlick at the eye, fading with roughness:

$$E_c = E_{\text{fit}}(r_c, \boldsymbol{n}\cdot\boldsymbol{\omega}_o, \eta_c)\cdot\text{lerp}\!\left(\frac{F(\boldsymbol{n}\cdot\boldsymbol{\omega}_o,\eta_c)}{F_{\text{Schlick}}},\ 1,\ \text{sat}(\alpha_c)\right),$$

and exactly F below $\alpha_c = 10^{-3}$. Against a quadrature of the lobe it is within 3 % at α 0.3 and 0.6 where the fit was 8-16 % under.

Layering. As MaterialX's `layer` node: the coat over the sheen over the base, each layer taking from what is below it the share it reflects at the eye. With E_{fit} the coat's fitted albedo, E_s the sheen's directional albedo and w_s the sheen colour's largest channel,

$$U = \big(1 - w_c E_{\text{fit}}\big)\big(1 - w_s E_s\big) \cdot \operatorname{lerp}\Big(1,\ \frac{1-K}{1-E_b K},\ w_c\Big), \qquad K = 1 - \frac{1 - F_0(\eta_c)}{\eta_c^2},$$

the last factor OpenPBR's coat darkening (light returning from the base is reflected back down by the coat's underside), with E_b the base albedo; the paint's ball read 1.5 times the mesh at its centre without it. The sheen (Imageworks, Conty and Kulla) is a lobe as broad as a body, so it is answered as one: its directional albedo times $\boldsymbol{B}$, under the coat.

4.7 Glass, transmission and thin walls

A Gaussian cannot refract. What it sends on is the light from behind it: $(1 - t)$ of the body faces the eye's side, t faces away, and what stands behind is seen through the Gaussian's opacity.

- **What passes is what is not reflected**: the transmitted half is weighted by $1 - w_{\text{spec}} E_{\text{fit}}(r, \eta)$, with w_{spec} the specular weight; one Fresnel per face, as the path tracer's transmission lobe pays it; a ball came back half again too bright with a rim 2.5 times too hot before.
- **Rasterised TX glass** sees through along $-\boldsymbol{\omega}_o$: the sky where the cells behind are open (`splatConeOpen` about $-\boldsymbol{n}$) and κF where they are closed, so a windscreen shows the cabin, blurred to degree 3. **A solid glass is a lens**: its own far face closes every cell behind it, so the field's white-sky value along $-\boldsymbol{\omega}_o$ decides openness (smoothstep$(0.5, 0.85)$ of its luminance) and the open part reads the sky sharp along the exit of a sphere guessed from the stored mean curvature (`lensExit`: bent in, across the chord, bent out), with the far face's Fresnel and second tint (`lensExitThrough`). Filling the lens with the field drew the pawn's clear head milky and flat (0.58 against the path-traced mesh).
- **Roughness through glass**: a microfacet tilt θ turns a mirror ray by 2θ but a transmitted one by far less, so the sky through both faces is read at $r_t = r\sqrt{\operatorname{sat}((\eta - 1)/\sqrt{2}\,)}$; the pawn's head went from 0.081 to 0.048 relMSE.
- **The far face's reflection** sends back $(1 - R)\,R\,(1 - R)$ times the tint of the light along the mirror (the Gaussian's colour carries the first tint); a clear ball under a uniform sky read 0.952 without it, 0.9998 with it (0.92 through both faces + 0.04 near polish + 0.037 this).
- **Thin walls.** A slab classified thin (open edges, or a thickness $2V/A$ under four cells or a fiftieth of its own size) is converted with opacity $2R_0/(1 + R_0)$, $R_0 = ((\eta - 1)/(\eta + 1))^2$, and its reflection scaled by the gain $g = 1/R_0$ at its own index (R_0 held at 10^{-3} at least; 1.5 only where no index was kept), so the blend leaves the reflection whole. The Corvette's headlight cover at 1.15 got 0.0097 of opacity and, at the old 1.5 floor, a gain of 25 where 205 makes it whole. A sheet's Gaussians are drawn at an opacity of at least 0.1 with their colour scaled down by as much (and its ramp with it): windshield relMSE 2.26 to 0.89 under the shop, 0.31 to 0.067 under goegap.

4.8 Reflection across the footprint

The paint's error is one of sharpness, not energy: against the path trace blurred by one pixel its relMSE falls from 0.80 to 0.28. A Gaussian showed its centre's reflection flat over its footprint.

4.8.1 The per-pixel normal turn

The conversion writes the shape operator of the centre triangle in the Gaussian's axes $(\boldsymbol{u}, \boldsymbol{v})$, $S = \mathrm{sym}\big([\Delta\boldsymbol{n}_1\ \Delta\boldsymbol{n}_2][\Delta\boldsymbol{p}_1\ \Delta\boldsymbol{p}_2]^{-1}\big)$ projected on $(\boldsymbol{u}, \boldsymbol{v})$ (`m2sShape`). The projection inverts its own linearisation on the splat's plane: with J the 2×2 map from plane coordinates to pixels, one pixel right and down are $\boldsymbol{e}_x = J^{-1}(1,0)$, $\boldsymbol{e}_y = J^{-1}(0,1)$, and the normal a pixel away is

$$\boldsymbol{n}_x = \mathrm{normalize}\big(\boldsymbol{n} + \boldsymbol{h}_x\big), \quad \boldsymbol{h}_x = \mathrm{clamp}_{0.5}\big((S\,\boldsymbol{e}_x)_u\,\boldsymbol{u} + (S\,\boldsymbol{e}_x)_v\,\boldsymbol{v}\big),$$

the turn held to half a unit so a splat seen edge on does not swing across the sky.

4.8.2 The slope, and the sharp lobe

For a rough reflection the projection reads the sky again along $\mathrm{reflect}(-\boldsymbol{\omega}_o, \boldsymbol{n}_x)$ and $\boldsymbol{n}_y$ and stores the colour's change per pixel, $\boldsymbol{g}_x, \boldsymbol{g}_y$ (`kSlopeMark`). The blend draws $\max(\boldsymbol{c} + \boldsymbol{g}_x d_x + \boldsymbol{g}_y d_y, 0)$ at offset $\boldsymbol{d}$. The ramp's largest fall over the three-sigma ellipse is held to the colour per channel, $3\sqrt{\boldsymbol{g}^\top \Sigma\, \boldsymbol{g}} \le \boldsymbol{c}$, so the shade never reaches zero inside the footprint (a dashed black line along a pawn's gold ring before). The one-sided slope beat the central difference and minmod on the Corvette (paint 0.393 against 0.584 and 0.564 relMSE).

A linear ramp cannot draw a point. A coat smoother than `kSharpRoughness` (0.2) and a base polish smoother than `kSharpPolishRoughness` (0.5) are instead handed to the blend (`kSharpMark`): the projection subtracts the centre's reading (sky and analytic sun) from the colour, unclamped, and writes the mirror's octahedral coordinate $\boldsymbol{a}_0$ and its steps $\Delta\boldsymbol{a}_x, \Delta\boldsymbol{a}_y$ (zeroed across the octahedron's fold), the lobes' weights (coat weight times O_c; polish rgb weight times O), their roughness and their sun weights. Per pixel the blend decodes $\boldsymbol{m}(\boldsymbol{d}) = \mathrm{octDecode}(\boldsymbol{a}_0 + \Delta\boldsymbol{a}_x d_x + \Delta\boldsymbol{a}_y d_y)$ and adds

$$w_c\,\tilde{L}(\boldsymbol{m}(\boldsymbol{d}), r_c') + \boldsymbol{w}_p\,\tilde{L}(\boldsymbol{m}(\boldsymbol{d}), r_p') + E_\odot\big(\boldsymbol{w}_{p\odot}\ p_\odot(\boldsymbol{m}(\boldsymbol{d}), \boldsymbol{s}; r_p') + w_{c\odot}\ p_\odot(\boldsymbol{m}(\boldsymbol{d}), \boldsymbol{s}; r_c')\big),$$

clamping after. A 1.5-degree sun mirrored in a curved coat is drawn where it falls inside the Gaussian rather than as a blob of the Gaussian's size. Cost: one sky read per pixel per sharp record, and the slope's group memory at 17,412 of WebGPU's 17,416 bytes.

4.8.3 The pixel's own spread

The sharp read is a point sample at the pixel's centre, while the normal turns by $\boldsymbol{h}_x, \boldsymbol{h}_y$ across the pixel: a box of slopes whose variance, doubled for the mirror, widens the lobe:

$$\alpha'^2 = \alpha^2 + 2 \cdot \frac{\|\boldsymbol{h}_x\|^2 + \|\boldsymbol{h}_y\|^2}{12},$$

held at $\alpha' \le 1$, applied after the unwidening and the fold widening of section 4.9. A panel's rolled lip no longer sparkles. In the WebGPU player's sharp lobe (§7), without this term, the door's and the sill's folds show bright specks: with it, 27/25 specks become 2/1 (on the 1280 px car the player reads the door at 35.53 dB and the sill at 32.36 against the path trace, against 33.77 and 31.87 without the term and with the 640 px bake).

4.9 Undoing the bake's widening

4.9.1 Toksvig at conversion

A Gaussian has one normal. Where a thin curved strip of chrome turns the reflection across a Gaussian, the path-traced pixel averages a moving highlight into a thin line, and a Gaussian drew a bead of its own size at full brightness (the Corvette's chrome read 1.6 times the path trace). The conversion widens the roughness, and the coat's, by the spread of normals under the Gaussian (Toksvig):

$$s = \lambda \max_{\text{edges } kj} \frac{\angle(\boldsymbol{n}_k, \boldsymbol{n}_j)}{\left\|\boldsymbol{p}_j - \boldsymbol{p}_k\right\|} \max(\sigma_u, \sigma_v), \qquad \alpha'^2 = \alpha^2 + 2s^2,$$

with $\lambda =$ `--specular-filter` (1 with `--transfer`) and $r' = \mathrm{sat}(\alpha'^{1/2})$.

4.9.2 Unwidened sharp lobes

The sharp lobes already turn the mirror per pixel; read at the widened roughness, the turn is counted twice. A door's lacquer, a mirror ($r_c = 0$), came out matte: stored at $p_{50} = 0.112$ on the door and 0.080 on the hood (640 px bake), saturating to 1 at folds. The projection recovers the spread from the stored curvature,

$$s \approx \kappa_{\max} \max(\sigma_u, \sigma_v), \qquad \kappa_{\max} = \left|\frac{S_{uu} + S_{vv}}{2}\right| + \sqrt{\left(\frac{S_{uu} - S_{vv}}{2}\right)^2 + S_{uv}^2},$$

and reads the sharp coat and polish at

$$r = \left(r'^4 - 2s^2\right)^{1/4} \quad \text{(held at 0)}.$$

4.9.3 The fold's coat

At a fold the widened lacquer reached nearly 1 and a joint drew a dark streak (door joints 0.69 of the path trace). The coat is held:

$$r_c \leftarrow \max\left(\text{unwidened}(r_c, s),\ \min(r_c,\ 0.3)\right) \quad \text{if } r_{\text{mat}} \geq 0.3,$$

`kCoatFoldRoughness` = 0.3. Applied everywhere, it lit chrome, glass and a rim's spokes the path trace shows dark (chrome -1.4 dB, spokes -0.5 dB by region in the web player), hence the paint-only gate. At 1280 px the bake no longer widens the coat (it comes out at exactly 0), so on the re-baked car this term changes nothing.

4.9.4 A tube turns, a fold widens

The per-pixel turn extrapolates the centre triangle's shape operator; past an edge it no longer holds. The stored normal and curvature are exact at the centre (normal p_{90} 0.009 degrees from the interpolated mesh normal), but 64 % of a hood's $\pm 2\sigma$ footprints cross an edge, and the extrapolated normal's RMS error reached p_{90} 46 and p_{99} 198 degrees. Let $t = 2s$ be the turn across $\pm 2\sigma$. In the 1280 px cloud a chrome tube's body sits at $t \in [0.3, 1]$, coherent with its neighbours (2 % of opposite sign); normals smoothed across a fold within triangles narrower than the Gaussian sit past 1 (a sill's 5-9 rad). With

$$w = \mathrm{smoothstep}(0.7,\ 1.5,\ t), \qquad S \leftarrow (1 - w)\, S, \qquad \alpha' = \min\left(\sqrt{\alpha^2 + w\,(t/2)^2},\ 0.5^2\right)$$

(`kFoldTurnLow` 0.7, `kFoldTurnHigh` 1.5, `foldWidened` capped at `kFoldRoughness` 0.5 in r, never narrower than before), a tube turns its pixels in full and draws its thin line of light, while a fold is integrated as the path tracer integrates it within a pixel. The smoothstep keeps a moving camera from seeing a step.

4.10 Measurements against the path trace

All car figures are against the mesh path trace (`/World/Camera`, 1920 x 1080, `autoshop_01_4k`). “Car crop” is PSNR on the 900 x 420 crop of the car, “frame” on the whole frame; class ratios are a region’s mean over the path trace’s. “Leaves” is the whole baked cloud drawn without a cut; “cut” is the error cut of §6.3 at 0.25 px; “cull” is the back-face culling of §6.6. The *widened* configurations read the sharp lobes at the bake’s stored roughness (§4.9.1), the *unwidened* ones apply §4.9.2.

4.10.1 The TX gate, per material (relMSE)

state	chrome	rough metal	paint
first transfer	0.536	0.0896	0.705
TX, cells and field everywhere	1.64	0.221	0.393
polish open by harmonics, cone ≥ cell, field on coat only	0.591	0.0835	0.341
later: + sharp coat and sharp polish per pixel	0.545	0.083	0.281

4.10.2 Playback cost

The whole TX Corvette (14.7 M Gaussians, no ground, 1920 x 1080) drew in 158 ms against about 83 for a relit cloud of its size; of the projection’s 84.5 ms the field and its coupling took 24, the cells read by the lobes 16.5, the sun’s share and bounce 12, the body’s transfer 7.5. Reading the field once for both lobes (§4.6.2) and caching the eye-independent terms (§4.5) address the first and the last.

4.10.3 The whole car, by bake and configuration

bake	configuration	car crop (dB)	frame (dB)
640 px	leaves (5.38 M), widened	16.37	21.92
640 px	leaves, unwidened sharp lobes	16.58	22.15
640 px	cut + cull, widened	16.29	–
640 px	cut, unwidened + coat held at folds (*cut A*)	16.51	22.15
640 px	*cut A* + polish field on rough materials (*cut B*)	16.55	22.18
640 px	leaves (cut at 0.01 px) with all of *cut B*’s terms, no cull / cull	16.73 / 16.75	22.40 / 22.43
640 px	*cut B* + turn capped at 0.2 rad / + coat fold on paint only	16.57 / 16.57	22.20 / 22.20
1280 px	**cut + cull, capped turn, coat fold on paint only (5.0 M)**	**17.88**	**23.90**
1280 px	the same without the cap	17.88	23.91

bake	configuration	car crop (dB)	frame (dB)
1280 px	capped + sharp polish gated on unwidened roughness	17.88	23.90
1280 px	capped + gate + tube/fold split	17.90	23.93
1280 px	capped + tube/fold split (retained)	17.88	23.91

PSNR over a crop moves little; the class ratios below show where the changes act.

4.10.4 The door and the hood, by class

The paint is a dark metal base (roughness 0.3421) under a coat of roughness 0 at 1.45. Each pixel is classed by where its mirror falls: the ground (closed; the field answers) or the HDRI (open). With the 640 px leaves and widened lobes, 86 % of the door's deficit was in pixels reflecting ground. Mean over the path trace (640 px bake):

door class	leaves, widened	cut + cull, widened	*cut A*	*cut B*	leaves, no cull	leaves, cull
mirror to ground < 3 m	0.78	0.89	0.92	**1.02**	0.90	1.02
mirror to ground 3-30 m	0.69	0.77	0.76	0.88	0.79	0.88
mirror to HDRI (open)	0.92	1.04	1.03	1.13	1.01	1.13
joints	0.69	0.83	0.95	**1.04**	0.90	1.02

Unwidening alone leaves the ground classes at 0.78/0.67 and the joints at 0.69; the polish field (*cut B*) lifts the near ground to 1.02 and the coat fold the joints to 1.04, as an analytic estimate of the missing term predicts (under-coat 0.94 x polish albedo 0.078 x reflected ground 0.11 = 0.008 against a measured deficit of 0.0056-0.0085). Back culling lifts the whole door by about 12 %. The open class's remaining 1.13 splits by whether the mirror meets the car itself: 68 % of the excess is in "car" pixels (2.04 of the path trace), where the 16 x 16 cells of about 11 degrees leave 25 % of sky open behind a mirror or pillar.

Local contrast (std of the high-pass over the mean, σ = 1.5 / 4 px), relative to the path trace:

	640 px leaves, widened	640 px leaves, unwidened	640 px *cut B*	1280 px cut + cull
door, open reflections	0.12 / 0.26	0.40 / 0.61	0.35 / 0.53	0.56 / 0.74
hood, open reflections	0.27 / 0.38	0.41 / 0.52	0.41 / 0.51	–

A simulation of the coat's reflection with the path trace's mirror over the prefiltered HDRI shows that the stored coat roughness alone explains the blur: at r_c = 0.11 the door keeps 0.24 / 0.31 of the r_c = 0 contrast, at 0.08 the hood 0.21 / 0.28.

4.10.5 The hood's highlights

Tiles of 24 px with highlights (168), aligned by high-pass correlation within $$6 px:

render	offset p_{50}	σ to match	peak (PT 5.8)
640 px leaves, widened	4.1 px	2.5	1.9
640 px, no per-pixel turn	4.1 px	1.5	2.1
640 px, no slope	4.2 px	2.5	1.8
640 px leaves, unwidened	3.2 px	1.5	3.0
640 px *cut B*	3.2 px	1.5	2.6
1280 px cut + cull	2.0 px	0.75	3.3

The hood's mean goes from 0.89 to 0.98 of the path trace at 1280 px. The normal's extrapolation error scales with the square of the cell (0.146 degrees median at 640 px, 0.060 at 960, 0.032 at 1280).

4.10.6 The extrapolated normal, and the tube/fold split

RMS error of the extrapolated normal on the hood (degrees):

variant	p_{50}	p_{90}	p_{99}
no turn	0.575	22.7	35.8
centre's shape operator	0.146	46	198
+ cap $\|S\| 2\sigma \le 0.2$ rad	0.147	20.2	32.0
least squares on mesh samples	0.088	9.8	14.6

At 640 px the cap moves the car crop from 16.55 to 16.57 dB but worsens the front rim's joints (1.48 to 1.54); at 1280 px removing it changes nothing (§4.10.3). The tube/fold split of §4.9.4 replaces it. On the 1280 px cut:

	capped turn	+ sharp polish gate	+ gate + split	+ split (retained)
bright specks on the sill's fold	10	11	1	1
chrome strip, RMS	0.0358	0.0535	0.0415	0.0359
chrome strip, high-pass / PT	0.41	1.12	0.58	0.41

The retained split also takes the door's specks from 405 to 389 and the rim's from 204 to 190, leaving frame and crop unchanged. At 1280 px the bake no longer widens the coat, and without the split the Gaussians on the sill's fold reflect as mirrors with spinning normals, bright dashes over the path trace's dark groove; the split removes them without touching the strip. Alternatives considered: gating the sharp polish on the unwidened roughness triples the strip's high-pass error; `--specular-filter 0` draws a dashed dark line on the sill's fold and sparkling chrome. The polish needs its widening there.

4.11 Limitations

- **Rank-one field.** The coupling puts a lit ground's colour, not its shape, in a reflection under a sky lit from one side; at degree 3 the field resolves a horizon to about twenty degrees and blurs the car's shadow on the ground into the far ground (in simulation: correlation 0.56 at degree 3, 0.63 at degree 5, 0.85 with a 32 x 32 octahedral map).
- **Cell resolution.** 16 x 16 cells of about 11 degrees let sky through where a sharp coat mirrors the car itself (the door's open class at 1.13); 32 x 32 cells (+96 B per Gaussian) would halve it.

- **One sun per dome**; lamps inside a scene are shadowed by bits that say which ways leave the scene, not which reach the lamp. The bake has no lamp emission: the headlights are dark against the path trace.
- **Remaining defects** on the 1280 px car: the chrome strip on the front bumper 38 % brighter than the path trace with a lower peak; the rim's near-ground reflection at 0.72; the hood highlights 2.0 px off, which a curvature fitted over the footprint from the mesh would roughly halve.
- **Zonal transfers** keep the direct half only, and know only the pose that was traced; occlusion between limbs in other poses is not multiplied in.

5. Animated assets: skeletons, skinning and per-part visibility (the pawn and the sparrow)

This section describes how athenea turns a rigged USD asset (UsdSkel) into a cloud of 3D Gaussians that a skeleton still drives. It covers how the cloud is converted, how it is posed on the GPU every frame, how its light and self-shadowing are kept valid under deformation, how levels of detail treat it, and, as an application, how it is played back in a browser and on a phone. Two assets run through the section. The Eurasian tree sparrow (`SparrowBird.usda`, 609 joints, 71 animation clips) is the animated test case. The chess pawn of The Open Chess Set is the rigid control. Every number below is taken from `docs/decisions.md`, the source files named alongside it, or the measurement files of the web player (`sparkwebGPU`). Numbers that were not measured are marked as not measured.

5.1 Design principle: carry the rig, not the frames

There are two ways to animate a Gaussian cloud. The first stores every gaussian's position, orientation and scale at every frame. The second stores the gaussians once, in the bind pose, together with the joints that carry each one, plus one transform per joint per frame. athenea uses the second, for three reasons, each measured:

1. **Size.** For a plausible character of 150 k gaussians over 120 frames, the per-frame arrays cost 960 MB at half precision (1.2 GB at float). The rig (four joints per gaussian, 60 joints per frame) costs 2.9 MB. A synthetic tube (200 rings × 64 segments, 40 joints, 184 320 gaussians) was measured as written. Its file is 9.27 MB at 25 instants and 9.50 MB at 121, against about 713 MB for the same range as per-frame arrays. Five times the frames cost 230 kB.
2. **Exactness between samples.** A rig is exact at every instant of the range. A sampled cloud is only as good as the reader's interpolation.
3. **Stable identity.** The posed cloud is written into buffers that outlive it, so a renderer sees the same cloud from one frame to the next. That is what lets the ray tracer *refit* its acceleration structure instead of rebuilding it (§5.6).

The cost of this choice is that whatever was baked in world space (radiance, spherical-harmonic transfer, visibility) stops being valid as soon as a limb moves. §5.7 and §5.8 deal with this.

5.2 Reading a rigged stage

The converter (`athenea mesh2splat`) does not go through Hydra. It reads meshes through its own `MeshStage`, which leaves two ways to handle a skinned mesh:

- **Posed.** Without `--skinned`, the stage is posed before it is read. `UsdSkelBakeSkinning` writes the posed points as time samples into an in-memory override layer above the root layer, over `GfInterval(t, t)`. This costs one pose, leaves the file on disk unchanged, and leaves a stage with no `SkelRoot` untouched. Test: a square carried by one joint that slides (1.2, 0.4, 0) between t = 0 and t = 1 converts to bounds, folded on the device, that differ by exactly that slide.
- **Carried.** With `--skinned`, the **bind** pose is read and `UsdSkelBakeSkinning` is skipped. Posing first would skin the geometry twice. For each mesh, `MeshStage` resolves its skel binding with a `UsdSkelCache` over every `SkelRoot` and one `UsdSkelSkinningQuery` per skinnable prim. It returns the joint influences, `geomBindTransform`, the skeleton path and the skeleton's joint order.

Joint order. A mesh may list its own subset of joints in `skel:joints`. UsdSkel's mapper only maps skeleton order to mesh order, so it is run over the identity to learn the skeleton index of each mesh index. Influences are always written in skeleton order, so one cloud has one joint order no matter how many meshes it was converted from. A mesh in a skinned stage that no joint carries (the sparrow stage holds a cylinder and a plane beside the bird) still gets one influence record per gaussian: four joints of zero weight, which the skinner reads as "stay in the bind pose". This keeps rig and cloud aligned one to one.

`--skinned` forces `--no-bake`. A radiance bake holds the scene's environment and bounces (the ground under a foot), and carrying that up with a lifted limb amounts to rotating a lightmap. A skinned cloud carries its material and is relit every frame. Its light transfer, when one is asked for, is kept in a form that rotates with the gaussian (§5.7).

5.3 Influences per gaussian, and their gradients

`mesh_pack.slang` writes each triangle corner's four heaviest influences into a second picture laid out exactly like the mesh's, so the AOFX effect addresses both the same way. The effect knows where inside the triangle each gaussian stands, so it blends the corners there (`blendInfluences`, `mesh2splat.slang`). With barycentric coordinates $\lambda_a, \lambda_b, \lambda_c$ and corner weights w_{cj}, the raw weight of joint j is

$$\tilde{w}_j = \sum_{c \in \{a,b,c\}} w_{cj}\,\lambda_c .$$

Twelve (joint, weight) pairs go in and four come out. A joint already present gains the weight, an empty slot takes a new joint, and otherwise the new joint replaces the lightest. Once the four joints are chosen, each kept weight is recomputed as its full blend and the four are renormalised, $w_k = \tilde{w}_k / W$ with $W = \sum_{i \in K} \tilde{w}_i$. A gaussian is therefore always fully carried.

Gradients. $\tilde{w}_j$ is linear on the triangle, with $\nabla\tilde{w}_j = \sum_c w_{cj}\nabla\lambda_c$, where $\nabla\lambda_b = (e_2 \times n)/|n|^2$, $\nabla\lambda_c = (n \times e_1)/|n|^2$ and $\nabla\lambda_a = -\nabla\lambda_b - \nabla\lambda_c$. The gradient of the renormalised weight follows from the quotient rule:

$$\nabla w_k = \frac{\nabla\tilde{w}_k - w_k \sum_{i \in K} \nabla\tilde{w}_i}{W}.$$

This is exact at the gaussian's centre. Only the two components along the gaussian's rest axes $(\widehat{u}, \widehat{v})$ act on a disc, and the gradients sum to zero, so only the first three joints' gradients are stored. That is three words of two halves, **12 bytes per gaussian**, in `primvars:athenea:splat:jointWeightGradients` (`half[]`, `elementSize` 6). The fourth gradient is minus the sum of the other three. With gradients, the effect's record for a carried gaussian grows to nine entries.

Measured. A square bound entirely to one joint converts to 1056 gaussians, all naming that joint with full weight. Through the effect, 272 gaussians of a quad whose corners all name joint 3 come out on joint 3, with the other slots empty and the weights summing to one. With one joint everywhere, the stored gradient is below 10^{-3}.

Truncation. Four influences is a truncation. On the sparrow, 27 % of wing vertices carry more than four (6686 vertices, `elementSize` 25), and a gaussian takes the union of three corners' joints, so the share of truncated gaussians is at least as large. That share has not been measured.

5.4 Skinning a Gaussian on the device

`scene::SplatSkinner` dispatches `shaders/athenea/scene/splat_skin.slang`, one thread per gaussian. Its inputs are the rest cloud, the (joint, weight) pairs, the joints' skinning transforms at the current instant, and three matrices: the bind transform G, `skelToWorld` and `worldToPrim`. USD matrices are row-major and act on row vectors, so the host transposes them as `geom::Skinner` does for meshes (Figure 11).

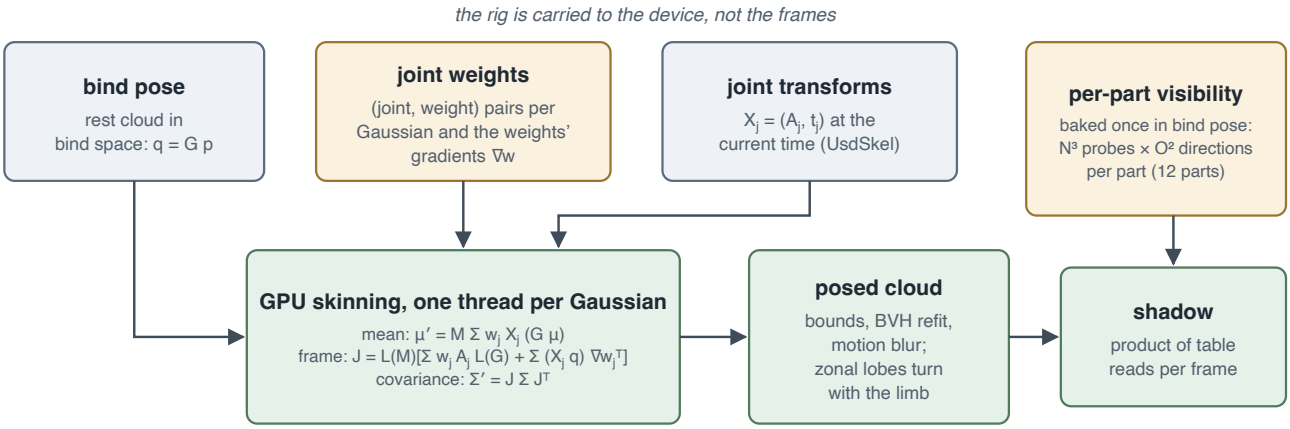


Figure 11. Skinning on the device: the bind-pose cloud, the per-Gaussian joint weights and their gradients, and the joint transforms of the current time give the posed mean and covariance; per-part visibility tables baked once in the bind pose shadow the posed cloud.

5.4.1 The mean

Let $q = G\,p$ be the rest point in bind space, $X_j q = A_j q + t_j$ joint j's skinning transform, and $M = \texttt{worldToPrim} \cdot \texttt{skelToWorld}$. The posed mean is linear blend skinning, with exactly the arithmetic the mesh skinner uses:

$$\mu' = M \sum_j w_j X_j (G\mu).$$

5.4.2 The frame: the whole Jacobian

The posed covariance should be the rest covariance pushed through the derivative of the position map. When the weights vary in space, that derivative is

$$J = L(M)\Big[\sum_j w_j A_j\, L(G) \;+\; \sum_j (X_j q)\,\nabla_p w_j^{\mathsf{T}}\Big],$$

where $L(\cdot)$ is the linear part. The first term is the usual "blend the joints' linear parts". The second, the *elastic* term, is what stretches a gaussian across a bend where two joints disagree about where its point goes. With two joints a relative angle θ apart and a weight transition of length L, its size relative to the first term is about $2\tan(\theta/2)\,d/L$, which is of order one at a right angle.

The kernel builds the elastic term in the same loop that forms $X_k q$. It accumulates $\sum_k (X_k q)\,g_k$ on each rest axis, *relative to the blended point*. The gradients sum to zero, so this is the same sum, but it is computed as a difference of nearby points, which stays exact in float far from the origin. A joint with zero weight but a non-zero gradient still contributes.

The posed covariance in the gaussian's plane is

$$\Sigma' = J E S^2 E^{\mathsf{T}} J^{\mathsf{T}}, \qquad E = [\widehat{u}\,\widehat{v}],\; S = \mathrm{diag}(s_u, s_v).$$

The kernel takes an orthonormal basis (e_1, e_2) of the plane spanned by $a = J\widehat{u}\,s_u$ and $b = J\widehat{v}\,s_v$, writes $[a\ b] = [e_1\ e_2]K$ with K upper triangular, and diagonalises $C = KK^{\mathsf{T}}$ in closed form. The larger eigenvalue is $m + r$, with $m = \frac{1}{2}(c_{00} + c_{11})$ and $r = \sqrt{\frac{1}{4}(c_{00} - c_{11})^2 + c_{01}^2}$. The smaller is computed as $(k_{00}k_{11})^2/(m + r)$ rather than $m - r$, which would cancel to nothing for a thin gaussian in float. The new axes are the eigenvectors, the new sizes the square roots of the eigenvalues. The first axis is the eigenvector closer to the first carried axis, and where c_{01} is pure rounding the basis is kept exactly, so a round gaussian is not spun. The third axis is the plane's normal, and the third size (a disc's thickness) is left as converted. The shading normal turns with the cofactor of the whole J, because $(Ja) \times (Jb) = \mathrm{cof}(J)(a \times b)$.

The kernel writes only position, rotation and the two in-plane scales. Opacity, colour, harmonics, PBR channels and the word holding the third size are left untouched. A frame therefore costs one kernel over the cloud and nothing is decoded again.

5.4.3 Measured

The strip test in `tests/scene/test_loading.cpp` uses 4096 gaussians across a linear ramp of weights, bent a quarter turn. It compares each gaussian's decoded posed covariance with the analytic $J = (1 - w)I + wR + (Rp - p)\nabla w^{\mathsf{T}}$, and counts relative Frobenius errors over 3 %. The tolerance comes from the format: a ten-bit smallest-three quaternion cannot pin an axis closer than about $\sqrt{2}$ /1023.

case	blend only, Gram–Schmidt (before)	whole Jacobian, eigenframe (after)
bent, axes 0.1 rad from the ramp	2048 of 4096	0
bent, axes turned 30°	2048	0
rigid turn and slide (control)	0	0
a joint that shears, constant weights	4096	0
bent, no gradients stored	2048	1984

An earlier test placed 4096 gaussians on a helix, each with its own frame and size. At rest every gaussian stays where it was. Under a rigid turn of 1.2 rad about a slanted axis plus a slide, every gaussian is carried rigidly. Only four counters are read back.

On the sparrow's `Flight` clip (61 frames, 1500 wing triangles, the posed mesh as reference), the blend term alone departs from the mesh's own map by a median of 4–6 %, a 90th percentile of 14–19 % and a 99th of 37–50 %, on the wing hand and the feather fans. With the elastic term the 90th and 99th percentiles fall to 5–6.5 % and 11–13 %. In path-traced images (1 125 022 gaussians, 1024×768, frames 5, 31, 36) the two kernels differ by a few 8-bit levels in about a tenth of the pixels (p99 14–16), and the wings look the same. The whole-image relMSE against the mesh gets 2–3 % *worse*, but an ablation shows this is the metric's doing: negated, swapped, halved and doubled gradients (0.0900, 0.0907, 0.0887, 0.0983 against 0.0904) rank by the size of the change, not its correctness, because conversion error dominates. Correctness rests on the strip test, and on a diagnostic over 174 305 wing gaussians: between neighbours with the same four joints, the stored gradients predict the change of the stored weights with an RMS residual of 12.5 %, against 158 % with the axes swapped and 100 % with no gradient.

5.5 The file: the rig as USD

A skinned cloud is a `ParticleField3DGaussianSplat`. The rig was first written under `AtheneaSplatSkinningAPI`, and is now written under UsdSkel's own names. `mesh2splat --skinned` applies `SkelBindingAPI` and writes `skel:jointIndices` and `skel:jointWeights` (`elementSize` 4),

`skel:geomBindTransform`, `skel:joints` and `skel:skeleton`. It also writes `primvars:athenea:splat:skinningXforms`: one matrix per joint, in skeleton order, **time sampled**. This is the only property of an animated cloud that changes from frame to frame. The transforms are `UsdSkelSkeletonQuery::ComputeSkinningTransforms` at each instant of `--range START:END[:STEP]`.

Binding to the Skeleton. usdSkelImaging builds ext computations only for meshes, curves and points. It does, however, resolve any Skeleton's `skinningTransforms` at any instant. `HdAtheneaSkelSplatSceneIndex` (`modules/usd/src/SkelSplat.cpp`) puts those resolved transforms on the cloud as `athenea:splat:skinningXforms`, remapped by `UsdSkelAnimMapper` when the cloud names a subset of joints, and forwards a Skeleton's dirty notices to its clouds. The primvar that the conversion writes is then a **cache**: a stage that composes the cloud without its Skeleton still moves, and where both are present the cache wins. Dual-quaternion skinning (`skel:skinningMethod`) is not implemented for clouds; such a prim is skinned by linear blend, with a warning printed once.

Measured. 512 gaussians bound to a one-joint Skeleton whose animation puts the joint at t = 1 where the cache has it at t = 2 draw, with the cache blocked, the cache's t = 2 frame at t = 1 pixel for pixel (p99 0, max 0). On the sparrow, the 609 joints the engine receives through the Skeleton agree with the rig layer's; USD's own transforms against the Blender-written cache differ by at most 5×10^{-4}, and the frames differ by a mean of 8×10^{-6}.

Time scale. Without `timeCodesPerSecond` the cloud was read at 24 under a 30 fps stage and lagged the mesh (mean error 0.031 at frame 16, 0.049 at 61); with the rate written, 1.4×10^{-5} and 1.1×10^{-5}.

Clips. The sparrow arrives as five FBX files with 71 clips. Each clip changes only `skinningXforms` (609 matrices a frame), so a clip is a layer over one shared cloud: 377 MB for all 71, against 22 GB for 71 converted clouds. `SparrowClips.usda` offers them as a USD **variant set**. The cloud comes in as a payload, because a sublayer's local opinion would beat the variant, and each clip comes in as a reference inside its variant. The engine stays generic: it lists variant sets, selects one in an in-memory override layer, and reports `animationRange()`, the span of authored samples (for example 1..25 for `land_hop_1`, 1..33 for `air_fly_A0`). Choosing a clip reads about 3 MB, not 213 MB.

asset (`mesh2splat --skinned`)	gaussians	joints	instants	file
rigged tube, resolution 512	184 320	40	25 / 121	9.27 / 9.50 MB
Khronos fox, resolution 512	204 517	24	28	13.35 MB
sparrow, resolution 1400 (first conversion)	4 991 908	609	33	—

5.6 The engine: posing, bounds, refit and motion blur

`Engine::carryCloud` pairs the file's two arrays into the (joint, weight) layout, keeps the uploaded cloud as the bind pose, and runs the skinner into a separate `GpuSplats` that shares the harmonics and PBR buffers, since the skinner writes neither. Measuring this exposed two errors. First, a posed cloud's extent is not the bind pose's, so the extent is now folded again on the device after skinning. Second, `Engine::bounds()` folded the uploaded cloud instead of the drawn one.

A time step is not a new cloud. At first, every time-code change re-decoded every array and rebuilt the CPU-side influence pairs (137 MB on the 4.27 M-gaussian sparrow). Arrays are now compared by identity (`skinningXforms` excluded), pairs are rebuilt only on upload, and arrays with no time samples are held (`_held`), since the crate file's compressed integer arrays arrive in new buffers on every read. Viewer, 400 frames at 1280×720:

	before	**after**
cloud uploads over 400 frames	400	**1**
raster, median / warm	134.96 / 31.56 ms	**56.87 / 13.40 ms**
`rt`, median / warm	268.66 / 55.31 ms	**178.99** / 55.05 ms

Refit, not rebuild. A pose changes neither the number of particles nor their order, so `GaussianRayTracer::sync` refits. On the hardware route this recomputes proxies and updates each chunk's BLAS in place. On the compute BVH route it refits one tree height at a time, bottom up, with no read-back. A full rebuild runs every `refitsPerRebuild` = 240 refits. Against a rebuild every pose, the images agree to p99 0, max 1. A stale tree reads p99 255. Sparrow at resolution 256 (300 862 gaussians, compute BVH): 140.9 / 143.1 / 135.6 ms per frame before, 84.7 / 83.9 / 83.7 ms after (−40 %).

Motion blur from the skeleton. `ParticleField` samples `skinningXforms` over the shutter. The skinner blends the second pose in the same loop (points only, about a third of a kernel) and writes each gaussian's displacement as two words of halves. The rasteriser turns that displacement into a rank-one covariance term. Hydra brackets rather than clips the shutter, so the displacement is scaled by $(t_c - t_o)/(t_1 - t_0)$; without that factor the wings blur twice as far. On the 5.9 M sparrow at 1280×800, a frame takes 90.0 ms with shutter 0 and 100.5 ms with shutter 0.5 (+11 %).

5.7 Light that turns with a limb: zonal transfer lobes

A static cloud can carry a precomputed transfer of nine spherical harmonics in world space. Under a skeleton those harmonics would keep the sky of the conversion pose, and rotating them costs a 9×9 matrix per gaussian. athenea instead stores one or two **zonal lobes** per gaussian, with each lobe's axis expressed in the gaussian's own frame. A zonal function rotates by rotating its axis, a single 3×3, so whatever frame the skinner produces carries the lobes with it, and no gaussian has to be assigned to a part.

A lobe with axis a and coefficients z_l is, in harmonics, $g_{lm} = z_l \, k_l \, Y_{lm}(a)$ with $k_l = \sqrt{4\pi/(2l+1)}$. For a fixed axis the best coefficients are the projection $z_l = k_l \sum_m f_{lm} Y_{lm}(a)$. By the addition theorem the residual is

$$\| f - g \|^2 \;=\; \| f \|^2 \;-\; \sum_{l=0}^{2} z_l^2,$$

so the axis is chosen to maximise $\sum_l z_l^2$. On the device the search tries the band-1 direction, the normal, and 64 Fibonacci directions on a hemisphere, then refines by a walk that halves its step. The second lobe is fitted to the first lobe's residual, and the two are refitted against each other twice (Sloan's ZH fit). The data is ten floats per gaussian, `primvars:athenea:splat:transferZonal` (`elementSize` 10): per lobe, the octahedral (u, v) of the axis in the frame and z_0, z_1, z_2. On the device it is the ordinary `transfer` buffer with `transferCount` = 10. At read time, `splatTransferFrame` turns each axis into world space using the gaussian's current rotation and rebuilds the nine harmonics.

The bake pose. At first the bake posed the cloud at `--time` using four influences per gaussian, while Hydra skinned the mesh with all of its own (ten per vertex on the feathers, 25 on the wings). Where the two disagreed, the bake's ray missed its surface, and 566 603 of 1 504 160 gaussians came out transparent.

The bake now holds the skeleton in its bind pose (`StageRenderer::holdBindPose`) and poses the cloud with identity transforms: 1 504 156 of 1 504 160 find their surface. The transfer therefore holds the bind pose's self-shadowing (wings spread). Occlusion that another limb casts in another pose is the job of §5.8.

Checked. In `[zonal]`, a two-lobe transfer survives the fit and f16 storage within 0.5 %. Turning a gaussian's frame by R and reading along Rw gives what the unturned gaussian gives along w. In a USD test, 512 gaussians carried by a joint that turns 60° match the same cloud under an Xform of that rotation, with rel-MSE below 10^{-4} on both raster and traced routes. The fit's error on the sparrow has not been measured. The indirect half of the transfer is not kept zonal.

5.8 Per-part visibility fields: baked once, a product of table reads per frame

A relit cloud also has to shadow itself, and the measurements ruled out the obvious routes. In the traced route `--splat-shadows` changed nothing, because a traced splat had no shadow pass. In the raster route it cost 36× the frame (49.58 → 1810.24 ms) and crushed the bird to black. Sweeping the self-offset from 1 to 800 own sigmas moved the mean image monotonically from 0 % to 99 % of the unshadowed value, with no plateau. The literature agrees that baked visibility does not survive articulation unless it is factored by body part: Lin et al. use directional MLPs per part, and DNF-Avatar a scalar AO grid per bone. athenea combines the two ideas: a **directional table per part**.

Partition. `partitionJoints` takes the largest subtree not yet a part, with at least `minJoints` (6) joints, and makes it a part, until there are `--parts` (12). On the sparrow this gives root, centre, chest, pelvis, head, and each wing in three segments. Each gaussian belongs to the part of its dominant joint.

Bake (`athenea visibility`, `splat_visibility_bake.slang`). This is done once, in the bind pose. For part p, a grid of N^3 probes covers the cloud's box plus a 5 % margin. Each probe holds an octahedral map of O^2 directions. A texel stores the transmittance of a ray leaving the probe through part p's gaussians alone. The part's gaussians are gathered into a cloud of their own and traced with inline rays, using the same product of $(1-\alpha)$ that the shadow kernels use. Each probe also stores an ambient term, its mean over directions, used for domes.

Read (`splat_visibility_read.slang`). Every frame, `visPartFrames` computes one affine map per part from the posed space back to the bind space where the field lives:

$$x_{\text{rest}} = UA_{k_p}^{-1}\,(x' - t_{k_p}) - Ug, \qquad U = L(G)^{-1},$$

where k_p is the part's head joint, $X_{k_p} = (A_{k_p}, t_{k_p})$ its skinning transform and g the translation of G; call this map T_p^{-1}. Before this map was precomputed it cost two 3×3 inverses per gaussian per part per light, about 260 million a frame on the sparrow; precomputing it took the read from 12.7 to 8.0 ms (5 887 323 gaussians, 1600×1000). The factor for gaussian i and light direction ω_ℓ is then the product over the other parts:

$$V_{i\ell} = \prod_{p \neq \pi(i)} F_p\Big(T_p^{-1}x_i,\; L(T_p)^{-1}\omega_\ell\Big),$$

where $\pi(i)$ is the gaussian's own part, F_p is read trilinearly over the probes and bilinearly over the octahedral map, and is 1 outside the part's box. A dome uses the ambient term instead. The factors go into the same `shadowFactors` that both routes read, so the traced route, which had no splat shadows at all, gets them too. No ray is traced per frame.

Found while building it. An empty part's unwritten texels read as total shadow and turned the bird black; a grid of 0 now marks it and it is skipped. The bake traces inline rays, which the CUDA target lacks, so it is a separate module and CUDA devices read fields baked elsewhere. A part does not shadow itself: its field,

read at its own gaussian, is the crowd it stands in. An 8×8 map (about 22° per texel) cannot resolve a wing's edge; 16 and 32 are exact outside a band of one probe cell.

Measured on the sparrow: 4 269 858 gaussians, 609 joints, 12 parts, grid 24, octave 16.

bake (M5 Pro)	22 s, 3 538 944 texels per part ($24^3 \times 16^2$)
fields in the file	81 MB (two f16 per word)
factors under ½, frame 31, sun + dome	21.3 %
viewer raster, 400 frames, median / warm	71.82 / 9.69 ms (56.87 / 13.40 without)
viewer `rt`, median / warm	193.88 / 51.65 ms (178.99 / 55.05 without)
L4 (CUDA) raster, 8 frames 1280×720	73.04 ms (67.48 without)
L4 (CUDA) `rt`	128.77 ms (121.48 without)

On the L4 the fields cost 5–7 ms per frame for 4.27 M gaussians × 2 lights × 11 parts of lookups. The shadow lands where the mesh's does (the far wing, the flank under the raised wing, the tail root) and follows the wings across frames. With every feather card included (9.5 M gaussians at resolution 1400), the bake ran out of memory at the largest part on both the M5 Pro (Metal) and an L4 through Vulkan (a 1.17 GB BLAS for about 3.3 M gaussians). The bird is therefore baked at resolution 1100.

Limits. A part's own local shadowing (feather on feather) is left out. The ambient term is a mean over the sphere rather than a cosine lobe. Parts are assumed rigid and to occlude independently. The floor under the bird is shadowed by a separate light-space map, which is stabilised against the "breathing" of a posed bounding box (sized from the rest sphere, snapped to the world texel grid). A density-grid alternative is not implemented.

5.9 Levels of detail and decimation for skinned clouds

The `.athc` octree merges Morton-ordered cells. That cannot apply to a skinned cloud, because a cell that took wing and body together would not know which joint to follow. Two mechanisms replace it.

Discrete levels by reconversion. `mesh2splat --lod-levels N` converts at the requested resolution, then at half of it, and so on, each level a cloud with its own rig. `Engine::lodLevelsFor` draws the coarsest level whose cell spans no more than `athenea:lod:threshold` pixels (default 1) at the nearest point of its *posed* bounds. Undrawn levels are not posed (`deferredPose`). Sparrow clips at 1100 / 550 / 275 cells (5.9 M / 1.5 M / 0.4 M gaussians), 1600×1000: 49.6 → 51.5 ms when the bird fills the frame; **71.3 → 24.7 ms** at four times the distance. Every level stays resident with its own visibility fields (81 MB each), and switching between levels is a step, not a blend.

Decimation that keeps the rig. `usd::decimateStage` finds every per-gaussian array and merges each by kind (`lod_attributes.slang`). When `skel:jointIndices` has a matching `skel:jointWeights`, the pair is merged as a rig (mode 3). Over the run of records R that a kept gaussian stands for, each with merge weight ω_r,

$$c_j = \sum_{r \in R} \omega_r \sum_k w_{rk} \, [\, j_{rk} = j \,],$$

collecting up to 16 distinct joints. The `width` heaviest are kept and renormalised, $w'_j = c_j / \sum_{\text{kept}} c$. A record's dominant joint is also folded into the merge key, so a merge never crosses a joint's edge. Checked: a floor whose halves belong to two joints goes from 4096 to 256 gaussians, each kept gaussian wholly on

its side's joint. The sparrow's finest level (5 887 323 gaussians, with ids, material, visibility parts and rig) keeps 87.7 % in 28 s. Posed at time 20, it draws as the undecimated cloud at an image RMS of 0.002.

5.10 The sparrow

The sparrow and a BMW were the first two assets converted that were not authored for this renderer, and most of what it uncovered was outside the skinning maths (Figure 12).

- **Animation export.** Blender's exporter wrote no joint samples for this FBX-imported rig, so `scripts/sparrow-anim.py` samples the evaluated pose. Solving a change of basis against `restTransforms` mangled the head, bill and toes, because rest and bind are different poses on this bird (worst 1.989 at `Toe.L.010`). It now writes $W(t) = P(t)\,L^{-1}B$, so that $W(t)B^{-1}$ is Blender's own skinning matrix; what remains is quantisation, 4.9×10^{-5}–7.1×10^{-5} at frames 1, 31, 61.
- **Corrected stage.** `SparrowBird.usda` carries the asset fixes: no metallic on feathers or eyes, an unpremultiplied feather normal map (`UV2_treesparrow_normal_fixed.png`), and the second UV set (`UVMap_001`) for the feather alpha. Every later measurement uses it.
- **Cut-outs and budget.** Honouring the feathers' connected alpha `opacity` cut their budget from 15 378 695 to 5 898 220 gaussians at resolution 1400. Meshes past the 2 M-per-run ceiling are converted in slices; with every card the bird is 9.5 M gaussians.
- **Mesh reference.** Making the raster and path-traced mesh renders agree took the raster's relMSE against the tracer from 7.28 to 3.67 (960×540).

Measured cloud against mesh (`--no-bake`, wing camera, alpha without dome, relMSE with it), after coverage was made to follow opacity:

	alpha before → after (mesh)	relMSE before → after
sparrow wing, raster	0.2395 → 0.2367 (0.2253)	0.1742 → 0.1635
sparrow wing, traced	0.2385 → 0.2358 (0.2462)	0.1331 → 0.1247



Figure 12. The sparrow: the mesh path traced (left) and the converted Gaussians rasterized (right).

For the web, the bird was converted with `--skinned --transfer --resolution 512`, with zonal lobes baked at 256 paths and 3 bounces in the bind pose and the 12-part visibility added (`sparrow_tx_vis.usdc`: 1 504 160 gaussians, 1 504 096 with content). Against the mesh path traced at 1024 paths (1024^2, `air_fly_A0`, frame 16), the cloud's raster render reads relMSE 0.0025. The cloud's body comes out slightly darker and browner, because the zonal transfer keeps only direct light.

5.11 The pawn: the rigid control

The chess pawn (The Open Chess Set, ASWF, CC BY 4.0) has **no skeleton**. It is two meshes: a black-marble body `M_Pawn_Body_B` and a head `M_Pawn_Top_B` with `transmission = 1`, a glass ball on a gold ring. It is in this section because it is the asset every animation-related change had to leave unchanged:

- When `MeshStageOptions` gained a time, the pawn converted to a file identical byte for byte to the one before.
- When the flat axis became $\min(\text{along}_U, \text{along}_V) \times 0.1$ to fix the fox's ghosting under the ray tracer, the pawn's marble moved from 0.0644/0.0798/0.0778 to 0.0645/0.0799/0.0780.
- Every skinned path, with identity transforms, reduces to the rigid one, and the pawn is the reference for the rigid one.

Its first end-to-end conversion produced 729 073 gaussians at resolution 512. Its material and transfer results belong to Sections 3 and 4. The high-quality white-pawn validation (`pawn_mesh.usda`, 2048^2, 2048 reference paths) gives, per material:

material	gaussians	relMSE	p99	mean (cloud)	mean (path-traced mesh)
`M_Pawn_Body_W`	1 568 499	0.0218	0.273	0.167 / 0.165 / 0.133	0.164 / 0.162 / 0.132
`M_Pawn_Top_W` (glass)	307 296	0.1418	0.354	0.236 / 0.260 / 0.221	0.365 / 0.397 / 0.279

Three later conversions of the same pawn, measured at 1024^2, give glass relMSE 0.149, 0.081 and 0.043, with the body unchanged at 0.025. The body is within a few percent of the path-traced mesh. The error is concentrated in the glass head (Figure 13).



Figure 13. The white pawn at 2048^2: the path-traced mesh with 2048 paths per pixel (left) and the converted Gaussians (right).

5.12 Application: playback on the web and on a phone

The web player (`sparkwebGPU`, a WebGPU fork of Spark) reuses athenea's kernels compiled to WGSL. athenea's `.athc` writer refuses `--skinned`, so the web side reads the USD cloud (`usd-athc`, built on the `openusd` crate) and writes a `.athc` v3. That file adds a `SKIN` section per element, with the influences packed as joint u16 | weight unorm16 and the gradients as halves, and an `ATSK` skeleton blob holding joint names, the bind transform and the first clip. Every other clip is an `ATCL` file fetched when it is played. Merged LoD nodes carry the opacity-weighted blend of their splats' influences (the four heaviest, renormalised) and no gradient.

The skinning kernel (`athenea_adapter/skin.slang`) restates `splatSkin` line for line, because the original binds eleven storage buffers and Chrome on a Mac allows ten per stage. A GPU test runs athenea's own kernel on the same splats and compares the two. Against athenea's raster the result is relMSE 0.0015–0.0053 with a brightness ratio of 1.01. No shutter motion is written.

page	content	measured
`/pawn` (published 2026-10-06)	body 727 061 + head 673 711 splats, transfer, 6 CC0 HDRIs	60 fps in Chrome; v8 data set: download 281.3 → 82.7 MB; desktop GPU memory 1127 → 800 MB
`/sparrow` (published 2026-10-07)	1 504 096 splats, 125.7 MB, 12 clips (6.4 MB gzip), zonal transfer, HDRI	desktop GPU memory 1084 → 684 MB
`/sparrow` on phones	`sparrow-mobile`: 488 656 splats, 31.7 MB, cut from the LoD tree (`--keep-splats 600000`)	60 fps headless in mobile mode, animated
sparrow-v2 (candidate)	1 100 334 splats (−26.8 %), 75.9 MB; mobile 371 138, 26.1 MB	same error as published (4.46e-3)

Pruning a moving asset. Gaussians hidden in the bind pose cannot simply be dropped, because other poses reveal them. Measured over 12 clips × 4 samples = 48 poses plus the bind pose, 61 views each: 61.2 % of the sparrow is hidden in the bind pose, but only **32.5 %** is hidden in every pose. Removing what the bind pose hides costs relMSE up to 4.5×10^{-2} in close views of unsampled poses. Removing what every pose hides costs at most 2.9×10^{-4}. sparrow-v2 is the latter. For comparison, the static pawn has only 2.7 % hidden.

5.13 What is not done

- More than four influences per gaussian. This limits accuracy at the sparrow's wing tips, both in the animation and in the bake's agreement with the mesh.
- Dual-quaternion skinning for clouds. Retargeting and clip blending, which a cloud holding baked transforms cannot do.
- Several skinned meshes at different prim transforms (only the first one's is kept), and skeletons with their own animated transform.
- The traced route poses once per frame, so it does not blur a skinned cloud. LoD instances carry no motion.
- The part's own self-shadowing term. Cross-part occlusion in poses other than the bind pose, inside the zonal transfer. Visibility bakes beyond about 3 M gaussians per part.
- The zonal fit's error on the sparrow, and a geometric measure of coverage at a close bend, which is what would show the gain of the elastic Jacobian term.

6. Levels of detail: merge trees, cuts by error, pages and occlusion

A cloud baked from a mesh at 1280 px holds 8.7 million Gaussians for one car. The close-up level baked at 5120 px holds 74.5 million. Views range from the whole asset far away to a badge close up, and a cloud of that size can be neither drawn nor kept resident whole: what is drawn and what is resident must follow the view, under a node budget and a memory budget, with the data streamed in units. This section first describes the engine's original level of detail (§6.1), which was built for captured scenes, and explains why it does not scale to such clouds. It then covers the structure that replaces it: per-part pair-merge trees (§6.2), one cut by a bounded screen error (§6.3), the opacity a merge is given (§6.4) and a spill metric that was evaluated and not adopted (§6.5), two view-dependent culls (§6.6, §6.7), pages ordered by error (§6.8), the 5120 px close-up level (§6.9) and the measurements (§6.10). Every kernel named here is a Slang compute kernel. The CPU reads files and does bookkeeping only (§2).

Notation. A node n is a leaf (a baked Gaussian) or a merge. μ_n is its centre and Σ_n its covariance, with scales $s_n = (s_x, s_y, s_z)$ and opacity o_n. par(n) is its parent. τ is the threshold in pixels, f is the focal length in pixels at unit distance (`pixelsPerUnit`), and e is the eye.

6.1 The engine's original level of detail

Octree by Morton code. `LodBuilder` (`modules/lod`, `shaders/athenea/lod/lod_common.slang`) sorts the splats by a 30-bit Morton code. The octree cell of a splat at level r is the top $3r$ bits of its code, so the cells of a level are runs of equal prefixes. A boundary pass and a prefix sum find them, and no tree is walked. Each cell carries moments, following Kerbl et al. 2024. With $w = o \cdot A_2(s)$, where A_2 is the area of the two longest axes,

$$W = \sum w, \quad S_1 = \sum w\,\mu, \quad S_2 = \sum w\,(\Sigma + \mu\mu^{\top}), \quad C = \sum w\,c.$$

Moments add, so each coarser level is summed from its children. A cell becomes one Gaussian with $\mu = S_1/W$, $\Sigma = S_2/W - \mu\mu^{\top}$ (Jacobi sweeps in the shader), $c = C/W$ and $o = \min(W/A_2, 0.99)$. The stored levels run from `coarsestLevel` down to the deepest level whose cell count is at most half the splat count.

Cut. `CutSelector` draws a group when its cell's edge, projected at the cell's nearest distance, is at most τ and its parent cell's projection is above τ. A child cell lies inside its parent, so the test is monotone along any root-to-leaf line, and every place is drawn at exactly one level. The only readback is a count per level and a read per instance per frame.

Chunks, `.athc` and streaming. The splats are cut into chunks of 65 536 consecutive Morton-ordered splats. The merged levels stay on the device, and the chunks are streamed. The `.athc` format (`.lrtc` in the lucabRTrender tree) has a 104 B header, level and chunk tables, and blocks aligned to 4 KiB so that a chunk can be mapped and faulted in alone. `StreamingPool` maps the file. Two loader threads copy the wanted chunks, most wanted first. Each slot is taken from a free slot, from the chunk wanted least recently, or from a chunk wanted less than half as much. A finest-level group whose chunks are missing draws its own merged Gaussian, so the image never has a hole. On train_7k (742 k splats, M5 Pro), the build took 47 ms and produced 146 k merged Gaussians. At 8 px the cut drew 83 k splats (11 %), rendering in 3.3 ms against 12.6 ms, with a mean absolute error of $3 \cdot 10^{-3}$. On train_30k, the `.athc` streamed into 4 of 16 slots wrote byte-identical EXRs and settled in 2 cuts.

Why it does not scale to baked assets. Four properties of this structure do not fit clouds of tens of millions of material Gaussians under a shared budget.

1. The threshold is set per instance and is not a budget. Each instance costs two blocking submits and a readback per frame, and no budget is shared across instances.
2. The octree cells ignore the object. A cell merges whatever falls inside it, including a badge with the panel beneath it and glass with paint.
3. A streamed splat costs 132 B, against ~55 B for the compact record (§7.4).
4. Only the splats are chunked. The merged levels, down to one with up to half as many cells as splats, are always resident, and for millions of leaves they alone exceed the budget.

6.2 Per-part pair-merge trees

`athenea decimate --tree` builds a binary merge tree by mutual nearest-neighbour pair merging. It is NanoGS (Xiong et al. 2026) on the device (`lod_pairs.slang`, `lod_pairs_final.slang`, `Lod.cpp`), with guards that suit baked material clouds (Figure 14).

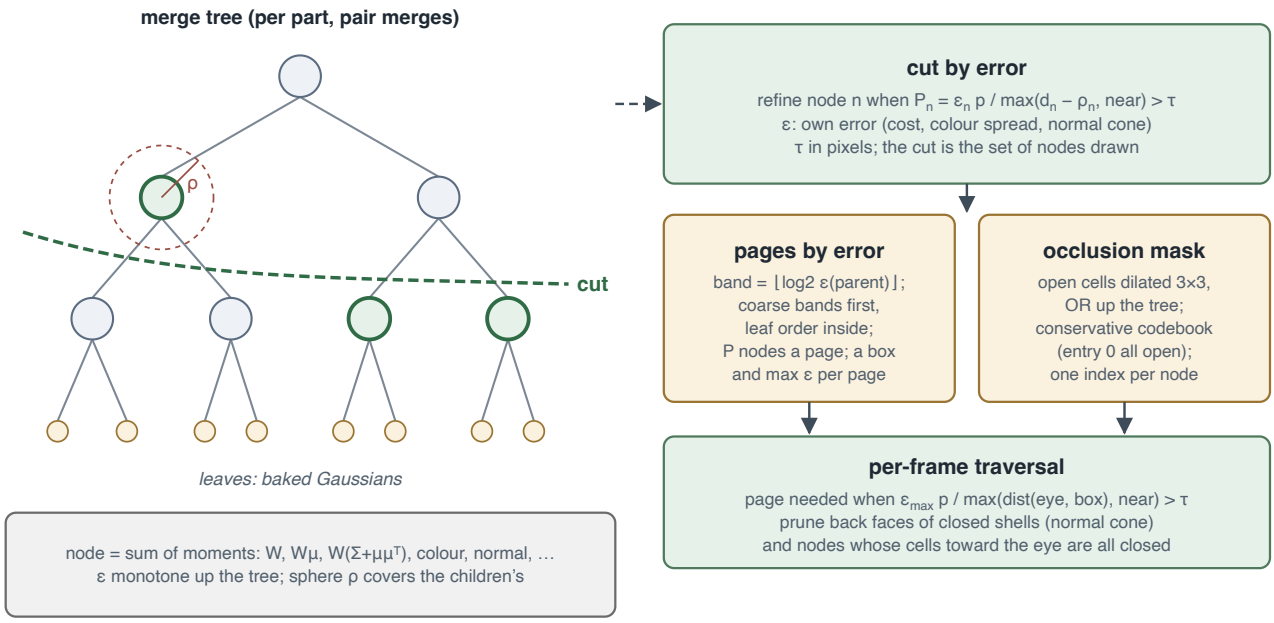


Figure 14. A merge tree and its cut. Nodes are sums of moments; the error ε is monotone up the tree and each node's sphere ρ covers its children's. Nodes are paged by their parent's error and carry an index into a conservative occlusion codebook; the per-frame traversal tests pages, back faces and occlusion.

6.2.1 Nodes as moments

A node is a vector of 24 floats: W, $W\mu$, $W(\Sigma + \mu\mu^\top)$ (6), $W \cdot \mathrm{sRGB}(c)$, Wn, $W\alpha$, $W \cdot$ metallic, $W \cdot$ transmission, $W \cdot$ specular, the key, the record count, the weighted thickness ratio and the node's index in the tree. Here $w = o \cdot \max(s_x s_y, s_y s_z, s_x s_z)$ (`discArea`), and α is the GGX roughness squared. A merge sums the two vectors. Because sums are associative, a node's moments are those of all its leaves whatever the merge order. Every other attribute (transfer, shadow bits, material, joints) is merged once at the end, mass-weighted over the node's run of leaves (`lodMergeAttribute`).

The merged Gaussian (`pairFinalize`) takes the moment-matched μ and Σ, eigen-decomposed by Jacobi with the thinnest axis third. Merging by moments preserves the second moment. A 2×2 block of discs as wide as their spacing therefore merges into one disc 0.58 of the spacing wide, which leaves visible holes when seen up close. Along each wide axis u, the merge is widened as follows. Let v_u be the area-weighted variance of the member centres, c_u their mean width, and $d^2 = 12\sqrt{v_u v_v}\ /(n-1)$ the spacing of a grid of n members over that spread. Then

$$s'_u = \mathrm{clamp}\left(\frac{c_u}{d}\sqrt{12v_u + d^2},\ s_u,\ 2s_u\right).$$

The opacity under the default rule is the members' area-weighted mean, capped at 0.99. The mass coverage $W/A(s')$ is kept as `coverageMass`. A merge of discs whose thinnest axis is under 0.12 of its middle axis is written as a surfel (thinnest scale exactly 0).

6.2.2 The cost

For nodes a, b with $\pi = W_a/(W_a + W_b)$ and merged Σ, the shape term is Runnalls' KL bound per unit mass. The disc's area replaces the volume, which a surfel does not have:

$$c_{\text{shape}} = \max\left(0,\ \tfrac{1}{2}\left[\log I_2(\Sigma) - \pi\log I_2(\Sigma_a) - (1-\pi)\log I_2(\Sigma_b)\right]\right),$$

where I_2 is the second invariant (sum of pairwise eigenvalue products, computed without eigenvalues). The full cost is

$$c = \sigma_{\text{px}}\left(c_{\text{shape}} + w_f\,\phi^2 + w_n(1-\cos\theta_{ab})(1+F_0/\alpha)\right) + w_c\left\|\Delta c_{\text{sRGB}}\right\|^2 + w_m\,\Delta_{\text{mat}},$$

where ϕ is the thickness the merge adds over its thicker member. The term $\sigma_{\text{px}} = \mathrm{smoothstep}(0.5, 2, \text{px})$ removes the shape terms of merges that are under half a pixel on the conversion camera. Defaults: $w_c = 20$, $w_n = 1$, $w_m = 4$, $w_f = 4$.

6.2.3 Guards

A pair whose members differ in any of the ways below costs ∞ (`kNoMerge`):

guard	rule	default
part	key differs (prim, joint, sheet, thin wall, Schlick metal)	always
glass	transmission > 0.5 on one side only	always
glass with glass	only with `--merge-glass`	off
mirror	$\alpha < 0.01$ on one side only	always
fold	$\cos\theta_{ab} < \cos 60°$	60°
highlight	$\alpha < 0.1$ and $1-\cos\theta_{ab} > \frac{1}{2}\kappa^2\alpha$	$\kappa = 1$
colour edge	$\left\|\Delta c_{\text{sRGB}}\right\| > 0.15$	0.15
needle	plane aspect above both `maxAspect` and 1.25× the members'	1000 (off)
stroke	merge wider than `maxPixels` on the camera	off

6.2.4 Passes and the relaxed top

In each pass, every node computes the cost of merging with the 8 neighbours on either side along a Morton order of the centres. It then repeats this along a second order shifted by half the finest cell, and keeps the cheaper partner. Pairs that chose each other are binned into a $\log_{10}$ cost histogram, and the host picks the cost threshold that merges as many pairs as the pass may, which is at most half the nodes. Merging stops at `treeRatio` (4096×) or when nothing more may merge. With `relaxTop`, merging then continues with the colour edge (2.0), the angle, the glass-among-itself guard, the aspect and the pixel limit lifted. The key, glass-with-non-glass and mirror guards stay. Each such merge records its cost **negated**, which marks it as past the guards. This gives a far view coarse levels to cut to: one headlight cover goes from 76 486 roots to 123.

Layout. The tree is written as a ParticleField. The leaves come first, in depth-first order, so that every node is a contiguous run of leaves. The merges follow in the order they were made. Each node carries `lodParent` (a root is its own parent), `lodStart` (its first leaf), `lodSize` (its number of leaves) and `lodCost`.

6.2.5 Many parts: `--passes`, `--merge-glass`, `--object-groups`

On the whole Corvette (5.38 M leaves), the tree can be built in one pass, but writing its attributes exceeds device memory. `--object-groups K --object-group i` keeps the Gaussians whose `cryptoObject` mod K is i. Merges never cross parts, so each group's tree is exactly the subtree the whole asset would hold for those parts. Each pass merges at most half of what is left, and a group of many small parts merges few pairs at once. The default cap of 60 passes leaves a group with 116 549 roots out of 323 613 leaves. At `--passes 1000` it reaches 1 298 roots in 310 passes. `--tree-report` shows two causes of the remaining roots: solid glass never merges (roots equal leaves), and the tyres merge a trickle per pass until the cap.

group (640 px car)	roots before	with `--merge-glass --passes 4000`
group 2	190 490	4 237
group 5	36 508	817
whole car (8 groups)	—	17 133 roots, 5.38 M leaves, 3 582 pages

6.3 The cut by error, with bounds

Alternative considered: the size cut. `--cut-pixels` keeps a node when its on-screen width $2\max(s)f/d$ is at most τ and no ancestor's is. Merges past the guards pass only at 0.25τ. That cut is still available. The error cut below, after Nanite, gives higher PSNR with fewer nodes (table at the end of this section).

Own error. Let $r_n = \left\|s_n\right\|$ (the square root of $\operatorname{tr}\Sigma_n$) and $\alpha_n = \max(\text{roughness}, 0.02)^2$. A merge's own error in metres is

$$\varepsilon_0(n) = r_n\Big(\,|\text{cost}_n| + w_c\, s_c(n) + w_n \frac{\theta(n)^2}{\alpha_n} + w_s \max(E_n - E_0, 0)\Big), \qquad w_c = 10,\ w_n = 1,$$

and $\varepsilon_0 = 0$ for a leaf. Merges past the guards ($\text{cost}_n < 0$) are not measured by their cost, which misses the moiré it causes on glass and the colour it drags across an edge. They are therefore held to

$$\varepsilon_0(n) \leftarrow \max\big(g\,\varepsilon_0(n),\ 2r_n\big), \qquad g = 4\ (\texttt{-cut-guard}).$$

The spill term w_s is 0 by default (§6.5).

Bounds. A merge's cost covers only the last pair it joined. A chain of merges, each within the colour edge, can drift across that edge. So a colour spread s_c and a normal cone θ are climbed up the tree by the triangle inequality (`treeBoundsUp`), starting at 0 for the leaves:

$$s_c(p) = \max_{k \in \mathrm{ch}(p)} \big(s_c(k) + \left\| c_k - c_p \right\| \big), \qquad \theta(p) = \max_k \big(\theta(k) + \angle(n_k, n_p) \big).$$

Both are therefore bounds on every leaf under p.

Monotone error and covering spheres. The error is made monotone, and every node gets a sphere that contains its children's spheres:

$$\varepsilon(n) = \max\big(\varepsilon_0(n),\ \max_k \varepsilon(k)\big), \qquad \rho(n) = \max_k \big(\left\| \mu_n - \mu_k \right\| + \rho(k) \big), \quad \rho(\text{leaf}) = 2\max(s).$$

Each climb is one kernel launched over all nodes, in which every node raises its parent with `InterlockedMax` on the float's bits. The kernel is relaunched until a counter reads zero, which is the tree's depth in rounds. The error is projected at the sphere's nearest point:

$$P(n) = \frac{\varepsilon(n)\, f}{\max\big(\left\| \mu_n - e \right\| - \rho(n),\ 10^{-3} \big)}.$$

Proposition. If $P(n) \le \tau < P(\mathrm{par}(n))$ selects n (a root needs only the first inequality), then every leaf is drawn exactly once. *Proof.* ε does not decrease going up. The parent's sphere contains the child's, so the distance to the sphere decreases going up. P therefore does not decrease along any leaf-to-root path. Leaves have $P = 0 \le \tau$, so the predicate "$P \le \tau$" holds on a prefix of the path starting at the leaf, and exactly one node is the last of that prefix. □

So the cut is a single parallel test per node (`treeErrorKeep`), with no walk up the ancestors. `--cut-budget N` bisects $\log\tau$ over $[10^{-4}, 10^{4}]$ in 40 steps for the smallest threshold whose cut holds at most N nodes.

configuration	scene	size cut	error cut
far, 393 px, 0.25 px	six pieces	83 k, 34.4 dB	34 k, 34.9 dB
near, 426 k nodes	six pieces	42.92 / 31.22 dB (frame / grille)	43.61 / 34.50 dB (reference prototype 43.61 / 34.47)

6.4 The opacity of a merge: least-squares coverage

Mass over the moment ellipse counts the leaves' overlapping tails twice. The overcount measures 27 % at the median and 2.3× at p10. On the grille at 393 px, merges draw darker than their leaves. `--cut-coverage` (at cut time) and `--tree-coverage` (when the tree is written) set every merge's opacity from what its leaves actually cover, as follows.

1. The node's plane is spanned by its two widest axes u, v, with scales s_u, s_v. An 8×8 grid covers the core $[-1.25, 1.25]^2$ in units of (s_u, s_v).
2. Each leaf l under the node (its run, up to 4096 leaves when writing the tree and 2048 in a cut) is projected onto the plane. Its in-plane covariance is $(u^\top \Sigma_l u,\ u^\top \Sigma_l v,\ v^\top \Sigma_l v)$ and its centre offset is m_l. A longer run keeps its mass-based opacity. The leaves are composited as the rasteriser would: $C(x) = 1 - \prod_l \big(1 - \min(o_l, 0.99)\, G_l(x)\big)$.
3. The node's own footprint on the grid is $G(x) = \exp(-\frac{1}{2}\|x\|^2)$. The opacity is the least-squares fit of aG to C:

$$a^\star = \frac{\sum_x G(x)\,C(x)}{\sum_x G(x)^2}, \qquad o_n = \mathrm{clamp}(a^\star, 0, 0.99).$$

Two alternatives were measured. Normalising by $\sum G$ instead of $\sum G^2$ gives 0.5 for a merge as wide as one opaque leaf. A fit over $\pm 2\sigma$ instead of $\pm 1.25\sigma$ gives a panel's merges 0.84, because the fringe is where the neighbours carry the surface, and a fender comes out mottled. Leaves edge-on to the plane, such as a grille's wires seen square on, cover nothing, so the merge goes to 0. A loader drops anything under 1/255, which would break the tree, so a merge is written at least at

$o_n \geq 2/255$ (`treeCoverageFloor`).

The tree is rewritten in its own file order, and only the merges' opacities change (`treeCoverageToFile`), because `lodParent`/`lodStart` are file indices that a rewrite through the Morton-ordered store would scramble. The fit reads a Blob's axes as rows. Read as columns, they give the inverse rotation and turn every Gaussian the wrong way: a fender's merges then come out at 0.59 of their mass where a reference fit gives 1.00.

configuration	what	coverage	mass
fit at cut time (48 k)	grille, 393 px	39.79 dB (from 33.87)	— ; leaves 41.59
refitted tree (49.4 k)	grille, 393 px	41.28 dB	39.79 at 48.1 k (`--cut-coverage`)
1 px, bounds, axes read as columns	six pieces near at 1920	40.96 frame / 32.09 grille	old tree 41.11 / 30.09
axes read as rows	six pieces near at 1920	**43.86**	42.55
axes read as rows	six pieces far at 393	39.95	39.66
axes read as rows	grille near at 393	39.71	**41.16**

The grille is the exception. Neighbouring merges overlap and composite as $1 - \prod$, so a little extra opacity darkens the whole patch. The web player (Spark) draws each node's mass (min(coverageMass, 0.99)), and the engine's rasteriser keeps the coverage.

6.5 The spill metric, and why it failed for the grille

At a coarse cut the grille's holes close: in the 0.25 px cut of the 640 px car at 1920 px with back-face culling, the near-ground region reads 2.24× the path trace, against 1.18 in the leaves. The cause is two-leaf merges of the sub-pixel strips of `Chrome_Part_Grill`, seen face on and 3–6 px wide at $\tau = 0.25$. Their cost is ∼0 because a pair's coverage, measured on its own plane, does not change. A spill is defined on the coverage fit's own grid:

$$E_n = \frac{\sum_x \max\big(aG(x) - C(x),\, 0\big)}{\sum_x aG(x)} \in [0, 1],$$

that is, the share of what the merge paints that its leaves do not. It matches a NumPy reference implementation to a correlation of 0.994, and is written as `lodSpill`. The cut adds $r\,w_s \max(E - E_0, 0)$ to the own error.

640 px car, group 0, 0.25 px	nodes	car crop	frame
$w_s = 0$	307 665	16.55 dB	22.18 dB
$w_s = 2$	561 242 (+82 %)	16.79	22.38
$w_s = 5$	599 983	16.79	22.38

640 px car, group 0, 0.25 px	nodes	car crop	frame
$w_s = 10$	612 963 (+99 %)	16.79	22.38

The spill closes the holes: from 1.90 to 1.09 of the leaves in the render, and from 4.66× to 1.0 in an image-space hole proxy. The price is 82–99 % more nodes. Every merge spills about 0.16, and the merges that fill the holes spill no more than a panel's merges do, so no floor E_0 separates the two groups. The hole is a property of the **projection**: the merge covers nothing in 3D that its neighbours do not cover, and the gap opens only where the strips' screen footprints fall apart. No view-independent per-node measure sees that. The default is $w_s = 0$. The 1280 px bake (§6.10) resolves the problem in the data instead: the grille's holes in the cut go from 2.24 to 1.04.

6.6 Back-face culling of closed shells

At small output sizes, a cut shows back faces as black. On the grille at 393 px, culling back faces raises the uncut leaves from 38.2 to 41.6 dB. `--cut-cull-back` drops a node whose whole normal cone faces away:

$$\theta(n) < \frac{\pi}{2} \;\wedge\; n_n \cdot \frac{e - \mu_n}{\left\| e - \mu_n \right\|} < -\sin\theta(n).$$

The cone bounds every leaf under the node (§6.3), so the node's whole subtree faces away too. Two measurements set where it applies. Applied to the six test pieces with the headlight's glass cover among them, it costs 6 dB at 1920: glass is seen through, so its back faces are visible. On the whole car at 393 px with the 1 px error cut, the doors and panels, which are two skins, show a net of dark cracks: the inner skin's merges, facing away and lit by the dark interior, sort in front of the outer skin in patches. The net is the same with mass or coverage opacity, and `--cut-guard 1000` does not change it. On a closed shell, what lies behind a face that looks away is the part's own inside, so culling it is safe and removes the net. The cull is therefore applied, to leaves and merges alike, only on **closed shells that are not glass** (`primvars:athenea:splat:closed`, transmission ≤ 0.5).

The open-edge rule. `mesh2splat` counts the edges not shared by exactly two triangles, O. Two simpler rules were considered: $O = 0$, and a share of open edges under 2 % (111 of 192 meshes closed). An open sheet's border grows as the square root of its area, so a fine tube passes the share rule: a rim's barrel at 1 % is culled, and light shows between the spokes. The rule used is

$$\text{closed} \Leftrightarrow O < 2\sqrt{E}\ , \qquad E = 1.5\,T,$$

for a mesh of T triangles. Under it, doors (0), logos (1.4) and calipers (1.5) are closed, and rims (4.0), tyres (3.1), the hood (3.1) and the bumper (4.4) are not, which gives 102 of the Corvette's 192 meshes. With all leaves at 1920 px against the path trace, the cull lifts the whole door ∼12 % evenly (door 1.01, near ground 1.02 of the path trace), with 4.35 M → 3.70 M Gaussians and car crop 16.73 → 16.75 dB.

6.7 Occlusion masks

A TX transfer stores 256 shadow bits per Gaussian, one per cell of a 16×16 octahedral grid over the world's sphere, set where a ray from the Gaussian reached the sky. A direction whose cells are all closed is one from which an eye outside the asset cannot see the Gaussian. The shading bits cannot be reused for this: a merge holds its leaves' bits by majority, and the player's codebook (`shadow_vq`, §7.3) is also a majority, so both would close directions from which some leaf is seen. `--occlusion-mask N` (`occlusion_mask.slang`, `Occlusion.cpp`) builds a conservative mask per node.

Masks. A leaf's mask M_l is its open cells dilated by one cell (3×3), mirrored across the octahedral square's edge at the fold. One cell is ~11°, and without the margin an occluder's edge closes directions from which a grille's chrome is still visible. The mask is all open on glass (transmission > 0.5, since what lies behind glass is seen through it) and on a leaf with no bits. A merge's mask is the OR of its children's, climbed until no mask changes:

$$M_p = \bigvee_{k \in \mathrm{ch}(p)} M_k.$$

Conservative codebook. The masks are replaced by an index into N entries of 256 bits each. Entry 0 is all open. A node's **worth** is the number of closed cells among the *relevant* ones, the directions with $d \cdot \widehat{up} > -0.25$ (an eye down to ~15° below the horizon). The nodes in the lowest $\log_2$-worth bins, together under 2 % of the total worth, take entry 0 and stay out of the fit. The remaining entries are seeded from fitted nodes spread through the file. Round 1 assigns each node to the entry nearest in Hamming distance over the relevant cells. Rounds 2–8 assign a node only to an entry that **contains** its open set, choosing the one with the fewest relevant open cells. After each round an entry becomes the OR of its members. Every entry is therefore a superset of each member's open set, and the codebook can never close a direction a node has open. The codebook's cost is the cells it reopens ($M_{\text{entry}} \wedge \neg M_n$) as a share of all closed cells.

Shared by need. One asset's trees share one codebook (`--occlusion-group`). Each unit is fitted alone at the full size and the entries it uses are counted. The codebook is then shared out as $\text{share}_u = 16 + \lfloor \text{spare} \cdot \text{used}_u / \sum \text{used} \rfloor$, and each unit is refitted within its share. Sharing by fit quality and by part (`--occlusion-by-part`) were both measured and are worse:

configuration	trees	codebook	reopened	culling vs ideal
one codebook per tree	6 pieces	256	3.2–10 %	—
one fit over all trees	6 pieces	4096 / 1024	3.2 % / 6.5 %	—
+ worth, relevant cells only	6 pieces	4096 / 1024	21.9 % / 24.7 %	—
worth = closed cells, not leaves	6 pieces	4096	14.7 %	—
shared by need	6 pieces	4096	7.2 % (grille 2.8 %, 1 249 entries)	86 % near / 82 % far
8 groups by need	car 640	4096	16.2 %	—
8 groups by need	car 640	8192	11.9 %	84 %
by part (192)	car 640	4096 / 8192	—	66 % / 74 %
8 groups by need	car 1280	8187	12.3 %	—
24 trees by need	car 5120	8182	0–23.6 % per tree	—

(From the third row onward, the fit counts only relevant cells, so the total reopened over *all* cells rises. Those percentages are not comparable with the first two rows.) On the close-up level, the sum of closed cells over ~150 M nodes exceeds a 32-bit counter, so the totals are summed in 64 bits. The per-tree results are: Windshield 0.0 % (glass, 1 entry used), Part_2 5.3 %, Hood 7.0 %, the pilot door 7.7 %, the fine groups 8.9–15.9 % and the small-panel group pan2 23.6 %.

The player's rule (and `occludedFrom` in the cut). This rule applies only when the eye is outside the asset's bounding box. Let $d = \text{normalize}(e - \mu_n)$ and $q = 16\,\text{oct}(d) - 0.5$. The four cells $\lfloor q \rfloor + \{0,1\}^2$, clamped to $[0,15]^2$, are the bilinear 2×2 footprint toward the eye. The node is culled when all four are closed in its en-

try. With the car as occluder (never in a partial scene), the rule removes 12 % of the near cut and 8 % of the far cut for 0.06 dB. On the whole car at 393 px, with back culling, occlusion takes the near cut from 522 931 to 360 333 nodes.

6.8 Pages by error

A player cannot hold a whole tree. It needs units that it fetches whole and tests without reading their nodes.

Alternative considered: pages by level. Pages taken as breadth-first tops of sibling subtrees still need 114 of the hood's 263 pages for a far cut. The bounds keep small nodes wherever a highlight or a colour turns, so the cut spans nodes from single leaves to 2^{11} leaves.

Pages by error. A node is needed exactly when it is a root or its parent's projected error exceeds τ. So the layout follows the parent's error (`lod_pages.slang`). The **band** of a node is

$$b(n) = \mathrm{clamp}\big(\lfloor \log_2 \varepsilon(\mathrm{par}(n)) \rfloor + 40,\ 0,\ 63\big), \qquad b(\mathrm{root}) = 63.$$

The sort key, in the pages' own order, is (coarsest band first, then first leaf, then larger node first):

$$\mathrm{hi} = (63 - b) \ll 22 \mid \mathrm{bucket} \ll 19 \mid \mathrm{start} \gg 6, \qquad \mathrm{lo} = (\mathrm{start} \,\&\, 63) \ll 26 \mid (2^{26} - 1 - \mathrm{leaves}).$$

A GPU radix sort orders the nodes, and page p holds sorted positions $[pP, (p+1)P)$. The pages cross band boundaries, so a coarse band of a few nodes does not become a page of its own. A parent's band is never finer than its child's, its first leaf is never later, and when both tie the parent is the larger node. A parent is therefore never placed after its child. (A layout that runs each band's leaves backwards puts 92 parents after their children on the hood.) `--tree-pages P` writes, for each node, `lodPage`, and for each page:

- `lodPageInfo` = (coarsest band, nodes, first node in page order), and
- `lodPageBounds` = the box holding the **parents'** spheres (a node's need is measured from its parent) and the largest and smallest parent error.

The player projects the largest error from the box and loads the page when that projection exceeds τ. The bucket bits (`--pages-by-facing`: dominant normal axis and sign) make residency worse, 4.45× against 3.98× at $\tau = 2.5$, and are off.

Residency. A walk needs the cut and all its ancestors, about 2× the cut. Pages hold about 4× the cut whatever their size or order, because needed and culled nodes share every page while the culling depends on the view.

configuration	scene	τ	cut	nodes needed	pages holding (× cut)
P 3000	six pieces, near 1920	1	244 114	—	2.0–2.6×
P 3000	six pieces, far 393	—	—	—	2.2–4.9× (hood: 2 pages)
P 3000	car 640, near 393	2	293 629	647 707 (2.20×)	1 170 000 (3.98×)
P 3000	car 640, near 393	2.5	263 918	580 997 (2.20×)	1 050 000 (3.97×)
P 3000	car 640, near 393	3	198 702	445 195 (2.24×)	957 000 (4.81×)
P 3000	car 640, near 393	6.75	92 987	214 827 (2.31×)	423 000 (4.54×)

configuration	scene	τ	cut	nodes needed	pages holding (× cut)
P 1000 / 500	car 640, near 393	2	293 629	—	3.91× / 3.87×
P 1000 / 500	car 640, near 393	2.5	263 918	—	3.77× / 3.60×

`treeCulledMark` marks the nodes the culls leave out. A culled node's subtree is culled too, so a walk stops there. That brings the nodes needed to 2.2× (a walk into culled subtrees needs ∼4×), but the pages still hold 3.6–4.5×. A streaming player can therefore keep the traversal at 12 B per node for what is needed and the payload in deflate runs of 64 for what is drawn; in the WebGPU player of §7, near at $\tau = 3$, that comes to ∼60 MB and 226 k drawn at 60 fps (emulated).

6.9 The close-up level

Close up, the logo and grille stay soft at the 1280 px bake. A close camera comes to about 0.5 mm per output pixel, while the 1280 px leaves are 3–4 mm. A logo-and-grille bake from a close camera at 5120 px shows the checkered flag, the bowtie and the grille's mesh crisp (108 527 → 404 594 Gaussians). The close-up level (car5120) re-bakes the car in groups (Figure 9):

- **Detailed parts.** The 66 parts that the 1280 px bake gives at most 1.5 Gaussians per triangle (badges, grilles, lamps, calipers, mirrors, wheels, wipers, trims), at 5120 px, in 7 groups (`fine0–6`).
- **Panels.** Each part at 5120 px, or as fine as keeps it under a leaf cap of 4.5 M (a 5.01 M-leaf tree is at the limit of device memory when written), with the small ones grouped up to 4 M. `--max-cells 1048576` is required: at 4736 px, Infinite_Black loses 359 628 cells past the default 262 144 per triangle, which leaves holes.

Each group is converted with every other part hidden, but the bake traces the whole stage, so a part keeps the others' shadows. Trees are per part, so the close-up level has the same format as car1280: the coarse roots stay, and the deeper levels are paged in only close up.

groups	leaves	resolution
fine0–6 (66 detailed parts)	18 610 627	5120
door (pilot)	4 084 170	5120
Bottom	4 897 428	3392
Part_3	4 370 459	4480
Infinite_Black	4 481 738	4736
Panel_under_Door_001, Windshield, Front_Fender, Hood, Front_Bumper, Part_2, Part_23	15 994 566	5120
pan0–4 (small panels)	17 088 152	5120
Front_Bumper_001	5 008 986	4480
24 trees	**74 536 126 leaves, 74 328 031 merges, 208 095 roots**	

The 24 trees hold 148 864 157 nodes in 49 634 pages of 3000, and 87.4 GB on disk. Some trees keep many lone roots (fine5: 29 213; Infinite_Black: 92 773), because merges never cross parts or guards, by design. The player's packer (§7.5.4) adds a multi-level synthetic root above them, since a file holds about 3 000 roots.

Memory for trees of 4–5 M leaves. A front wheel with 4.45 M leaves and 8.88 M nodes (the tyre alone 2.56 M) builds its tree within device memory, but writing it needs two changes.

1. `mergeAttribute` needs a weights buffer only for a rig's joints. Allocated as large as each attribute, it costs 1.7 GB alongside each of a transfer's 48-float arrays over 8.9 M nodes. It is one word for every other attribute.
2. The write must not hold each array three times in unified memory (what was read, a copy, and the merged arrays started as another copy). The merged arrays start from the metadata alone, and the tree's write, the last step, releases each array once it is on the device.

With these changes, a 4.08 M-leaf door peaks at 10.5 GB resident and 4.90 M leaves are written. Front_Bumper_001 (5.01 M) is written with 2.23 GB of free memory at the lowest point; the 4.5 M cap for panels keeps that margin.

Occlusion written in place. At 87.4 GB for the 24 trees, a copy of each tree does not fit beside them on disk. `occlusionMasksShared` writes each tree to `<tree>.occlusion.usdc` beside it and renames it over the original. The shared codebook (8 182 entries) takes 645 s, with a peak resident set of 10.2 GB; per tree it reopens 0–23.6 % of the closed cells (§6.7).

6.10 Measured results

All renders are the engine's rasteriser. "Leaves at 4×" means every leaf drawn at 4× resolution and box-reduced, the reference for far cuts (a 1920 px bake drawn at 393 px is see-through). "PT" is the mesh path trace at 1920×1080 from `/World/Camera`, compared on the frame and on the car's 900×420 crop.

The 640 px car at 393 px, against the leaves at 4×.

configuration	cut	near nodes	near dB	far nodes	far dB
coverage trees	error 1 px, bounds	666 585	33.29	343 541	39.15
mass trees	error 1 px, bounds	666 585	33.27	—	—
coverage trees	$\tau = 0.5$	930 119	33.17	—	—
+ back culling	closed shells by the 2 % rule	522 931	33.63	—	—
+ occlusion	8192 shared by group	360 333	33.58	195 049	39.53
both culls	$O < 2\sqrt{E}$ rule (102 parts)	378 028	33.53	—	—

The far cut (195 k) fits a budget of ∼250 k nodes, the one the mobile player of §7 draws at 30 fps. The near cut is ∼1.4× that.

Against the path trace (1920 px).

bake	what	nodes	car crop	frame
640 px	all leaves	5.38 M	16.37	21.92
640 px	error 1 px, both culls	675 760	16.28 (31.91 vs leaves)	21.80
640 px	error 0.25 px, back culling	2 313 541	16.29 (34.01 vs leaves)	21.81

bake	what	nodes	car crop	frame
640 px	leaves, unwidened sharp lobes	5.38 M	16.58	22.15
640 px	0.25 px, back culling, shading fixes	—	16.55	22.18
640 px	leaves (0.01 px), no cull / cull	4.35 M / 3.70 M	16.73 / 16.75	22.40 / 22.43
640 px	group 0 with spill $w_s = 2$	561 242 (g0)	16.79	22.38
1280 px	0.25 px, back culling	4 998 976	**17.88**	**23.90**
1280 px	0.25 px, back culling, tube/fold split (§4.9.4)	—	17.88	23.91

The level of detail costs about 0.1 dB against the path trace (16.37 → 16.28 at 1/8 of the leaves). The bake accounts for the rest: its resolution, its lamps and the sharpness of its reflections. The 1280 px re-bake (8.69 M leaves, 8.67 M merges, 20 723 roots, 5 789 pages, occlusion reopening 12.3 %) adds 1.3 dB on the car. Against the 640 px 0.25 px cut with the shading fixes, the 1280 px cut takes the grille's holes from 2.24 to 1.04 of the path trace, the hood highlights from 3.2 to 2.0 px off, and the joints from 0.90 to 1.02 of the path trace.

6.11 Limits and open items

- **Projection-only holes.** A coarse cut of sub-pixel strips (the grille) fills holes that no per-node, view-free measure can see (§6.5). This is resolved in practice by finer leaves, not by the cut.
- **Occlusion weighting.** The codebook is shared by entries used, not by how often a view culls a node. Weighting by an orbit of views in the bake is expected to lift culling past 84 % of the ideal.
- **Residency at ~4×.** It is bound by pages that mix needed and view-culled nodes. A streaming player splits traversal from payload rather than shrinking pages.
- **Glass merges.** Every thin-sheet Gaussian is lifted to the 0.1 sheet floor, so merging fewer, larger sheets varies the transmission cell to cell. Spark declares sheet merges at 4× their size in the interim, and a floor per pixel rather than per Gaussian is pending.
- **Far, thin trims** still become fat dashes in every cut. The bounds miss this error.
- **Not ported.** The closed-shell back rule and the error cut live in `athenea-pieces`. The main engine's `CutSelector` is unchanged.

7. Application: compression and a WebGPU player on mobile

This section describes one application of the relightable clouds of §3–§6, developed as a parallel track: playback of a whole asset on a phone. Its design requirements are its own: the whole Corvette, with its full light transfer, drawn on an iPhone in a web page and judged against the mesh path trace; a budget of about 50–60 MB of car data while moving and about 100 MB at rest; small parts that look right. The player is a fork of Spark with a WebGPU backend (sparkwebGPU, branch `webgpu`); athenea writes the clouds, the merge trees, the pages and the occlusion. This section describes the formats, the streaming, the draw and the relight on the phone, and the measured results. The `.athc` v3 format is specified in the player repository (`spark/docs/docs/athc-v3.md`).

Three kinds of measurement appear, and they are not interchangeable:

- **Dawn / M5 Pro**: GPU timestamps from headless Dawn (Metal) on an Apple M5 Pro, at the phone's CSS resolution (393 × 852) or at 1920 × 1080. They are a proxy for the phone's GPU.
- **Emulated iPhone**: the live site in desktop Chrome on the same M5 Pro, with the page in its mobile mode at 393 × 852. The pool sizes, the cut and the memory are the phone's; the speed is not. The emulated frame rates are about twice the real iPhone's, so draw counts are judged on the device.
- **Real iPhone**: iOS 18.7, Safari 26.6.1, a 393 × 852 @3 screen (the iPhone 14 Pro–16 class), canvas 393 × 695 at pixel ratio 1. These are stress-page ramps and two whole-car runs on the device.

Image quality is given as PSNR in dB, or as relMSE ($\sum (x-y)^2 / \sum y^2$; where PSNR is written $-10\log_{10}$relMSE it is stated). The reference is either the path trace at athenea's `/World/Camera` (1920 × 1080) or the player's own uncompressed path. The reference used is named in each case.

7.1 The container: `.athc` version 3

athenea's `.athc` v2 is a flat array of words per splat. The web player rewrites it as **version 3**, which is laid out by sections and tiers. Inside each block, the data a tier needs is a prefix, so each (block, tier) pair is one HTTP Range request:

Tier	Sections	Contents
1	`CORE`, `SHRS`, and optionally `LODS`, `LODE`, `OCCL`, `LODM`, `TREE`, `SKIN`	splat geometry and colour; LoD metadata
2	`MATL`	normals, emission, pbr, lobes
3	`SHAD`, `CURV`, `TXDI`, `TXIN`, `TXFD`	open-direction bits, curvature, transfer (direct, indirect, reflected field)
4	`CMPC`	the compact record for the phone (§7.4)

A browser offers gzip and nothing better (`DecompressionStream`), so the lossless lever is what gzip is given. **Encoding 1** transposes each array into byte planes: byte 0 of every element, then byte 1, and so on. The high byte of an f16 or f32 changes little from one splat to the next in Morton order, so it forms runs that gzip finds. **Encoding 2** does the same after replacing each 16-bit lane with its wrapped difference from the previous element. `athc-convert --gzip --planes` picks the smaller encoding per section. Measured on the published clouds: Corvette light 68.1 → 60.9 MB (−10.6 %), Corvette HD 288.7 → 248.6 MB (−13.9 %), its paint −23 %, pawn parts −17 to −33 %. The shadow bits are random, and they never gain.

7.2 Transfer compression by clustered PCA

The transfer is 30–70 % of a relit cloud's bytes, and neighbouring transfers are close. Two clustered-PCA codings are used: a lossy encoding of the file (encoding 3), and a joint basis that the phone keeps resident.

7.2.1 Encoding 3 (file level)

Following Sloan et al. (2003), a transfer section of one block is stored as up to K clusters (k-means on a sample; $K \le n/128$). Each cluster c has a mean μ_c and $m_c \le M$ principal directions $\boldsymbol{d}_{c,j}$ in f16. Each element i has a cluster id and m_c quantised coefficients:

$$\hat{\boldsymbol{t}}_i = \mu_{c(i)} + \sum_{j < m_{c(i)}} \left(q_{ij}\, \Delta_{c(i)} \right) \boldsymbol{d}_{c(i),j}, \qquad q_{ij} \in \mathbb{Z},$$

where every product and sum is rounded to f32 in that order and the result is rounded to the nearest half. This makes the Rust/WASM and TypeScript decoders bit-identical. The step Δ_c is proportional to the cluster's RMS, which gives the same relative error in dim clusters as in bright ones. The coefficients are zig-zag integers of 1–3 bytes, stored as byte planes. m_c is the smallest count whose truncation takes at most 40 % of the error budget. The budget is relative per block:

$$E_{\text{block}} = \frac{\sum_i \left\| \hat{\boldsymbol{t}}_i - \boldsymbol{t}_i \right\|^2}{\sum_i \left\| \boldsymbol{t}_i \right\|^2} \le E, \qquad E = 10^{-5} \text{ by default},$$

and the step shrinks until the budget holds. Because the SH transfer coefficients are orthonormal, E is the L^2 error over the sphere. A block that cannot meet E, or that byte planes store smaller, keeps its values exactly. Measured: the transfer sections take 4–7× fewer bytes (the pawn top's TX 112 went from 157 to 28 MB); a relit splat moves by relMSE 4×10^{-8} to 3×10^{-5}; the Corvette light set went from 67.9 to 49.5 MB and the pawn page from 281 to 84 MB.

7.2.2 The joint basis kept on the GPU

For the phone, the transfer is never expanded. athenea's joint CPCA splits the 112 TX values into three blocks $b \in \{\text{direct } (16), \text{indirect } (48), \text{reflected } (48)\}$. It uses C clusters shared by the three blocks and $K = 8$ directions a block. A gaussian stores one cluster id c and 3×8 weights:

$$\boldsymbol{t}_i^{(b)} \;\approx\; \mu_c^{(b)} + \sum_{k=1}^{8} q_{ik}^{(b)}\, \frac{s_{c,k}^{(b)}}{127}\, \boldsymbol{B}_{c,k}^{(b)},$$

where $\boldsymbol{B}$ are the per-cluster axes and s their scales. On the GPU, each direction is stored premultiplied by $s/127$ in f16, 504 words a cluster. **Weights** are int8 or, on the phone, int4: encoded at 7 levels and decoded as $q_8 = \text{round}(q_4 \cdot 127/7)$. On the phone, int4 costs $+0.7 \times 10^{-4}$ of per-splat colour relMSE and 0.1–4.6×10^{-5} of image relMSE; the CPCA's own truncation dominates.

The projected route. Under one dome, only the reflected block depends on the eye. Each relight, a thread per cluster projects the cluster's mean and 8 axes of the direct and indirect blocks onto the light (the dome's 16 harmonics, the sun direction, the white-sky coupling): 144 words a cluster. The view-independent ("kept") terms of a gaussian are then dot products of its weights with those projections. Per frame, only the 48 reflected values are rebuilt, in registers. On real data this route reproduces the full rebuild to relMSE 3×10^{-11}.

Basis evolution.

basis	clusters	fitted on	relMSE of the transfer	notes
athenea's basis	256	athenea's sample of `base_tx`	1.69e-2 (whole car)	1.17 MB f32 file; the player's cluster assignment matches athenea's 100 %
player fit, six parts	256	six parts of the car	direct 2.7e-4, indirect 2.5e-3, reflected 1.1e-3	in-sample; held-out pieces 10–400× worse
player fit, whole car	256	every second node of the car	1.03e-2 (direct 1.6e-3, indirect 8.0e-3, reflected 1.06e-2)	image within relMSE 1e-3 of the uncompressed transfer
player fit, 512 clusters (v8 data)	512	car1280 + car5120	fine3_0: 0.0181 → 0.0136	every 1280 panel group −20 to −30 %

The fit (`fit-compact-basis`) normalises the energy per block, runs k-means (k-means++ seeds, Lloyd over every node) on the joint vectors, then a per-cluster PCA of each block, and chooses the scale per direction with the least error through the int4 code. The **512-cluster basis** keeps the record size: the 9th bit of the cluster id goes into the free bit 15 of the shadow-dictionary index, which caps the dictionary at 32 768 entries (the car's has 4096). Files with 256 clusters read as before. Cost: basis 516 KB → 1.03 MB (f16), projections 147 → 295 KB, tier 4 +1.6–1.8 %, and the same shade time on the M5 Pro (2.82–2.88 ms at 400k drawn). It has not been measured on the real phone.

7.3 The open-direction bits by vector quantisation

Each gaussian's 16 × 16 open-direction cells (256 bits) are replaced by an index into a dictionary (`athc-convert --shadow-vq`). The dictionary is k-means in Hamming space, each centroid the bitwise majority of its members with ties kept open, 12 rounds, seeded from member $\lfloor c\,n/C \rfloor$. With 1024 centroids a piece, the image error is 62.9–65.4 dB (athenea's figure: 62 dB), and the mean Hamming distance is 2.0–2.5 of 256 (p99 14–16). The raw masks as a dictionary would be 15.3 MB at 1M splats, against 1.51 MB for the VQ. Across six pieces, one asset dictionary of 4096 entries (131 KB, mean distance 1.07) beats six dictionaries of 1024 (197 KB, mean distance 1.00) on bytes for about the same error, so the shared dictionary is the default.

7.4 The compact record

The compact path holds a gaussian as byte planes in one storage buffer. Each plane is 64-byte aligned and read by `fieldAt(plane, width, i)`:

plane	bytes	encoding
position	6	$3 \times$ u16 in the page's box: $x = \ell + q\,(h - \ell)/65535$
albedo	3	u8 sRGB
opacity	1	$q \le 191$: $\alpha = (q/191)^2$; $q \ge 192$: $\alpha = 1 + (q - 192)/63$ (Spark's LoD opacity, > 1)
scale	3	$s = \exp(\ell + (q-1)\,\delta)$, $\delta = (h - \ell)/254$; $q = 0$ means exactly 0 (surfels)

plane	bytes	encoding
rotation	4	quaternion word: octahedral axis 10+10 bits, angle 12
normal	3	octahedral, 12 bits an axis
curvature	3	$s8$: $\kappa = \pm f\,(2^{\lvert q\rvert R/127} - 1)$, signed log in the page's range
material	2	u16 into an exact table (pbr + lobes, 4 words)
cluster	1	u8 (9th bit in the shadow index)
weights	12	3×8 int4
shadow	2	u16 into the dictionary
GPU total	**40**	

The planes are what the data forced. An 8+8-bit normal shifted the coat's reflections by relMSE 5.2×10^{-2}; at 12 bits the error is 2.7×10^{-4}. A linear opacity code destroyed the glass sheets' 0.01 alphas. The car has 17–22k distinct materials, so a u8 index could not hold them. In an ablation, position, opacity, albedo, scale and curvature each contribute under 10^{-6} of image error. Against the full relight on the same cloud, the compact path's per-splat colour is within relMSE $6.3\text{–}7.3 \times 10^{-4}$. The image is within 40.2–41.9 dB at 0.4–1.0M splats.

Against the earlier relight's 308 B a gaussian, this path keeps **42.4 B** resident (planes plus tables) and **16 B per drawn splat** (the sort buffers reused as draw records). The fixed cost drops from 29 MiB to about 3 MiB. At 1M splats that is 58.6 MiB, against 326 MiB before (Dawn, buffer sizes).

7.5 Error-band pages and the compact tier

7.5.1 The error cut

Each merged node n carries the bounds ε_n (error) and ρ_n (sphere radius) in f16 (`LODE`). With p the pixels per unit at unit distance, the traversal refines n when

$$P_n = \frac{\varepsilon_n\, p}{\max(d_n - \rho_n,\ \text{near})} > \tau.$$

τ is in pixels. athenea pages its merge trees by **error bands**, as Nanite does. A node is needed exactly when its parent's projected error exceeds τ, so each node goes into the band (an octave) of its parent's error. The bands are ordered coarse to fine, each in leaf order, and the order is cut into pages of $P = 3000$ nodes. A page-table row holds a box around the parents' spheres and the largest and smallest parent error. A page is needed when

$$\frac{\varepsilon^{\max}_{\text{page}}\, p}{\max(\text{dist}(\text{eye}, \text{box}),\ \text{near})} > \tau \quad \text{(page 0 always)}.$$

Because the box bounds the parents' spheres, this test is conservative against the traversal. On a set of car parts, the settled paged cut was identical to the fully resident cut; residency was 2.27–2.53× the cut.

Two culls are applied in the traversal:

- **Back faces.** A node of a closed part is pruned with its subtree when $\boldsymbol{n} \cdot \boldsymbol{v} < -\sin(\theta + \arcsin(\rho/d))$, with θ the half-angle of its normal cone.

- **Occlusion.** With the eye outside the asset's box, a node is pruned when the four bilinear cells toward the eye ($\mathrm{oct}(\boldsymbol{v}) \cdot 16 - 0.5$) are all closed in its occlusion-codebook entry.

athenea's codebook for the car has 8192 entries (84 % of the ideal, with no false positives), and the v8 codebook has 8182. With quantised centres, occlusion is tested over the box of possible centres (split by orthant, where both octahedral coordinates are monotone). Over the car's 10.7M nodes, at athenea's cameras and 24 random eyes, this hid 0 nodes that the exact centre keeps, at a cost of at most 0.06 % of the hidden nodes being kept. With every cull on, the drawn count is -51 % against every cull off, at 36.9 dB between the two images.

7.5.2 The compact tier and its versions

The tier-4 section `CMPC` holds the 40 GPU bytes plus the traversal's bytes, so a phone reads none of tiers 1–3. The tables shared by a piece (`CMPT`: dictionary, materials, basis) and by an asset (`.cmpa`: basis, one dictionary, one material table, and the occlusion codebook with the asset's box; identified by FNV-1a 64) are read once.

`CMPC`	bytes a node (file)	traversal	change
v1	64 = 40 + 24	LoD tree's 4 words + `LODE`'s 2	first compact pages; paged = resident bit for bit
v2	56 = 40 + 16	16 B file, 14 B WASM	`CMPT` v2 and the `.cmpa` asset table; same cut node for node; near-view download 23.7 → 16.8 MB, tables 3.62 → 0.97 MB
v3	64 = 40 + 24	+cone word, 13-bit occlusion index; 30 B in, 20 B WASM	back-cull and occlusion on by default; only the pages the traversal asks for are planned
v4	container `CMP4`	12-byte quantised records	payload as **raw-deflate runs of 64 nodes**, plane after plane; node pool; tier 4 +10 % (314 → 344 MB)

The **v4 page** is a container: a header, each run's end offset, the gzipped traversal planes, then each run of 64 nodes compressed by raw deflate. Runs can be fetched, kept and evicted one by one, so the CPU keeps a page's payload compressed and the GPU keeps records only for the **drawn** cut. The node pool's census motivated this: with whole pages resident, only about 26 % of each page's nodes were drawn, and smaller pages did not help (residency/cut 4.54× at $P = 3000$, 4.46× at $P = 500$) (Figure 15).

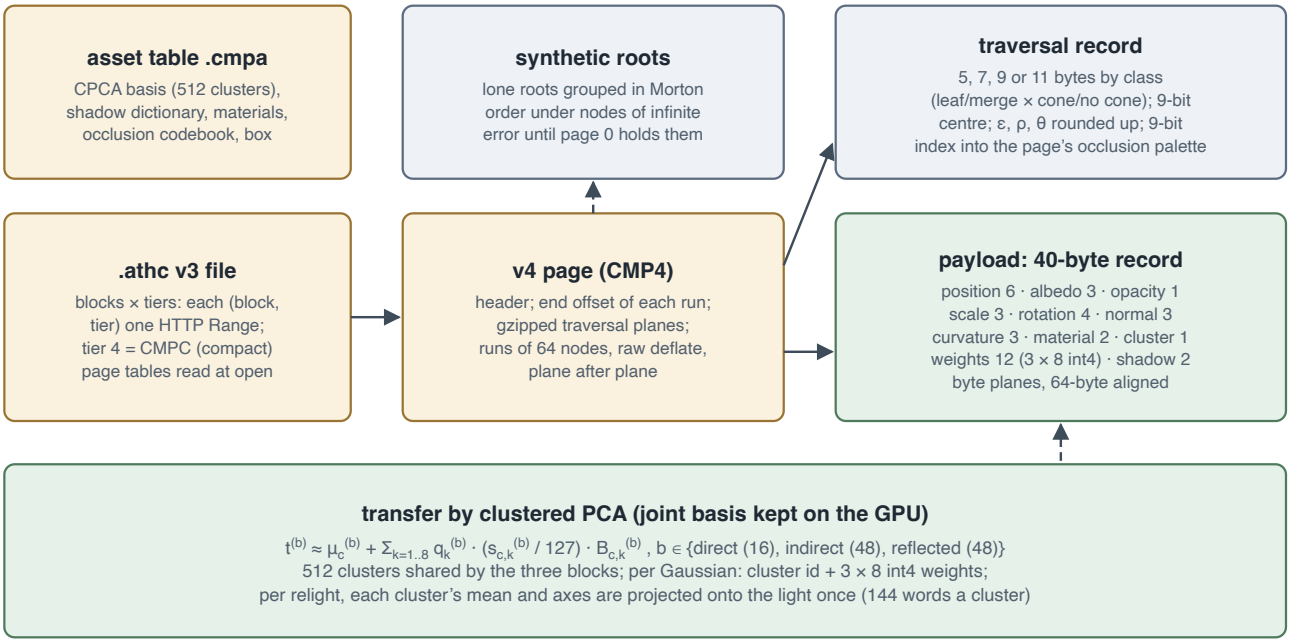


Figure 15. The compact format: an asset table, `.athc` v3 files whose tier 4 holds v4 pages of traversal records and 40-byte payload records, and the clustered-PCA transfer basis kept on the GPU.

7.5.3 Traversal records

The WASM traversal's per-node cost went down in steps, each producing the same cut:

stage	bytes a node	how
Spark `LodSplat`	24	full tree
`CMPC` v2 / v3 in WASM	14 / 20	f16 centre as an f16-order step from the compact position (exact), ε, ρ f16, children
v4 `QuantNode`	12	centre 3×10 bits in the page's box of centres; ε 8 and ρ 7 bits of log code over the page's range; θ 6; axis 5+5; occlusion 13; children as place + page delta
class records	5, 7, 9 or 11	2 bits pick leaf/merge × cone/no cone; 9-bit centres; **9-bit index into the page's occlusion palette** (511: none); a byte offset every 32 nodes

Rounding only refines more and culls less: $\varepsilon, \rho, \theta$ are rounded up, a merge's axis error is added to its θ, and the page's largest centre error is added to every sphere. Across the whole car, 1.15M resident nodes take 10.4 MB as class records, against 13.8 MB as 12-byte words. The page budget assumes 8 B a node. A further lossless step pointed each page's merges to their children's page through a **per-page palette** of 9 bits. This kept 436 nodes in full form instead of 181k on the 1280 car, and removed a 2.5 MB WASM map.

7.5.4 Synthetic multi-level roots

A page-0 slot holds 3072 nodes, but athenea's car5120 forests keep thousands of lone roots on purpose (merges never cross parts or athenea's guards): fine5 has 29 213 and fine6 21 065. Past a fan-out (2048), the packer cuts the roots, in Morton order of their centres, into runs, each the children of a synthetic node, and repeats until page 0 can hold the rest. Each synthetic node has infinite error (always refined), a sphere around its children, no cone and no occlusion index, so the cut is the flat forest's (tested at 5 eyes × 5 values of τ). fine5 went from 10 files to 3 and fine6 from 8 to 1.

7.6 Streaming and the per-frame cut

The reader (`CompactBandStreamer`) reads only the page tables at open. After each traversal:

1. The **needed set** comes from each tree's page table alone, with the eye in the tree's object space (the car's turn from athenea's Z-up stage) and foveated.
2. The **missing pages** are requested in order of the error a page removes per byte it costs; ties go to the nearer page. Consecutive pages are merged into one Range request (up to 8 MiB), within the pool's free room.
3. The pool **evicts** pages that are not needed from the finest band down. Page 0 and any page a resident page depends on are never evicted.
4. τ rises until the needed set fits (`tauToFit`). The current version finds, by bisection, the lowest τ that fits, which stopped the near-view oscillation (4.61–5.76 → 5.16).

A node whose children are absent is drawn as a leaf, or as its parent or children while a run is being decoded. With v4 runs, the cut's own runs and those of its parents and children are always kept, and the rest are dropped first under a run budget. An extrapolated camera (2.5 updates ahead) is traversed too, so its runs are prefetched. Up to 12 Range requests are in flight.

The work is split across the main thread, the WASM traversal and two workers (Figure 16):

where	what	measured
loader worker (WASM)	traversal over all trees, streaming the cut in chunks; stack instead of heap when the budget does not bind	Camera, $\tau = 3$: 128 → 40 ms (same cut)
node worker (`NodeCutter`)	cut to slots, run inflation (~0.3 μs a node), record packing, draw list with substitutes, prefetch	JS packing 32.8 → ~9 ms; orbit max / p99 from 31–35 / 26–31 to 17.5–19.0 / 17.4–18.7 ms
main thread	Range fetches, writes of records and draw lists (adopted, no GC)	—
GPU compute	record moves when the pool shrinks (one dispatch, a thread a word)	first frame after a still frame 88–100 → 34 ms

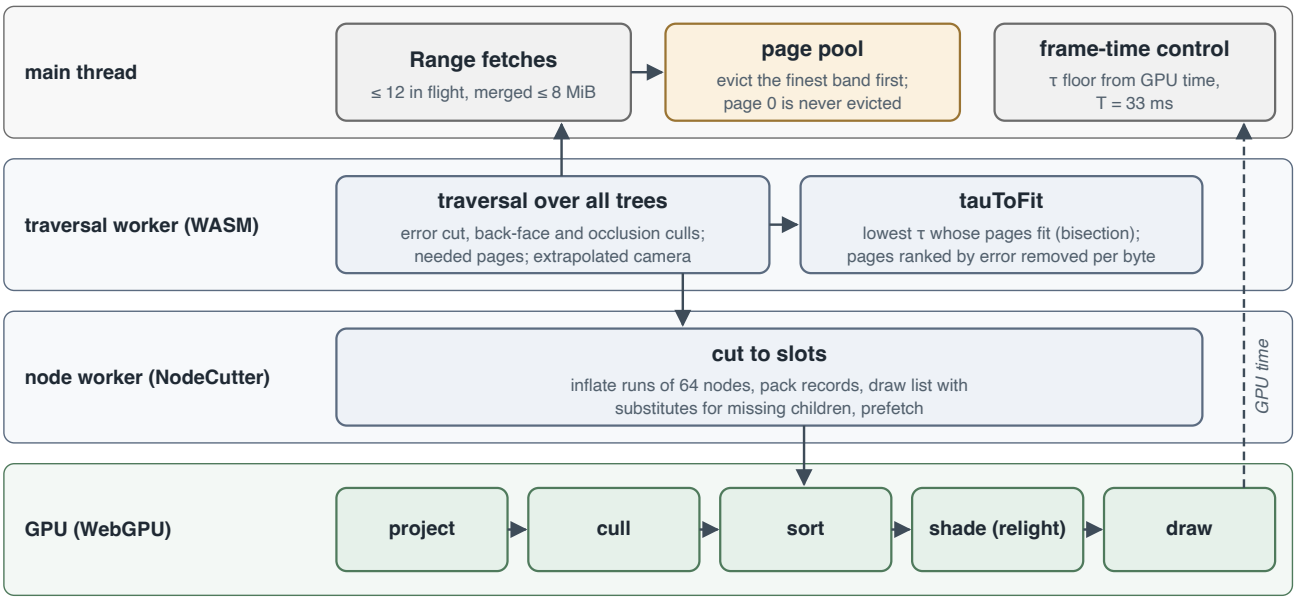


Figure 16. The player's threads: Range fetches and the page pool on the main thread, the traversal in a WASM worker, the cut packed into slots in a second worker, and five GPU passes a frame; the GPU time drives the τ floor.

On a fast orbit of 120° at 1° a frame on the emulated phone, the share of the cut missing from a frame fell from 26 % with the first v4 streamer to 4.3–6.4 %, which is 0.24–0.30 % of the screen area. Frames stayed at 60 fps with a maximum of 17.3–19.6 ms. The cut settled 2.6–3.1 s after load.

7.7 The draw

The compact path (`CompactRelitPath`, `compact.slang`) runs five passes a frame:

1. **Project** (one thread a cluster, only when the light changes): the per-cluster projections of §7.2.2.
2. **Cull** (one thread a listed node): frustum test; a 24-bit radial-distance key and the index appended with an atomic.
3. **Sort**: GPU radix sort of the drawn keys, back to front (6 passes over 24 bits).
4. **Shade** (one thread a drawn slot, binned by cluster): decode, relight (§7.8), project. A 16-byte record is written into the sort's own buffers: centre as u16 at 1/16 px, σ_1 (half) with the major-axis angle, σ_2 (half) with α (half; a negative sign marks LoD opacity), and colour in RGB9E5.
5. **Draw**: instanced quads, premultiplied over an rgba16float target.

Projection (EWA). With W the view rotation, R, S the splat's rotation and scale, and J the perspective Jacobian at the view-space centre (x, y, z):

$$J = \begin{pmatrix} f_x/z & 0 & -f_x x/z^2 \\ 0 & f_y/z & -f_y y/z^2 \end{pmatrix}, \qquad \Sigma_{2D} = JWRS\,(JWRS)^\top = \begin{pmatrix} a & b \\ b & d \end{pmatrix}.$$

Opacity compensation. The anti-aliasing blur adds $\beta = 0.3$ px^2 to the diagonal, and the opacity is scaled so the splat's mass is kept:

$$\Sigma' = \Sigma_{2D} + \beta I, \qquad \alpha' = \alpha_0 \sqrt{\det\Sigma_{2D} / \det\Sigma'}\,.$$

A merged node with LoD opacity $\alpha_0 > 1$ is drawn with $\alpha_0 \leftarrow \min(4\alpha_0 - 3, 5)$, and its quad is widened by $0.7(\alpha_0 - 1)$ deviations. A faint splat ($\alpha' < 4\,\alpha_{\min}$) keeps its mass $\alpha_0\sqrt{\det\Sigma_{2D}}$: its blur is chosen so its opacity reaches the faint target, or it is drawn at that target with probability $\alpha'/(4\alpha_{\min})$ by a stable hash dither. A thin glass sheet's floor of 0.1 is applied **after** the blur and faded by its area on screen. Applied before the blur, it had made the headlight cover 1.63× too bright. The fragment evaluates $\alpha = \alpha' e^{-z^2/2}$ inside the cut radius, discards under $\alpha_{\min}$, and adds the per-pixel sharp lobes (§7.8).

A thread per drawn slot reuses the sort buffers as records, so nothing else is per-splat. The compact path has no relit records (84 B), no pixel records (48 B), no accumulator (32 B) and no ext splats (32 B). On the phone it also has no depth texture: nothing reads it. A still view re-runs only the draw. Writing directly to 8 bits saved no GPU time (Apple blends in tile memory) and changed the image by up to about 100 codes in 1–3 % of channels, so the target stays rgba16float.

7.8 Relighting on the phone

The shade evaluates athenea's `relitByDome` (TX branch) for one dome. The code is a verbatim copy with the field reads taken from registers, because WebGPU cannot bind one buffer as both read and write in a dispatch. The kept terms come from the projections; the reflected field (16 coefficients × RGB) is read along the mirror. The following refinements were ported or added, and each was measured.

The reflected field. It is 42 % of the shade, and the cost is arithmetic, not memory. Two changes cut it: shading binned by cluster (a counting sort of the slots per cluster, with per-workgroup histograms), −6 to −9 %; and the field limited to degree 1 (4 coefficients) when a footprint is under 3 px, −23 % of the shade at 215k drawn. Against the path trace, the change in the door, sill, bumper and hood is under 0.01 dB.

Sharp per-pixel lobes. A coat or polish below the sharp roughness thresholds is not answered once at the centre. The shade subtracts the centre's sky reading and hands the draw a 5-word record: the mirror direction at the centre (unorm16 × 2), its steps one pixel right and one pixel down (f16 pairs), the coat weight and both roughnesses as square roots, and the polish weight in RGB9E5. The fragment then reads the sky along $\mathrm{oct}^{-1}(\boldsymbol{a}_0 + \boldsymbol{s}_x\,\delta x + \boldsymbol{s}_y\,\delta y)$. The roughness read is built in three steps:

- *Unwidening.* The bake widened the stored roughness by Toksvig, $r'^4 = r^4 + 2s^2$ with s the normal spread. Since the per-pixel turn already accounts for that spread, the lobe reads $r = \max(r'^4 - 2s^2, 0)^{1/4}$.
- *Fold rule.* With $t = 2s$ the turn over $\pm 2\sigma$ and $w = \mathrm{smoothstep}(0.7, 1.5, t)$, the pixel's turn is scaled by $1 - w$ and the lobe widens, $\alpha' = \sqrt{\alpha^2 + w\,(t/2)^2}$, held under 0.5.
- *Pixel spread.* The normal turns by $\boldsymbol{d}_x, \boldsymbol{d}_y$ across a pixel. That box of slopes, doubled for the mirror, widens the GGX $\alpha = r^2$:

$$\alpha^2 \leftarrow \alpha^2 + \frac{2\left(\left\|\boldsymbol{d}_x\right\|^2 + \left\|\boldsymbol{d}_y\right\|^2\right)}{12} = \alpha^2 + \frac{\left\|\boldsymbol{d}_x\right\|^2 + \left\|\boldsymbol{d}_y\right\|^2}{6}.$$

Without the pixel spread, v7 showed bright specks on folds. At τ 1 / 3, the specks went from 24 / 27 to 6 / 2 on the sill and from 19 / 25 to 2 / 1 on the door crease; the door went from 34.21 to 35.53 dB. A lobe whose mirror stays within 2 texels of its sky level across 3σ is read once at the centre (the draw −10 to −12 % close up, with no change against the path trace within 0.001 dB).

The environment as a filterable atlas. At first the fragment decoded two levels of four texels each from a storage buffer. The dome's eight prefiltered levels are now copied into one rgba16float atlas: level 0 at the origin, the others down a column to its right (0.9 MB for a base side of 256, 3.1 MB at 512). The atlas is sampled with a hardware bilinear sampler, two samples a lobe. The image is 77 dB of the storage reads.

The compact pass at 393 × 852, $\tau = 3$ (M5 Pro, wall time) went from 11.8 to 4.0 ms with sharp lobes from 1 px; at 1920, $\tau = 1$, from 75 to 17.2 ms. The shade reads its own env texels from the atlas as well, which freed the env buffer (−2.67 MB).

polishField and coatFold. Each material carries its own roughness in pbr bits 26–31, at no cost per node. Where that roughness is ≥ 0.3, the polish asks its own lobe's cells, and the reflected field answers what they close (the door's ratio to the path trace went from 0.786 to 0.857). Where the surface turns fast, the coat is held: $r_{\text{coat}} = \max(\text{unwidened}(c, s),\ \min(c, 0.3))$. This applies only where the material's roughness is ≥ 0.3. Applied everywhere, it lit chrome, glass and rim spokes (wheels +63 % relMSE). On the 1280 bake the coat comes out at 0, so the term no longer changes the car.

7.9 The contact shadow

A compact set draws without Spark's renderer, so the contact shadow is a pass of its own (`ground-shadow.js`), and it uses gaussians only (no mesh, no AO). athenea's shadow catcher is black gaussians on the ground plane, each with its open-direction cells. They are splatted over the dome before the car: EWA with the same 0.3 px² blur and opacity compensation, black at opacity $\alpha_i = o_i\,\kappa_i$. Black over black needs no sort. The darkening is

$$\kappa_i = \frac{-\ln r_i}{2\pi}, \qquad r_i = \frac{\sum_{\text{open cells } j} L_{\text{dome}}(\omega_j)\cos\theta_j\,\Delta\Omega_j \;+\; \text{sun through the cells}}{\text{irradiance of the whole sky above the floor}},$$

recomputed on the CPU from the dome's 16 harmonics whenever the dome changes. athenea's day catcher (309k gaussians at 1.26 cm) costs 9.9 MB of GPU drawn whole, and thinning it 1 in 4 shows a wavy grain. Instead, `floor_shadow --merge 2` moment-matches 2 × 2 neighbours (centre and covariance weighted by opacity × area, mass kept, cells by majority). The result is 78k gaussians, 0.65 MB gzipped, 2.5 MB of GPU, and the closest to the path trace in every region:

region (shadow factor)	PT	thinned 1 in 4, 77k	full 309k	merge 2 × 2, 78k
front wheel	0.607	0.565	0.568	**0.594**
rear wheel	0.713	0.617	0.618	**0.638**
front floor	0.211	0.168	0.164	**0.224**
under sill	0.437	0.390	0.387	**0.430**

Without the shadow the car floated, and the sill read 1.66× the path trace: it was the hall's floor at infinity seen past the car's edge. With the shadow the sill reads 1.08 (desktop: 1.07).

7.10 Frame-time control and the still frame

The draw target has had four designs:

1. **Draw target.** It started at 260k drawn and was cut at once when the GPU median frame passed 16.5 ms. It rose 6 % a second while under 14 ms.
2. **~227k drawn.** For a still eye, the τ floor was bisected until about 227k were drawn (v6's 60 fps cut).
3. **τ floor from frame time, plus a still frame**, with a 16.5 ms target.
4. **Quality first**: image quality takes precedence over frame rate, with a 33 ms target. This is the current version.

In the current version, the GPU time of a frame is measured from its start to `onSubmittedWorkDone`. Every 0.5 s, with $\bar{t}$ the median of at least 8 moving frames and $T = 33$ ms:

$$\tau_{\text{floor}} \leftarrow \begin{cases} \tau_{\text{floor}} \cdot \min\left(1.6,\ \sqrt{\bar{t}/(0.91\,T)}\right) & \bar{t} > T, \\ \tau_{\text{floor}}/1.04 & \bar{t} < 0.85\,T \text{ for two periods}, \\ \tau_{\text{floor}} & \text{otherwise}, \end{cases} \qquad \tau_{\text{floor}} \in [1.5,\ 8] \text{ px}.$$

The square root follows from the number drawn scaling about as $1/\tau^2$.

The still frame. After 300 ms with no input and no camera change, the moving cut is pinned in the node pool, τ drops to 0.5 px, and the page limit rises from the moving pool (14 MB of traversal) to the still pool (28 MB). One fine cut is drawn whatever it costs, and drawn again as more pages land. Then nothing is redrawn until a touch, which brings the pinned moving cut back in that same frame and evicts down to the moving budget at 32 pages an update. The budget is about 100 MB of car data at rest and about 55 MB while moving. Measured on the emulated phone, close up and still: 529k drawn at τ 1.23 with 95.6 MB of car data. A second measurement gave 531k at τ 1.22 in 118 ms, at a simulated 190 ms a million, with the first frame after a touch at 39–47 ms.

7.11 Memory accounting

The 50–60 MB budget applies to the **car data** only. Car data is the resident pages and runs, the WASM traversal, the node worker's heap and buffers, and the GPU records. The **render state** is counted apart: sharp records, atlas, shadow, targets, sort. It carries the image quality, and an iPhone 15 tab held about 1.2 GB in the stress probes. The **page** itself (JS heap, three.js, domes) is a third figure.

state	data	car data	render state	page
six parts, τ 1, emulated	CMPC v1	—	GPU 53 MB total (peak 59.6): data 24.7, fx 9, target + dome 7.9, env 2.7, sort 4.6, canvas 3.8	—
whole car, τ 3, first v4 streamer	v5	≈60 MB CPU + GPU (page pool at the same τ: ≈87)	—	—
whole car, τ 3, class traversal records	v6	Camera 55.1, close 59.0, far 49.1 (all together)		
whole car, τ 3, after the atlas	v6	Camera **39.0**, close **43.8**, far **33.0**	24.1 / 26.0 / 21.7	~39 → 25 → ~17 (two reductions)
whole car, τ 3	v7	31–36	—	—
still frame, close up	v7	**95.6** (529k, τ 1.23)	—	—

Several reductions were each byte-identical in the image:

- the relight sky's floats rebuilt from RGBE on upload (1.6 MB held instead of 8 MB for a 1k dome);
- the visible dome in rg11b10ufloat on the phone;
- one shared occlusion codebook (4.3 MB of JS arrays before);
- sharp records sized to the drawn slots (−0.3 to −2.2 MB);
- the WASM shipped as a separate file, with the layout parsed in a worker.

The earlier per-gaussian accounting explains why the compact path was needed. The previous relight kept 308 B a gaussian plus 29 MiB fixed: 147 / 206 / 323 MiB at 0.4 / 0.6 / 1.0M. The compact path keeps 42.4 B + 16 B per drawn splat + about 3 MiB.

7.12 Measured results

7.12.1 The real iPhone

Stress page, 10–11 October (synthetic ramps over real Corvette splats; medians):

path	200k	400k	600k	800k	~1M	notes
plain 3DGS, fps	58.8	58.8	45.5	34.5	26.3–27	cool phone; ~37 ms a million
transfer, earlier "by" path, fps	38.5	12.3	7.2	5.2	3.7	362–469 MB GPU
compact, synthetic basis, ms (fps)	30 (33.3)	63 (15.9)	101 (9.9)	137 (7.3)	154 (6.5)	20.8–79.8 MB GPU
compact, basis fitted on car data, optimised shade, ms (fps)	27 (37)	47 (21.3)	70 (14.3)	95 (10.5)	108 (9.3)	35 MB GPU at 400k

Model for the optimised shade: about 115 ms a million drawn + 4 ms, so 30 fps holds about 250k drawn and 60 fps about 110k. The first second of each step ran 2–3× faster than the sustained rate (clock or thermal state), and a run following four runs in 20 minutes was half as fast from 600k up. Sustained figures are used for planning.

Whole-car runs on the device (v7 data):

run	view	drawn	τ	fps	observation
first	default	106k	8.00	53	fluid; the τ 8.00 was the old frame-time "breath" stuck at its ceiling (fixed)
first	close up	160k	—	32	soft logo and grille
second	close up	—	—	—	still soft

The estimated shade cost on that iPhone is about 190 ms a million. The softness close up comes from the cloud, not the player: the logo baked at 1280 px is blurred even with all its leaves, and crisp at 5120 px. The car5120 close-up level (v8) addresses it.

7.12.2 The emulated iPhone

data	view	drawn / resident	τ	download	fps
six parts, paged	default	174k / 384k	1.55	25 MB	61
six parts, CMPC v1	default	199k / 555k	1	24.6 MB	60
same, CMPC v2 + `.cmpa`	default	199k / 555k	1	17 MB	60
whole car, v3 (24 MB pool, fov = h)	default	147k / 580k	6.75	20 MB	60
whole car, v5 (CMPC v4)	default	226k / 1016k	3	36 MB read	60
whole car, v7	default	216k / 1318k	2.22	—	60
whole car, v7, later streaming revisions	default	234k	1.95	—	60
whole car, v7, still frame	close up	529k	1.23	—	(one frame)

7.12.3 Against the path trace, by data version

`car-compact-tx` data versions are distinct from the `CMPC` format versions:

set	content
v3	occl8k, CMPC v3
v4	athenea's closed2 trees
v5	CMPC v4 runs
v6	athenea's mrough trees, per-material roughness, 1.35 GB
v7	car1280, 1.81 GB
v8	car5120: 24 trees, 74.5M leaves, 148.9M nodes, shared 8182-entry occlusion (§6.9)

Unless noted, the figures below are PSNR at athenea's camera at 1920 × 1080, by the validate-corvette harness's regions.

version	car	door	sill	bumper	hood	source / condition
v6	19.33	33.77	31.87	20.21	14.16	compact path
v6, Toksvig unwiden	19.68 → 20.58	+1.0	—	—	+1.3	per-pixel detail on
v7 (first conversion)	+0.5 vs v6	−2.3	−2.7	—	+0.9	specks on folds
v7 + pixel spread, τ 1	—	**35.53**	**32.36**	relMSE 0.199	14.46	pixel spread on

version	car	door	sill	bumper	hood	source / condition
v7, current defaults	19.03	35.52	32.36	22.28	14.68	degree-1 field
v8	—	—	—	—	—	not measured

The car crop metric of §1.6 (900 × 420 crop) is a different region. On it, the per-pixel sharp lobe in the compact path took the car from 16.49 to 17.46 dB (17.41 with lobes from 4 px), which is the desktop path's level. On the phone, τ 3 against τ 1 differs by 40 dB, so the remaining defects (hairs, streaks, blotches) come from the cloud, not from the cut. Lossless steps were checked against the previous image: the sharp records in 5 words, 60–80 dB; binned shading, 84 dB (run-to-run noise 78 dB); the glass-only terms, 188 dB.

7.13 Limitations and open work

- **v8** (car5120 with the 512-cluster basis and the shared 8182-entry occlusion codebook) has not been converted for the player at the time of writing; athenea's side (the 24 trees and their shared occlusion, §6.9) is complete. Its quality against the path trace, its download and its behaviour on the real phone are unmeasured.
- Real-device timing exists only for the stress ramps and the two whole-car runs. The emulated phone's speed is not the phone's.
- The front bumper's chrome strip is 38 % brighter than the path trace. The lamps have no emission in the bake. The door's open reflections (1.13) come from 16 × 16 cells of 11°.
- The compact traversal does not keep a merge's LoD size, so the combined size/error cut has nothing to guard with. The error cut alone did not beat the size cut on these data in every view.

8. Evaluation

This section gathers the measurements of Sections 2–7 into one account. It leads with the bake, validated by relighting the baked Gaussians against the path trace of the source mesh (§8.2, §8.3); the engine (§8.4), the levels of detail (§8.5) and the bake's costs (§8.6) follow, and the mobile application (§8.7–§8.9) comes last. It introduces no new numbers: every figure is quoted from the section named beside it. Where two sources give different figures for the same quantity, §8.10 says which is which.

8.1 Setting

Desktop measurements were taken on an Apple M5 Pro under Metal; Vulkan and CUDA results come from an NVIDIA L4 (§2.15). The Corvette's reference is the mesh path trace from `/World/Camera` at 1920 × 1080 (§1.6). The player was measured in three ways that are not interchangeable (§7): GPU timestamps from headless Dawn on the M5 Pro, the live site in desktop Chrome in its mobile mode at 393 × 852 (the *emulated iPhone*, about twice as fast as the real one), and a real iPhone (iOS 18.7, Safari 26.6.1, 393 × 852 @3).

8.2 The conversion against the path trace

asset or test	measure	result	§
unit quad lit head-on (mesh pixel 0.5032)	converted pixel at $\sigma = 1.0$ / material values passed	0.456 / 0.5009	3.4, 3.6
Corvette, 640 px, edges held vs control	relMSE / PSNR / SSIM, day frame	0.0887 / 21.93 dB / 0.8567 against 0.0902 / 21.86 dB / 0.8554	3.5
Corvette shadow catcher vs path trace	under the car / bumper contact / open ground	0.192 / 0.030 / 0.220 against 0.182 / 0.043 / 0.217	3.9
Corvette, 640 → 1280 px bake	car crop / frame	16.55 → 17.88 dB / 22.18 → 23.90 dB	3.11
white pawn, body / glass head (2048^2)	relMSE	0.0218 / 0.1418 (later runs: glass 0.149, 0.081, 0.043)	5.11
sparrow (web conversion) vs mesh path trace	relMSE, raster	0.0025	5.10
sparrow wing, raster / traced	relMSE after coverage follows opacity	0.1635 / 0.1247	5.10

8.3 The whole car along the pipeline

The same frame, relit under the path trace's sky and measured as each step of the bake and the draw was added, in the order they were added. Rows from §4.10.3 and §6.10; nodes are drawn Gaussians.

bake	configuration	nodes	car crop (dB)	frame (dB)
640 px	all leaves, before §4's later steps	5.38 M	16.37	21.92
640 px	same bake, error cut 1 px, both culls	675 760	16.28	21.80

bake	configuration	nodes	car crop (dB)	frame (dB)
640 px	error cut 0.25 px, back culling	2 313 541	16.29	21.81
640 px	leaves, sharp lobes unwidened	5.38 M	16.58	22.15
640 px	cut, + coat held at folds	—	16.51	22.15
640 px	cut, + polish field on rough materials	—	16.55	22.18
640 px	leaves (0.01 px), no cull / cull	4.35 M / 3.70 M	16.73 / 16.75	22.40 / 22.43
640 px	group 0 with spill $w_s = 2$	561 242 (group 0)	16.79	22.38
640 px	turn capped / coat fold on paint only	—	16.57 / 16.57	22.20 / 22.20
1280 px	cut 0.25 px, back culling	4 998 976	**17.88**	**23.90**
1280 px	without the cap / gated sharp polish	—	17.88 / 17.88	23.91 / 23.90
1280 px	gate + tube/fold split	—	17.90	23.93
1280 px	tube/fold split (retained)	—	17.88	23.91

Three conclusions follow. The level of detail costs about 0.1 dB against the path trace (all 640 px leaves against the 1 px cut with both culls, 16.37 → 16.28 at an eighth of the leaves; §6.10). The shading steps of §4 moved the 640 px car by about 0.2–0.4 dB. The bake's resolution moved it by 1.3 dB on the crop and 1.7 dB on the frame (the 640 px cut with the polish field against the 1280 px cut). PSNR over a crop moves little, so the class ratios below show where the changes act.

region or class (mean over the path trace's)	before	after	condition	§
door, mirror to ground < 3 m	0.78	1.02	640 px leaves → 640 px cut with polish field	4.10.4
door, mirror to ground 3–30 m	0.69	0.88	640 px leaves → 640 px cut with polish field	4.10.4
door, mirror to the HDRI (open)	0.92	1.13	640 px leaves → 640 px cut with polish field	4.10.4
door joints	0.69	1.04	640 px leaves → 640 px cut with polish field	4.10.4
door open reflections, local contrast (σ 1.5 / 4 px)	0.35 / 0.53	0.56 / 0.74	640 px cut with polish field → 1280 px cut	4.10.4
hood highlights, offset p_{50}	3.2 px	2.0 px	640 px cut with polish field → 1280 px cut	4.10.5
hood mean	0.89	0.98	640 → 1280 px	4.10.5
grille's holes in the coarse cut	2.24	1.04	640 → 1280 px	3.11, 6.5
sill's hairs (pixels)	987	459	640 → 1280 px	3.11

region or class (mean over the path trace's)	before	after	condition	§
sill, under the car (phone)	1.66	1.08	without → with the contact shadow	7.9
chrome strip high-pass / PT	0.41	0.41	1280 px cut → tube/fold split (gated sharp polish: 1.12)	4.10.6
bright specks on the sill's fold	10	1	1280 px cut → tube/fold split	4.10.6

material (relMSE against the path trace)	chrome	rough metal	paint	§
first transfer	0.536	0.0896	0.705	4.10.1
TX, cells and field everywhere	1.64	0.221	0.393	4.10.1
TX with the polish gate	0.591	0.0835	0.341	4.10.1
+ sharp coat and sharp polish per pixel	0.545	0.083	0.281	4.10.1

8.4 The engine's correctness and speed

Every check below is a GPU kernel against a closed form, an independent estimator, MaterialX, or Storm.

check	result	§
two half-opaque discs, raster / reference / ray tracer vs analytic over-operator	p99 0 (encoded blend: p99 26)	2.4
converted planes of coverage 0.25 / 0.5 / 0.75, raster at full size	0.251 / 0.501 / 0.752	2.5.3
shutter step 0.03 vs N instants	p99 2, energy ratio 1.0000	2.5.3
ray tracer vs its exact-order reference	p99 ≤ 1, at most 0.05 % of pixels over 2	2.15
mesh visibility vs Storm (Kitchen_set), raster / rays / bvh	segmentation 18 / 41 / 36 differing pixels of ~57 k	2.8
MaterialX graphs vs MaterialX's own genglsl, 65 536 directions	worst $3.2 \cdot 10^{-5}$ over eleven graphs	2.9
lights vs closed forms; χ^2 against `lightPdf`	0 of 8281 pixels beyond 2 %; $\lvert z \rvert \le 1.61$	2.10
path tracer: closed emissive shell; convergence exponent	0 of 3072 beyond 10^{-4}; -0.499	2.11
OIDN, 16 paths vs 4096-path reference	relMSE $3.65 \cdot 10^{-4} \rightarrow 7.17 \cdot 10^{-5}$	2.11
test suite on three devices (188 cases)	Metal 188 passed; Vulkan 180 passed, 8 skipped; CUDA 96 passed, 1 failed, 91 skipped	2.15

workload	figure	§
radix sort, 10 M 32-bit pairs	12.4 ms (39.9 ms before)	2.3

workload	figure	§
sparrow clips, 5 887 323 gaussians at 1600 × 1000, playing	102 → 49 ms a frame	2.5.3
train_7k (742 k), 1080p: tile raster / ComputeBvh / hardware tracer	13 / 343 / 435 ms	2.6
Kitchen_set visibility at 1920 × 1080: raster / rays / bvh	31.7 / 19.2 / 29.2 ms	2.8
Kitchen_set lit at 1920 × 1080: raster / path tracer (1 path, 1 bounce)	96.0 / 358.9 ms	2.11
whole TX Corvette (14.7 M gaussians) at 1920 × 1080, desktop rasteriser	158 ms (about 83 ms for a relit cloud of its size)	4.10.2

8.5 Levels of detail

level	leaves	merges	roots	pages	occlusion codebook	cells reopened	§
640 px car	5.38 M	—	17 133	3 582	8192	11.9 %	6.2.5, 6.7
1280 px car	8.69 M	8.67 M	20 723	5 789	8187	12.3 %	6.7, 6.10
5120 px close-up level	74 536 126	74 328 031	208 095	49 634	8 182	0–23.6 % per tree	6.7, 6.9

The close-up level is 24 trees and 148 864 157 nodes, 87.4 GB on disk, with its shared occlusion written in place (645 s, 10.2 GB peak resident; §6.9).

cut, at 393 px wide, against the leaves at 4×	near nodes	near dB	far nodes	far dB	§
error 1 px, bounds	666 585	33.29	343 541	39.15	6.10
+ back culling of closed shells	522 931	33.63	—	—	6.10
+ occlusion	360 333	33.58	195 049	39.53	6.10
both culls, $O < 2\sqrt{E}$ rule	378 028	33.53	—	—	6.10

Each cull lowers the node count, and no culled cut reads lower against the leaves than the error cut alone (§6.10). In the player, every cull on draws 51 % fewer nodes than every cull off, at 36.9 dB between the two images, and the occlusion codebook reaches 84 % of the ideal (§7.5.1). Residency is the weak point: a walk needs about 2.2× the cut, and pages hold 3.6–4.5× it whatever their size (§6.8), which is why the player keeps the traversal apart from the payload (§7.5.2).

8.6 Costs

step	figure	§
1280 px car, 8 groups, transfer each	212–304 s, about 34 min in all	3.11
shadow catcher, 309 k gaussians	79 s	3.9
sparrow’s per-part visibility fields (12 parts)	22 s	5.8

step	figure	§
sparrow decimation keeping the rig (5.89 M)	28 s	5.9
shared occlusion codebook over the close-up level	645 s	6.9
original LOD build, train_7k	47 ms	6.1

8.7 Compression

step	before	after	image cost	§
byte planes (lossless), Corvette light / HD	68.1 / 288.7 MB	60.9 / 248.6 MB	none	7.1
transfer encoding 3 (CPCA in the file), Corvette light set / pawn page	67.9 / 281 MB	49.5 / 84 MB	per splat relMSE 4×10^{-8}–3×10^{-5}	7.2.1
joint CPCA on the GPU, whole-car basis (256 clusters)	112 values	1 id + 24 int4 weights	transfer relMSE $1.03\cdot10^{-2}$; image within relMSE 10^{-3}	7.2.2
512-cluster basis (v8 trial), fine3_0	0.0181	0.0136	same shade time on the M5 Pro	7.2.2
open-direction bits by VQ, 1024 centroids a piece	256 bits	u16 index	62.9–65.4 dB	7.3
compact record (resident)	308 B a gaussian + 29 MiB	42.4 B + 16 B a drawn splat + ~3 MiB	40.2–41.9 dB against the full relight	7.4
traversal record	24 B (Spark)	5–11 B by class	same cut	7.5.3

8.8 The phone

Against the path trace, by data version (validate-corvette regions, PSNR in dB at 1920 × 1080; §7.12.3):

version	car	door	sill	bumper	hood
v6 (640 px trees)	19.33	33.77	31.87	20.21	14.16
v7 + pixel spread, τ 1 (1280 px)	—	35.53	32.36	relMSE 0.199	14.46
v7, current defaults	19.03	35.52	32.36	22.28	14.68
v8 (5120 px close-up level)	—	—	—	—	—

v8 had not been converted for the player at the time of writing and has not been measured. On the car crop, the compact path's per-pixel sharp lobe took the car from 16.49 to 17.46 dB (§7.12.3).

Speed (§7.12.1, §7.12.2):

device	data	drawn	result
real iPhone, stress page	plain 3DGS	200 k / ~1 M	58.8 / 26.3–27 fps
real iPhone, stress page	compact, real basis	200 k / 400 k / ~1 M	27 / 47 / 108 ms
real iPhone, model	compact	—	~115 ms a million + 4 ms: ~250 k at 30 fps, ~110 k at 60 fps
real iPhone, first device test	v7, default view	106 k	53 fps (τ 8.00)
real iPhone, first device test	v7, close up	160 k	32 fps, soft logo and grille
emulated iPhone	v5 / v7 / v7 after a later frame-time adjustment	226 k / 216 k / 234 k	60 fps at τ 3 / 2.22 / 1.95
emulated iPhone, still frame	v7, close up	529 k	one frame at τ 1.23
Dawn, M5 Pro, compact pass	393 × 852 τ 3 / 1920 τ 1	—	4.0 / 17.2 ms

By the model, the far LOD cut with both culls (195 049 nodes, §8.5) fits the ~250 k the phone draws at 30 fps, and the near cut is about 1.4× that (§6.10). A second device test found the close-up view still soft. The softness was traced to the 1280 px cloud, not to the player, which led to the 5120 px close-up level (§7.12.1, §6.9). For the sparrow, the mobile cut (488 656 splats, 31.7 MB) ran at 60 fps headless in mobile mode, animated (§5.12).

8.9 Memory

quantity	figure	§
car data, v6 after the atlas, τ 3: Camera / close / far	39.0 / 43.8 / 33.0 MB	7.11
render state, same views	24.1 / 26.0 / 21.7 MB	7.11
page (JS heap, three.js, domes)	~39 → 25 → ~17 MB	7.11
car data, v7, τ 3	31–36 MB	7.11
car data, still frame close up (budget ~100 MB)	95.6 MB	7.10, 7.11
per gaussian, earlier relight / compact path	308 B + 29 MiB / 42.4 B + 16 B a drawn splat + ~3 MiB	7.4, 7.11
at 1 M splats, earlier / compact	326 / 58.6 MiB	7.4
tree write, 4.08 M-leaf door (after the fix)	10.5 GB peak resident	6.9
shared occlusion over 24 trees	10.2 GB peak resident	6.9

The player's 50–60 MB target applies to car data only; render state is counted apart (§7.11). Moving, the car data was 49.1–59.0 MB in v6 before the atlas and 31–44 MB in v6 after it and in v7, inside the target (§7.11).

8.10 Where sources differ

- **1280 px leaves.** The eight bakes wrote 8 703 429 Gaussians (§3.11, "8.70 M"); the trees hold 8.69 M leaves (§6.10). The first is the conversion's count, the second the trees'.
- **Front_Bumper_001.** The transfer bake reports 5 009 059 Gaussians, the tree 5 008 986 leaves (§3.11); the totals use the tree's figure.

- **The close-up level's size on disk.** 87.4 GB for the 24 trees; "about 86 GB" was the estimate before the last tree was written (§6.9).
- **The close-up occlusion total.** A first count read 132.4 % reopened; it was a 32-bit overflow of the closed-cell total, since fixed. The per-tree figures (0–23.6 %) and the codebook were correct (§6.7).
- **The still frame.** One measurement gives 529 k drawn at τ 1.23 with 95.6 MB of car data; another gives 531 k at τ 1.22 in 118 ms (§7.10). They are two runs of the same policy.
- **Door joints at 1280 px.** §6.10 gives the joints as rising from 0.90 to 1.02 between the 640 px cut with the polish field and the 1280 px cut, whereas the class table of §4.10.4 gives the former's joints as 1.04 (0.90 is the 640 px leaves without culling). The two figures come from different measurement runs; both put the 1280 px joints within 2–4 % of the path trace.
- **Car crop on the phone.** 17.46 dB for the compact path (§7.12.3) and 17.88 dB for the 1280 px cut on the desktop rasteriser (§4.10.3) are separate measurements on different paths and data, and are not compared here.
- **Real versus emulated speed.** Emulated figures run at about twice the real iPhone's; draw counts are judged on the device (§7).

9. Limitations and future work

The limitations below are gathered from the sections' own lists (§2.16, §3.12, §4.11, §5.13, §6.11, §7.13) and grouped by where they act, the relit bake first and the mobile application last. Each is stated with the figure that measures it, where one exists.

9.1 Remaining defects on the Corvette

These are what the relit 1280 px car still shows against the path trace.

defect	measured	cause	remedy proposed	§
lamps dark	headlights dark against the path trace	the transfer bake carries no lamp emission	bake the lamps' emission and their light on the car	3.12, 4.11
chrome strip on the front bumper	38 % brighter than the path trace, with a lower peak	a tube a few pixels across: each Gaussian reads its own mirror with one normal	— (open)	3.12, 4.11
rim's near-ground reflection	0.72 of the path trace	— (not analysed further)	— (open)	4.11
door's open reflections	1.13 of the path trace; 68 % of the excess where the mirror meets the car	16 × 16 cells of about 11° let sky through behind a mirror or pillar	32 × 32 cells (+96 B a Gaussian) would roughly halve it	4.10.4, 4.11
hood highlights	2.0 px off their path-traced position	the per-pixel turn extrapolates the centre triangle's shape operator	a curvature fitted over the footprint from mesh samples (p_{90} 9.8° against 46°) would roughly halve it	4.10.6, 4.11
bumper's lower lip	ragged	one cell's footprint per Gaussian at a rolled edge	— (open)	3.12
grille's holes in coarse cuts	2.24 of the leaves at 640 px, 1.04 at 1280 px	merges of sub-pixel strips; the hole is a property of the projection	finer leaves (done at 1280 px); no view-free per-node measure sees it	6.5, 6.11
far, thin trims	fat dashes in every cut	the error bounds miss this error	— (open)	6.11

9.2 The representation and the transfer

- **Rank-one field.** The reflected field's coupling to the sky puts a lit ground's colour, not its shape, into a reflection under a sky lit from one side; at degree 3 it resolves a horizon to about twenty degrees (correlation 0.56 at degree 3, 0.63 at degree 5, 0.85 with a 32 × 32 octahedral map, in simulation) (§4.11).
- **Lights inside the scene.** One sun is extracted per dome. Lamps inside a scene are shadowed by bits that say which directions leave the scene, not which reach the lamp (§4.11).
- **One normal, one footprint.** A Gaussian has one normal and a cell's footprint; thin curved strips and rolled edges are where this shows (§9.1). The coverage mapping assumes the cell lattice's overlap $2\pi\sigma^2$,

which sub-cell triangles and edge-held Gaussians do not have (§3.12).

- **Glass.** A tinted thin glass is converted solid, and a cloud keeps a single index of refraction unless it carries layers (§3.12). Thin-sheet Gaussians are lifted to a 0.1 opacity floor per Gaussian, so merging fewer, larger sheets varies the transmission cell to cell; a floor per pixel is pending (§6.11).
- **Size on device.** Device memory bounds a part at about 5 M leaves for the merge tree; the 4.5 M cap made larger parts convert at a coarser camera resolution (Bottom at 3392 px, Part_3 and Front_Bumper_001 at 4480 px; §3.11, §6.9).
- **Instancing.** Nested point instancers are expanded at the innermost level only, and motion authored as velocities is read at the sample (§3.12).

9.3 Animated assets

- Four influences per Gaussian truncate the sparrow's wing tips (27 % of wing vertices carry more); the share of truncated Gaussians is unmeasured (§5.3, §5.13).
- No dual-quaternion skinning for clouds, no retargeting and no clip blending; several skinned meshes at different prim transforms keep only the first one's (§5.13).
- The traced route poses once per frame, so it does not blur a skinned cloud, and LoD instances carry no motion (§5.13).
- Per-part visibility leaves out a part's own self-shadowing, assumes rigid parts that occlude independently, and has not been baked beyond about 3 M Gaussians per part; zonal transfers keep only the direct half and the bind pose's self-shadowing (§4.11, §5.8, §5.13).
- The zonal fit's error on the sparrow, and a geometric measure of coverage at a close bend, are unmeasured (§5.13).

9.4 The engine

- Splats are not yet inside the path tracer's secondary rays: with meshes, splats are composited over path-traced surfaces by depth and cast shadows onto them, but a bounce ray does not see them (§2.16).
- The rasteriser's motion and lens blur are moment-matched Gaussians, up to 38 % too peaked for long streaks (§2.16).
- Gaussians a few hundredths of a pixel across vanish from the raster; levels of detail are the remedy (§2.16).
- Raster mesh visibility pays about 8 ms of host time per 1800 draws; material compilation is synchronous; the TLAS is rebuilt every frame; lens distortion beyond k_1, k_2 is not applied (§2.16).

9.5 Levels of detail, streaming and the player

- **Occlusion weighting.** The shared codebook is apportioned by entries used, not by how often a view culls a node; weighting by an orbit of views in the bake should lift culling past 84 % of the ideal (§6.11).
- **Not ported.** The closed-shell back rule and the error cut live in `athenea-pieces`; the main engine's `CutSelector` is unchanged (§6.11).
- **Residency.** Pages hold about 4× the cut because needed and view-culled nodes share every page; the player splits traversal from payload rather than shrinking pages (§6.8, §6.11).
- **Merge opacity in the player.** The player draws each merge's mass, while athenea's rasteriser keeps the least-squares coverage; the coverage fit is the better choice except on the grille (§6.4).

- **The cut in the player.** The compact traversal does not keep a merge's LoD size, so a combined size/error cut has nothing to guard with, and the error cut alone did not beat the size cut in every view (§7.13).

9.6 Gaps in the validation

- **One production asset.** The car-level figures are for one asset, the Corvette, with the sparrow and the pawn as the animated and rigid controls. Generality to other assets has not been measured.
- **One environment for the car.** The car-level comparisons against the path trace are under one sky (`autoshop_01_4k`, §1.6). Relighting fidelity under other skies and local lights has not been measured at the scale of the car.
- **The 5120 px close-up level against the path trace.** Its trees and shared occlusion are written (§6.9); the close-up softness it addresses (§7.12.1) has been shown resolved only on a logo-and-grille test (§3.11), not on the whole car.
- **v8 on the phone.** The close-up level's conversion for the player was not complete at the time of writing. Its quality against the path trace in the player, its download and its behaviour on the real iPhone are unmeasured (§7.13). The 512-cluster basis has not been measured on the real phone (§7.2.2).
- **Real-device timing** exists only for the stress ramps and two device tests (§7.13); the emulated phone's speed is not the phone's.

9.7 Future work

In the order the measurements suggest:

1. Bake the lamps' emission into the transfer, so that the headlights and their light on the car match the path trace.
2. Fit curvature over each Gaussian's footprint from mesh samples, to halve the hood highlights' offset and steady the per-pixel turn.
3. Move to 32×32 open-direction cells where a sharp coat mirrors the car itself (the door's open class).
4. Address thin chrome tubes and rolled lips (the bumper's strip and lower lip), the rim's near-ground reflection and far thin trims.
5. Measure the relit car against the path trace under varied environments and local lights, and the 5120 px close-up level over the whole car.
6. For animated assets: more than four influences, cross-part occlusion in poses other than the bind pose, and visibility bakes beyond 3 M Gaussians per part.
7. Weight the occlusion codebook by an orbit of views, and port the error cut and the closed-shell rule into the main engine's `CutSelector`.
8. In the mobile application, measure v8 against the path trace and on the real iPhone, close up and far, with the 512-cluster basis.

10. References

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Appendix A. Repositories and commands

A.1 Repositories

repository	where	what it holds
athenea	github.com/jesusluque/athenea	the engine and renderer of §2: modules, shaders, the `hdAthenea` delegate, the `athenea` CLI, the manuals `docs/operations.md` and `docs/development.md` with their Spanish translations, and `docs/decisions.md`
athenea-pieces	github.com/jesusluque/athenea-pieces, branch `pieces` (private)	a fork of athenea holding the per-part merge trees, the error cut, closed shells, the shared occlusion, the 1280 and 5120 px bakes (§3.11, §6); `docs/pieces.md` and this paper under `docs/paper/`
sparkwebGPU	`~/luc/sparkwebGPU`, branch `webgpu`	the WebGPU fork of Spark of §7: the `.athc` v3 reader, the compact path, the streamer, the player pages

A.2 Commands and options

The manual for every subcommand and option is `docs/operations.md` (Spanish: `docs/operations.es.md`), which gives each option's default, unit and constraint. The commands this paper relies on:

command	used for	manual section	§
`athenea mesh2splat STAGE` (`--cell-from-camera`, `--camera-pixels`, `--transfer`, `--hide`, `--max-cells`, `--specular-filter`, `--edge-reach`, `--shadow-catcher`, `--light-catcher`, `--validate`, `--skinned`, `--lod-levels`)	conversion and bake	2.8, 3.1	3, 4, 5
`athenea decimate` (`--tree`, `--passes`, `--merge-glass`, `--object-groups`, `--tree-coverage`, `--tree-pages`, `--occlusion-mask`, `--occlusion-group`, `--cut-budget`, `--cut-cull-back`)	merge trees, cuts, pages, occlusion	2.4.1, 3.3	5.9, 6
`athenea visibility`	per-part visibility fields	2.9	5.8
`athenea stage`, `athenea view`	timings (`--frames`) and the viewer	2.5, 2.6	2
`athenea compare`	GPU image comparison	2.11	2.15
`athenea aofx`	the AOFX effects (mesh2splat, Measure)	2.10, 7	2.14, 3.1

command	used for	manual section	§
`athc-convert (--gzip --planes, --shadow-vq), fit-compact-basis, floor_shadow --merge 2, usd-athc`	the player's formats (sparkwebGPU)	—	7

Environment variables (`ATHENEA_GPU_BUDGET`, `ATHENEA_STAGES`) are listed in `docs/operations.md` §8.1, the formats (`.athc`, the codeless schemas) in §8.2 and §4.3.